



The Communist Manifesto

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Manifesto of the Communist Party

[From the English edition of 1888, edited by Friedrich Engels]

A spectre is haunting Europe—the spectre of Communism. All the Powers of old Europe have entered into a holy alliance to exorcise this spectre: Pope and Czar, Metternich and Guizot, French Radicals and German police-spies.

Where is the party in opposition that has not been decried as Communistic by its opponents in power? Where is the Opposition that has not hurled back the branding reproach of Communism, against the more advanced opposition parties, as well as against its reactionary adversaries?

Two things result from this fact.

I. Communism is already acknowledged by all European Powers to be itself a Power.

II. It is high time that Communists should openly, in the face of the whole world, publish their views, their aims, their tendencies, and meet this nursery tale of the Spectre of Communism with a Manifesto of the party itself.

To this end, Communists of various nationalities have assembled in London, and sketched the following Manifesto, to be published in the English, French, German, Italian, Flemish and Danish languages.

I. Bourgeois and Proletarians

The history of all hitherto existing societies is the history of class struggles.

Freeman and slave, patrician and plebeian, lord and serf, guild-master and journeyman, in a word, oppressor and oppressed, stood in constant opposition to one another, carried on an uninterrupted, now hidden, now open fight, a fight that each time ended, either in a revolutionary reconstitution of society at large or in the common ruin of the contending classes.

In the earlier epochs of history, we find almost everywhere a complicated arrangement of society into various orders, a manifold gradation of social rank. In ancient Rome, we have patricians, knights, plebeians, slaves; in the Middle Ages, feudal lords, vassals, guild-masters, journeymen, apprentices, serfs; in almost all of these classes, again, subordinate gradations.

The modern bourgeois society that has sprouted from the ruins of feudal society has not done away with class antagonisms. It has but established new classes, new conditions of oppression, new forms of struggle in place of the old ones. Our epoch, the epoch of the bourgeoisie, possesses, however, this distinctive feature: it has simplified the class antagonisms. Society as a whole is more and more splitting up into two great hostile camps, into two great classes, directly facing each other: Bourgeoisie and Proletariat.

From the serfs of the Middle Ages sprang the chartered burghers of the earliest towns. From these burgesses, the first elements of the bourgeoisie were developed.

The discovery of America, the rounding of the Cape, opened up fresh ground for the rising bourgeoisie. The East-Indian and Chinese markets, the colonization of America, trade with the colonies, the increase in the means of exchange and in commodities generally, gave to commerce, to navigation, to industry, an impulse never before known, and thereby, to

the revolutionary element in the tottering feudal society, a rapid development.

The feudal system of industry, under which industrial production was monopolized by closed guilds, now no longer sufficed for the growing wants of the new markets. The manufacturing system took its place. The guild-masters were pushed on one side by the manufacturing middle class; division of labour between the different corporate guilds vanished in the face of division of labour in every single workshop.

Meantime the markets kept ever growing, the demand ever rising. Even manufacture no longer sufficed. Thereupon, steam and machinery revolutionized industrial production. The place of manufacture was taken by the giant, Modern Industry, the place of the industrial middle class, by industrial millionaires, the leaders of whole industrial armies, the modern bourgeois.

Modern industry has established the world-market, for which the discovery of America paved the way. This market has given an immense development to commerce, to navigation, to communication by land. This development has, in its time, reacted on the extension of industry; and in proportion as industry, commerce, navigation, railways extended, in the same proportion the bourgeoisie developed, increased its capital, and pushed into the background every class handed down from the Middle Ages.

We see, therefore, how the modern bourgeoisie is itself the product of a long course of development, of a series of revolutions in the modes of production and of exchange.

Each step in the development of the bourgeoisie was accompanied by a corresponding political advance of that class. An oppressed class under the sway of the feudal nobility, an armed and self-governing association in the medieval commune; here independent urban republic (as in Italy and Germany), there taxable "third estate" of the monarchy (as in France), afterwards, in the period of manufacture proper, serving either the semi-feudal or the absolute monarchy as a counterpoise against the nobility, and, in fact, the corner-stone of the great monarchies in general, the

bourgeoisie has at last, since the establishment of Modern Industry and of the world market, conquered for itself, in the modern representative State, exclusive political sway. The executive of the modern State is but a committee for managing the common affairs of the whole bourgeoisie.

The bourgeoisie, historically, has played a most revolutionary part.

The bourgeoisie, wherever it has got the upper hand, has put an end to all feudal, patriarchal, idyllic relations. It has pitilessly torn asunder the motley feudal ties that bound man to his “natural superiors,” and has left remaining no other nexus between man and man than naked self-interest, than callous “cash payment.” It has drowned the most heavenly ecstasies of religious fervour, of chivalrous enthusiasm, of philistine sentimentalism, in the icy water of egotistical calculation. It has resolved personal worth into exchange value, and in place of the numberless and indefeasible chartered freedoms, has set up that single, unconscionable freedom—Free Trade. In one word, for exploitation, veiled by religious and political illusions, naked, shameless, direct, brutal exploitation.

The bourgeoisie has stripped of its halo every occupation hitherto honoured and looked up to with reverent awe. It has converted the physician, the lawyer, the priest, the poet, the man of science, into its paid wage labourers.

The bourgeoisie has torn away from the family its sentimental veil and has reduced the family relation to a mere money relation.

The bourgeoisie has disclosed how it came to pass that the brutal display of vigour in the Middle Ages, which Reactionists so much admire, found its fitting complement in the most slothful indolence. It has been the first to show what man’s activity can bring about. It has accomplished wonders far surpassing Egyptian pyramids, Roman aqueducts, and Gothic cathedrals; it has conducted expeditions that put in the shade all former Exoduses of nations and crusades.

The bourgeoisie cannot exist without constantly

revolutionizing the instruments of production, and thereby the relations of production, and with them the whole relations of society. Conservation of the old modes of production in unaltered form, was, on the contrary, the first condition of existence for all earlier industrial classes. Constant revolutionizing of production, uninterrupted disturbance of all social conditions, everlasting uncertainty and agitation distinguish the bourgeois epoch from all earlier ones. All fixed, fast-frozen relations, with their train of ancient and venerable prejudices and opinions, are swept away, all new-formed ones become antiquated before they can ossify. All that is solid melts into air, all that is holy is profaned, and man is at last compelled to face with sober senses, his real conditions of life, and his relations with his kind.

The need for a constantly expanding market for its products chases the bourgeoisie over the whole surface of the globe. It must nestle everywhere, settle everywhere, establish connexions everywhere.

The bourgeoisie has through its exploitation of the world-market given a cosmopolitan character to production and consumption in every country. To the great chagrin of Reactionists, it has drawn from under the feet of industry the national ground on which it stood. All old-established national industries have been destroyed or are daily being destroyed. They are dislodged by new industries, whose introduction becomes a life and death question for all civilized nations, by industries that no longer work up indigenous raw material, but raw material drawn from the remotest zones; industries whose products are consumed, not only at home but in every quarter of the globe. In place of the old wants, satisfied by the productions of the country, we find new wants, requiring for their satisfaction the products of distant lands and climes. In place of the old local and national seclusion and self-sufficiency, we have intercourse in every direction, universal inter-dependence of nations. And as in material, so also in intellectual production. The intellectual creations of individual nations become common property. National one-sidedness and narrow-mindedness become

more and more impossible, and from the numerous national and local literatures, there arises a world literature.

The bourgeoisie, by the rapid improvement of all instruments of production, by the immensely facilitated means of communication, draws all, even the most barbarian, nations into civilization. The cheap prices of its commodities are the heavy artillery with which it batters down all Chinese walls, with which it forces the barbarians' intensely obstinate hatred of foreigners to capitulate. It compels all nations, on pain of extinction, to adopt the bourgeois mode of production; it compels them to introduce what it calls civilization into their midst, i.e., to become bourgeois themselves. In one word, it creates a world after its own image.

The bourgeoisie has subjected the country to the rule of the towns. It has created enormous cities, has greatly increased the urban population as compared with the rural, and has thus rescued a considerable part of the population from the idiocy of rural life. Just as it has made the country dependent on the towns, so it has made barbarian and semi-barbarian countries dependent on the civilized ones, nations of peasants on nations of bourgeois, the East on the West.

The bourgeoisie keeps more and more doing away with the scattered state of the population, of the means of production, and of property. It has agglomerated production and has concentrated property in a few hands. The necessary consequence of this was political centralization. Independent, or but loosely connected provinces, with separate interests, laws, governments and systems of taxation, became lumped together into one nation, with one government, one code of laws, one national class-interest, one frontier and one customs-tariff. The bourgeoisie, during its rule of scarce one hundred years, has created more massive and more colossal productive forces than have all preceding generations together. Subjection of Nature's forces to man, machinery, application of chemistry to industry and agriculture, steam navigation, railways, electric telegraphs, clearing of whole continents for cultivation, canalization

of rivers, whole populations conjured out of the ground—what earlier century had even a presentiment that such productive forces slumbered in the lap of social labour?

We see then: the means of production and of exchange, on whose foundation the bourgeoisie built itself up, were generated in feudal society. At a certain stage in the development of these means of production and of exchange, the conditions under which feudal society produced and exchanged, the feudal organization of agriculture and manufacturing industry, in one word, the feudal relations of property became no longer compatible with the already developed productive forces; they became so many fetters. They had to be burst asunder; they were burst asunder.

Into their place stepped free competition, accompanied by a social and political constitution adapted to it, and by the economical and political sway of the bourgeois class.

A similar movement is going on before our own eyes. Modern bourgeois society with its relations of production, of exchange and of property, a society that has conjured up such gigantic means of production and of exchange, is like the sorcerer, who is no longer able to control the powers of the nether world whom he has called up by his spells. For many a decade past the history of industry and commerce is but the history of the revolt of modern productive forces against modern conditions of production, against the property relations that are the conditions for the existence of the bourgeoisie and of its rule. It is enough to mention the commercial crises that by their periodical return put on its trial, each time more threateningly, the existence of the entire bourgeois society. In these crises, a great part not only of the existing products but also of the previously created productive forces, are periodically destroyed. In these crises there breaks out an epidemic that, in all earlier epochs, would have seemed an absurdity—the epidemic of over-production. Society suddenly finds itself put back into a state of momentary barbarism; it appears as if a famine, a universal war of devastation had cut off the supply of every means of subsistence; industry and commerce seem to

be destroyed; and why? Because there is too much civilization, too much means of subsistence, too much industry, too much commerce. The productive forces at the disposal of society no longer tend to further the development of the conditions of bourgeois property; on the contrary, they have become too powerful for these conditions, by which they are fettered, and so soon as they overcome these fetters, they bring disorder into the whole of bourgeois society, endanger the existence of bourgeois property. The conditions of bourgeois society are too narrow to comprise the wealth created by them. And how does the bourgeoisie get over these crises? On the one hand enforced destruction of a mass of productive forces; on the other, by the conquest of new markets, and by the more thorough exploitation of the old ones. That is to say, by paving the way for more extensive and more destructive crises, and by diminishing the means whereby crises are prevented.

The weapons with which the bourgeoisie felled feudalism to the ground are now turned against the bourgeoisie itself.

But not only has the bourgeoisie forged the weapons that bring death to itself; it has also called into existence the men who are to wield those weapons—the modern working class—the proletarians.

In proportion as the bourgeoisie, i.e., capital is developed, in the same proportion is the proletariat, the modern working class, developed—a class of labourers, who live only so long as they find work, and who find work only so long as their labour increases capital. These labourers, who must sell themselves piece-meal, are a commodity, like every other article of commerce, and are consequently exposed to all the vicissitudes of competition, to all the fluctuations of the market.

Owing to the extensive use of machinery and to division of labour, the work of the proletarians has lost all individual character, and consequently, all charm for the workman. He becomes an appendage of the machine, and it is only the most simple, most monotonous, and most easily acquired knack, that is required of

him. Hence, the cost of production of a workman is restricted, almost entirely, to the means of subsistence that he requires for his maintenance, and for the propagation of his race. But the price of a commodity, and therefore also of labour, is equal to its cost of production. In proportion, therefore, as the repulsiveness of the work increases, the wage decreases. Nay more, in proportion as the use of machinery and division of labour increases, in the same proportion the burden of toil also increases, whether by prolongation of the working hours, by increase of the work exacted in a given time or by increased speed of the machinery, etc.

Modern industry has converted the little workshop of the patriarchal master into the great factory of the industrial capitalist. Masses of labourers crowded into the factory, are organized like soldiers. As privates of the industrial army, they are placed under the command of a perfect hierarchy of officers and sergeants. Not only are they slaves of the bourgeois class, and of the bourgeois State; they are daily and hourly enslaved by the machine, by the over-looker, and, above all, by the individual bourgeois manufacturer himself. The more openly this despotism proclaims gain to be its end and aim, the more petty, the more hateful and the more embittering it is.

The less the skill and exertion of strength implied in manual labour, in other words, the more modern industry becomes developed, the more is the labour of men superseded by that of women. Differences of age and sex have no longer any distinctive social validity for the working class. All are instruments of labour, more or less expensive to use, according to their age and sex.

No sooner is the exploitation of the labourer by the manufacturer, so far at an end, that he receives his wages in cash than he is set upon by the other portions of the bourgeoisie, the landlord, the shopkeeper, the pawnbroker, etc.

The lower strata of the middle class—the small tradespeople, shopkeepers, retired tradesmen generally, the handicraftsmen and peasants—all these sink gradually into the

proletariat, partly because their diminutive capital does not suffice for the scale on which Modern Industry is carried on, and is swamped in the competition with the large capitalists, partly because their specialized skill is rendered worthless by the new methods of production. Thus the proletariat is recruited from all classes of the population.

The proletariat goes through various stages of development. With its birth begins its struggle with the bourgeoisie. At first, the contest is carried on by individual labourers, then by the workpeople of a factory, then by the operatives of one trade, in one locality, against the individual bourgeois who directly exploits them. They direct their attacks not against the bourgeois conditions of production, but against the instruments of production themselves; they destroy imported wares that compete with their labour, they smash to pieces machinery, they set factories ablaze, they seek to restore by force the vanished status of the workman of the Middle Ages.

At this stage, the labourers still form an incoherent mass scattered over the whole country, and broken up by their mutual competition. If anywhere they unite to form more compact bodies, this is not yet the consequence of their own active union, but of the union of the bourgeoisie, which class, in order to attain its own political ends, is compelled to set the whole proletariat in motion, and is moreover yet, for a time, able to do so. At this stage, therefore, the proletarians do not fight their enemies, but the enemies of their enemies, the remnants of absolute monarchy, the landowners, the non-industrial bourgeois, the petty bourgeoisie. Thus the whole historical movement is concentrated in the hands of the bourgeoisie; every victory so obtained is a victory for the bourgeoisie.

But with the development of industry, the proletariat not only increases in number; it becomes concentrated in greater masses, its strength grows, and it feels that strength more. The various interests and conditions of life within the ranks of the proletariat are more and more equalized, in proportion as machinery obliterates all distinctions of labour, and

nearly everywhere reduces wages to the same low level. The growing competition among the bourgeois, and the resulting commercial crises, make the wages of the workers ever more fluctuating. The unceasing improvement of machinery, ever more rapidly developing, makes their livelihood more and more precarious; the collisions between individual workmen and individual bourgeois take more and more the character of collisions between two classes. Thereupon the workers begin to form combinations (Trades Unions) against the bourgeois; they club together in order to keep up the rate of wages; they found permanent associations in order to make provision beforehand for these occasional revolts. Here and there the contest breaks out into riots.

Now and then the workers are victorious, but only for a time. The real fruit of their battles lies, not in the immediate result, but in the ever-expanding union of the workers. This union is helped by the improved means of communication that are created by modern industry and that place the workers of different localities in contact with one another. It was just this contact that was needed to centralize the numerous local struggles, all of the same character, into one national struggle between classes. But every class struggle is a political struggle. And that union, to attain which the burghers of the Middle Ages, with their miserable highways, required centuries, the modern proletarians, thanks to railways, achieve in a few years.

This organization of the proletarians into a class, and consequently into a political party, is continually being upset again by the competition between the workers themselves. But it ever rises up again, stronger, firmer, mightier. It compels legislative recognition of particular interests of the workers, by taking advantage of the divisions among the bourgeoisie itself. Thus the ten-hours' bill in England was carried.

Altogether collisions between the classes of the old society further, in many ways, the course of development of the proletariat. The bourgeoisie finds itself involved in a

constant battle. At first with the aristocracy; later on, with those portions of the bourgeoisie itself, whose interests have become antagonistic to the progress of industry; at all times, with the bourgeoisie of foreign countries. In all these battles it sees itself compelled to appeal to the proletariat, to ask for its help, and thus, to drag it into the political arena. The bourgeoisie itself, therefore, supplies the proletariat with its own instruments of political and general education, in other words, it furnishes the proletariat with weapons for fighting the bourgeoisie.

Further, as we have already seen, entire sections of the ruling classes are, by the advance of industry, precipitated into the proletariat, or are at least threatened in their conditions of existence. These also supply the proletariat with fresh elements of enlightenment and progress.

Finally, in times when the class struggle nears the decisive hour, the process of dissolution going on within the ruling class, in fact within the whole range of society, assumes such a violent, glaring character, that a small section of the ruling class cuts itself adrift, and joins the revolutionary class, the class that holds the future in its hands. Just as, therefore, at an earlier period, a section of the nobility went over to the bourgeoisie, so now a portion of the bourgeoisie goes over to the proletariat, and in particular, a portion of the bourgeois ideologists, who have raised themselves to the level of comprehending theoretically the historical movement as a whole.

Of all the classes that stand face to face with the bourgeoisie today, the proletariat alone is a really revolutionary class. The other classes decay and finally disappear in the face of Modern Industry; the proletariat is its special and essential product. The lower middle class, the small manufacturer, the shopkeeper, the artisan, the peasant, all these fight against the bourgeoisie, to save from extinction their existence as fractions of the middle class. They are therefore not revolutionary, but conservative. Nay more, they are reactionary, for they try to roll back the wheel of history. If by chance they are revolutionary, they are so only in view of their impending transfer into the proletariat,

they thus defend not their present, but their future interests, they desert their own standpoint to place themselves at that of the proletariat.

The “dangerous class,” the social scum, that passively rotting mass thrown off by the lowest layers of old society, may, here and there, be swept into the movement by a proletarian revolution; its conditions of life, however, prepare it far more for the part of a bribed tool of reactionary intrigue.

In the conditions of the proletariat, those of old society at large are already virtually swamped. The proletarian is without property; his relation to his wife and children has no longer anything in common with the bourgeois family relations; modern industrial labour, modern subjection to capital, the same in England as in France, in America as in Germany, has stripped him of every trace of national character. Law, morality, religion, are to him so many bourgeois prejudices, behind which lurk in ambush just as many bourgeois interests.

All the preceding classes that got the upper hand, sought to fortify their already acquired status by subjecting society at large to their conditions of appropriation. The proletarians cannot become masters of the productive forces of society, except by abolishing their own previous mode of appropriation, and thereby also every other previous mode of appropriation. They have nothing of their own to secure and to fortify; their mission is to destroy all previous securities for, and insurances of, individual property.

All previous historical movements were movements of minorities, or in the interests of minorities. The proletarian movement is the self-conscious, independent movement of the immense majority, in the interests of the immense majority. The proletariat, the lowest stratum of our present society, cannot stir, cannot raise itself up, without the whole superincumbent strata of official society being sprung into the air.

Though not in substance, yet in form, the struggle of the proletariat with the bourgeoisie is at first a national

struggle. The proletariat of each country must, of course, first of all, settle matters with its own bourgeoisie.

In depicting the most general phases of the development of the proletariat, we traced the more or less veiled civil war, raging within existing society, up to the point where that war breaks out into open revolution, and where the violent overthrow of the bourgeoisie lays the foundation for the sway of the proletariat.

Hitherto, every form of society has been based, as we have already seen, on the antagonism of oppressing and oppressed classes. But in order to oppress a class, certain conditions must be assured to it under which it can, at least, continue its slavish existence. The serf, in the period of serfdom, raised himself to membership in the commune, just as the petty bourgeois, under the yoke of feudal absolutism, managed to develop into a bourgeois. The modern labourer, on the contrary, instead of rising with the progress of industry, sinks deeper and deeper below the conditions of existence of his own class. He becomes a pauper, and pauperism develops more rapidly than population and wealth. And here it becomes evident, that the bourgeoisie is unfit any longer to be the ruling class in society, and to impose its conditions of existence upon society as an over-riding law. It is unfit to rule because it is incompetent to assure an existence to its slave within his slavery, because it cannot help letting him sink into such a state, that it has to feed him, instead of being fed by him. Society can no longer live under this bourgeoisie, in other words, its existence is no longer compatible with society.

The essential condition for the existence, and for the sway of the bourgeois class, is the formation and augmentation of capital; the condition for capital is wage-labour. Wage-labour rests exclusively on competition between the labourers. The advance of industry, whose involuntary promoter is the bourgeoisie, replaces the isolation of the labourers, due to competition, by their revolutionary combination, due to association. The development of Modern Industry, therefore, cuts from under its feet the very foundation on which the bourgeoisie produces and appropriates products. What

the bourgeoisie, therefore, produces, above all, is its own grave-diggers. Its fall and the victory of the proletariat are equally inevitable.

II. Proletarians and Communists

In what relation do the Communists stand to the proletarians as a whole?

The Communists do not form a separate party opposed to other working-class parties.

They have no interests separate and apart from those of the proletariat as a whole.

They do not set up any sectarian principles of their own, by which to shape and mould the proletarian movement.

The Communists are distinguished from the other working-class parties by this only: (1) In the national struggles of the proletarians of the different countries, they point out and bring to the front the common interests of the entire proletariat, independently of all nationality. (2) In the various stages of development which the struggle of the working class against the bourgeoisie has to pass through, they always and everywhere represent the interests of the movement as a whole.

The Communists, therefore, are, on the one hand, practically, the most advanced and resolute section of the working-class parties of every country, that section which pushes forward all others; on the other hand, theoretically, they have over the great mass of the proletariat the advantage of clearly understanding the line of march, the conditions, and the ultimate general results of the proletarian movement.

The immediate aim of the Communist is the same as that of all the other proletarian parties: formation of the proletariat into a class, overthrow of the bourgeois supremacy, conquest of political power by the proletariat.

The theoretical conclusions of the Communists are in no way based on ideas or principles that have been invented

or discovered, by this or that would-be universal reformer. They merely express, in general terms, actual relations springing from an existing class struggle, from a historical movement going on under our very eyes. The abolition of existing property relations is not at all a distinctive feature of Communism.

All property relations in the past have continually been subject to historical change consequent upon the change in historical conditions.

The French Revolution, for example, abolished feudal property in favour of bourgeois property.

The distinguishing feature of Communism is not the abolition of property generally, but the abolition of bourgeois property. But modern bourgeois private property is the final and most complete expression of the system of producing and appropriating products, that is based on class antagonisms, on the exploitation of the many by the few.

In this sense, the theory of the Communists may be summed up in the single sentence: Abolition of private property.

We Communists have been reproached with the desire of abolishing the right of personally acquiring property as the fruit of a man's own labour, which property is alleged to be the groundwork of all personal freedom, activity and independence.

Hard-won, self-acquired, self-earned property! Do you mean the property of the petty artisan and of the small peasant, a form of property that preceded the bourgeois form? There is no need to abolish that; the development of industry has to a great extent already destroyed it and is still destroying it daily.

Or do you mean modern bourgeois private property?

But does wage-labour create any property for the labourer? Not a bit. It creates capital, i.e., that kind of property which exploits wage-labour, and which cannot increase except upon condition of begetting a new supply of wage-labour for fresh exploitation. Property, in its present form, is based on the antagonism of capital and wage-labour. Let us examine both sides of this antagonism.

To be a capitalist is to have not only a purely personal, but a social

status in production. Capital is a collective product, and only by the united action of many members, nay, in the last resort, only by the united action of all members of society, can it be set in motion.

Capital is, therefore, not a personal, it is a social power.

When, therefore, capital is converted into common property, into the property of all members of society, personal property is not thereby transformed into social property. It is only the social character of the property that is changed. It loses its class character.

Let us now take wage-labour.

The average price of wage-labour is the minimum wage, i.e., that quantum of the means of subsistence, which is absolutely requisite in bare existence as a labourer. What, therefore, the wage-labourer appropriates by means of his labour, merely suffices to prolong and reproduce a bare existence. We by no means intend to abolish this personal appropriation of the products of labour, an appropriation that is made for the maintenance and reproduction of human life, and that leaves no surplus wherewith to command the labour of others. All that we want to do away with is the miserable character of this appropriation, under which the labourer lives merely to increase capital, and is allowed to live only in so far as the interest of the ruling class requires it.

In bourgeois society, living labour is but a means to increase accumulated labour. In Communist society, accumulated labour is but a means to widen, to enrich, to promote the existence of the labourer.

In bourgeois society, therefore, the past dominates the present; in Communist society, the present dominates the past. In bourgeois society, capital is independent and has individuality, while the living person is dependent and has no individuality.

And the abolition of this state of things is called by the bourgeois, abolition of individuality and freedom! And rightly so. The abolition of bourgeois individuality, bourgeois independence, and bourgeois freedom is undoubtedly aimed at.

By freedom is meant, under the present bourgeois conditions of production, free trade, free selling and buying.

But if selling and buying disappears, free selling and buying disappears also. This talk about free selling and buying, and all the other “brave words” of our bourgeoisie about freedom in general, have a meaning, if any, only in contrast with restricted selling and buying, with the fettered traders of the Middle Ages, but have no meaning when opposed to the Communistic abolition of buying and selling, of the bourgeois conditions of production, and of the bourgeoisie itself.

You are horrified at our intending to do away with private property. But in your existing society, private property is already done away with for nine-tenths of the population; its existence for the few is solely due to its non-existence in the hands of those nine-tenths. You reproach us, therefore, with intending to do away with a form of property, the necessary condition for whose existence is the non-existence of any property for the immense majority of society.

In one word, you reproach us with intending to do away with your property. Precisely so; that is just what we intend.

From the moment when labour can no longer be converted into capital, money, or rent, into a social power capable of being monopolized, i.e., from the moment when individual property can no longer be transformed into bourgeois property, into capital, from that moment, you say individuality vanishes.

You must, therefore, confess that by “individual” you mean no other person than the bourgeois, than the middle-class owner of property. This person must, indeed, be swept out of the way, and made impossible.

Communism deprives no man of the power to appropriate the products of society; all that it does is to deprive him of the power to subjugate the labour of others by means of such appropriation.

It has been objected that upon the abolition of private property all work will cease, and universal laziness will overtake us.

According to this, bourgeois society ought long ago to have gone to the dogs through sheer idleness; for those of its members who work, acquire nothing, and those who acquire anything, do not work. The whole of this objection is but another expression of the tautology: that there can no longer be any wage-labour when there is no longer any capital.

All objections urged against the Communistic mode of producing and appropriating material products, have, in the same way, been urged against the Communistic modes of producing and appropriating intellectual products. Just as, to the bourgeois, the disappearance of class property is the disappearance of production itself, so the disappearance of class culture is to him identical with the disappearance of all culture.

That culture, the loss of which he laments, is, for the enormous majority, a mere training to act as a machine.

But don't wrangle with us so long as you apply, to our intended abolition of bourgeois property, the standard of your bourgeois notions of freedom, culture, law, etc. Your very ideas are but the outgrowth of the conditions of your bourgeois production and bourgeois property, just as your jurisprudence is but the will of your class made into a law for all, a will, whose essential character and direction are determined by the economical conditions of existence of your class.

The selfish misconception that induces you to transform into eternal laws of nature and of reason, the social forms springing from your present mode of production and form of property—historical relations that rise and disappear in the progress of production—this misconception you share with every ruling class that has preceded you. What you see clearly in the case of ancient property, what you admit in the case of feudal property, you are of course forbidden to admit in the case of your own bourgeois form of property.

Abolition of the family! Even the most radical flare up at this infamous proposal of the Communists.

On what foundation is the present family, the bourgeois

family, based? On capital, on private gain. In its completely developed form, this family exists only among the bourgeoisie. But this state of things finds its complement in the practical absence of the family among the proletarians, and in public prostitution.

The bourgeois family will vanish as a matter of course when its complement vanishes, and both will vanish with the vanishing of capital.

Do you charge us with wanting to stop the exploitation of children by their parents? To this crime, we plead guilty.

But, you will say, we destroy the most hallowed of relations when we replace home education by social.

And your education! Is not that also social, and determined by the social conditions under which you educate, by the intervention, direct or indirect, of society, by means of schools, etc.? The Communists have not invented the intervention of society in education; they do but seek to alter the character of that intervention and to rescue education from the influence of the ruling class.

The bourgeois clap-trap about the family and education, about the hallowed co-relation of parent and child, becomes all the more disgusting, the more, by the action of Modern Industry, all family ties among the proletarians are torn asunder, and their children transformed into simple articles of commerce and instruments of labour.

But you Communists would introduce community of women, screams the whole bourgeoisie in chorus.

The bourgeois sees in his wife a mere instrument of production. He hears that the instruments of production are to be exploited in common, and, naturally, can come to no other conclusion than that the lot of being common to all will likewise fall to the women.

He has not even a suspicion that the real point is to do away with the status of women as mere instruments of production.

For the rest, nothing is more ridiculous than the virtuous indignation of our bourgeois at the community of women which,

they pretend, is to be openly and officially established by the Communists. The Communists have no need to introduce a community of women; it has existed almost from time immemorial.

Our bourgeois, not content with having the wives and daughters of their proletarians at their disposal, not to speak of common prostitutes, take the greatest pleasure in seducing each other's wives.

Bourgeois marriage is, in reality, a system of wives in common and thus, at the most, what the Communists might possibly be reproached with, is that they desire to introduce, in substitution for a hypocritically concealed, an openly legalized community of women. For the rest, it is self-evident that the abolition of the present system of production must bring with it the abolition of the community of women springing from that system, i.e., of prostitution both public and private.

The Communists are further reproached with desiring to abolish countries and nationality.

The working men have no country. We cannot take from them what they have not got. Since the proletariat must, first of all, acquire political supremacy, must rise to be the leading class of the nation, must constitute itself the nation, it is, so far, itself national, though not in the bourgeois sense of the word.

National differences and antagonisms between peoples are daily more and more vanishing, owing to the development of the bourgeoisie, to freedom of commerce, to the world-market, to uniformity in the mode of production and in the conditions of life corresponding thereto.

The supremacy of the proletariat will cause them to vanish still faster. United action, of the leading civilized countries at least, is one of the first conditions for the emancipation of the proletariat.

In proportion as the exploitation of one individual by another is put an end to, the exploitation of one nation by another will also be put an end to. In proportion as the antagonism between classes within the nation vanishes, the hostility of one nation to another will come to an end.

The charges against Communism made from a religious, a philosophical, and, generally, from an ideological standpoint, are not deserving of serious examination.

Does it require deep intuition to comprehend that man's ideas, views and conceptions, in one word, man's consciousness, changes with every change in the conditions of his material existence, in his social relations and in his social life?

What else does the history of ideas prove, than that intellectual production changes its character in proportion as material production is changed? The ruling ideas of each age have ever been the ideas of its ruling class.

When people speak of ideas that revolutionize society, they do but express the fact, that within the old society, the elements of a new one have been created, and that the dissolution of the old ideas keeps even pace with the dissolution of the old conditions of existence.

When the ancient world was in its last throes, the ancient religions were overcome by Christianity. When Christian ideas succumbed in the 18th century to rationalist ideas, feudal society fought its death battle with the then revolutionary bourgeoisie. The ideas of religious liberty and freedom of conscience merely gave expression to the sway of free competition within the domain of knowledge.

"Undoubtedly," it will be said, "religious, moral, philosophical and juridical ideas have been modified in the course of historical development. But religion, morality philosophy, political science, and law, constantly survived this change."

"There are, besides, eternal truths, such as Freedom, Justice, etc. that are common to all states of society. But Communism abolishes eternal truths, it abolishes all religion, and all morality, instead of constituting them on a new basis; it, therefore, acts in contradiction to all past historical experience."

What does this accusation reduce itself to? The history of all past society has consisted in the development of class antagonisms, antagonisms that assumed different forms at different epochs.

But whatever form they may have taken, one fact is common to all past ages, viz., the exploitation of one part of society by the other. No wonder, then, that the social consciousness of past ages, despite all the multiplicity and variety it displays, moves within certain common forms, or general ideas, which cannot completely vanish except with the total disappearance of class antagonisms.

The Communist revolution is the most radical rupture with traditional property relations; no wonder that its development involves the most radical rupture with traditional ideas.

But let us have done with the bourgeois objections to Communism.

We have seen above, that the first step in the revolution by the working class, is to raise the proletariat to the position of ruling as to win the battle of democracy.

The proletariat will use its political supremacy to wrest, by degrees, all capital from the bourgeoisie, to centralize all instruments of production in the hands of the State, i.e., of the proletariat organized as the ruling class; and to increase the total of productive forces as rapidly as possible.

Of course, in the beginning, this cannot be effected except by means of despotic inroads on the rights of property, and on the conditions of bourgeois production; by means of measures, therefore, which appear economically insufficient and untenable, but which, in the course of the movement, outstrip themselves, necessitate further inroads upon the old social order, and are unavoidable as a means of entirely revolutionizing the mode of production.

These measures will of course be different in different countries.

Nevertheless, in the most advanced countries, the following will be pretty generally applicable.

1. Abolition of property in land and application of all rents of land to public purposes.
2. A heavy progressive or graduated income tax.

3. Abolition of all right of inheritance.
4. Confiscation of the property of all emigrants and rebels.
5. Centralization of credit in the hands of the State, by means of a national bank with State capital and an exclusive monopoly.
6. Centralization of the means of communication and transport in the hands of the State.
7. Extension of factories and instruments of production owned by the State; the bringing into cultivation of waste-lands, and the improvement of the soil generally in accordance with a common plan.
8. Equal liability of all to labour. Establishment of industrial armies, especially for agriculture.
9. Combination of agriculture with manufacturing industries; gradual abolition of the distinction between town and country, by a more equable distribution of the population over the country.
10. Free education for all children in public schools. Abolition of children's factory labour in its present form. Combination of education with industrial production, &c., &c.

When, in the course of development, class distinctions have disappeared, and all production has been concentrated in the hands of a vast association of the whole nation, the public power will lose its political character. Political power, properly so-called, is merely the organized power of one class for oppressing another. If the proletariat during its contest with the bourgeoisie is compelled, by the force of circumstances, to organize itself as a class, if, by means of a revolution, it makes itself the ruling class, and, as such, sweeps away by force the old conditions of production, then it will, along with these conditions, have swept away the conditions for the existence of class antagonisms and of classes generally, and will thereby have abolished its own supremacy as a class.

In place of the old bourgeois society, with its classes and class

antagonisms, we shall have an association, in which the free development of each is the condition for the free development of all.

III. Socialist and Communist Literature

I. Reactionary Socialism

A. Feudal Socialism

Owing to their historical position, it became the vocation of the aristocracies of France and England to write pamphlets against modern bourgeois society. In the French revolution of July 1830, and in the English reform agitation, these aristocracies again succumbed to the hateful upstart. Thenceforth, a serious political contest was altogether out of the question. A literary battle alone remained possible. But even in the domain of literature, the old cries of the restoration period had become impossible.

In order to arouse sympathy, the aristocracy was obliged to lose sight, apparently, of their own interests, and to formulate their indictment against the bourgeoisie in the interest of the exploited working class alone. Thus the aristocracy took their revenge by singing lampoons on their new master and whispering in his ears sinister prophecies of coming catastrophe.

In this way arose Feudal Socialism: half lamentation, half lampoon; half echo of the past, half menace of the future; at times, by its bitter, witty and incisive criticism, striking the bourgeoisie to the very heart's core; but always ludicrous in its effect, through total incapacity to comprehend the march of modern history.

The aristocracy, in order to rally the people to them, waved the proletarian alms-bag in front for a banner. But the people, so often as it joined them, saw on their hindquarters the old

feudal coats of arms, and deserted with loud and irreverent laughter.

One section of the French Legitimists and “Young England” exhibited this spectacle.

In pointing out that their mode of exploitation was different to that of the bourgeoisie, the feudalists forget that they exploited under circumstances and conditions that were quite different, and that are now antiquated. In showing that, under their rule, the modern proletariat never existed, they forget that the modern bourgeoisie is the necessary offspring of their own form of society.

For the rest, so little do they conceal the reactionary character of their criticism that their chief accusation against the bourgeoisie amounts to this, that under the bourgeois regime a class is being developed, which is destined to cut up root and branch the old order of society.

What they upbraid the bourgeoisie with is not so much that it creates a proletariat, as that it creates a revolutionary proletariat.

In political practice, therefore, they join in all coercive measures against the working class; and in ordinary life, despite their high falutin phrases, they stoop to pick up the golden apples dropped from the tree of industry, and to barter truth, love, and honour for traffic in wool, beetroot-sugar, and potato spirits.

As the parson has ever gone hand in hand with the landlord, so has Clerical Socialism with Feudal Socialism.

Nothing is easier than to give Christian asceticism a Socialist tinge. Has not Christianity declaimed against private property, against marriage, against the State? Has it not preached in the place of these, charity and poverty, celibacy and mortification of the flesh, monastic life and Mother Church? Christian Socialism is but the holy water with which the priest consecrates the heart-burnings of the aristocrat.

B. Petty-Bourgeois Socialism

The feudal aristocracy was not the only class that was ruined by the bourgeoisie, not the only class whose conditions of existence pined and perished in the atmosphere of modern bourgeois society. The medieval burgesses and the small peasant proprietors were the precursors of the modern bourgeoisie. In those countries which are but little developed, industrially and commercially, these two classes still vegetate side by side with the rising bourgeoisie.

In countries where modern civilization has become fully developed, a new class of petty bourgeois has been formed, fluctuating between proletariat and bourgeoisie and ever renewing itself as a supplementary part of bourgeois society. The individual members of this class, however, are being constantly hurled down into the proletariat by the action of competition, and, as modern industry develops, they even see the moment approaching when they will completely disappear as an independent section of modern society, to be replaced, in manufactures, agriculture and commerce, by overlookers, bailiffs and shopmen.

In countries like France, where the peasants constitute far more than half of the population, it was natural that writers who sided with the proletariat against the bourgeoisie, should use, in their criticism of the bourgeois regime, the standard of the peasant and petty-bourgeois, and from the standpoint of these intermediate classes should take up the cudgels for the working class. Thus arose petty-bourgeois Socialism. Sismondi was the head of this school, not only in France but also in England.

This school of Socialism dissected with great acuteness the contradictions in the conditions of modern production. It laid bare the hypocritical apologies of economists. It proved, incontrovertibly, the disastrous effects of machinery and division of labour; the concentration of capital and land in a few hands; overproduction and crises; it pointed out the inevitable ruin

of the petty-bourgeois and peasant, the misery of the proletariat, the anarchy in production, the crying inequalities in the distribution of wealth, the industrial war of extermination between nations, the dissolution of old moral bonds, of the old family relations, of the old nationalities.

In its positive aims, however, this form of Socialism aspires either to restoring the old means of production and of exchange, and with them the old property relations, and the old society, or to cramping the modern means of production and of exchange, within the framework of the old property relations that have been, and were bound to be, exploded by those means. In either case, it is both reactionary and Utopian.

Its last words are: corporate guilds for manufacture, patriarchal relations in agriculture.

Ultimately, when stubborn historical facts had dispersed all intoxicating effects of self-deception, this form of Socialism ended in a miserable fit of the blues.

C. German, or “True,” Socialism

The Socialist and Communist literature of France, a literature that originated under the pressure of a bourgeoisie in power, and that was the expression of the struggle against this power, was introduced into Germany at a time when the bourgeoisie, in that country, had just begun its contest with feudal absolutism.

German philosophers, would-be philosophers, and beaux esprits, eagerly seized on this literature, only forgetting, that when these writings immigrated from France into Germany, French social conditions had not immigrated along with them. In contact with German social conditions, this French literature lost all its immediate practical significance and assumed a purely literary aspect. Thus, to the German philosophers of the eighteenth century, the demands of the first French Revolution

were nothing more than the demands of “Practical Reason” in general, and the utterance of the will of the revolutionary French bourgeoisie signified in their eyes the law of pure Will, of Will as it was bound to be, of true human Will generally.

The world of the German literate consisted solely in bringing the new French ideas into harmony with their ancient philosophical conscience, or rather, in annexing the French ideas without deserting their own philosophic point of view.

This annexation took place in the same way in which a foreign language is appropriated, namely, by translation.

It is well known how the monks wrote silly lives of Catholic Saints over the manuscripts on which the classical works of ancient heathendom had been written. The German literate reversed this process with the profane French literature. They wrote their philosophical nonsense beneath the French original. For instance, beneath the French criticism of the economic functions of money, they wrote “Alienation of Humanity,” and beneath the French criticism of the bourgeois State, they wrote “dethronement of the Category of the General,” and so forth.

The introduction of these philosophical phrases at the back of the French historical criticisms they dubbed “Philosophy of Action,” “True Socialism,” “German Science of Socialism,” “Philosophical Foundation of Socialism,” and so on.

The French Socialist and Communist literature was thus completely emasculated. And, since it ceased in the hands of the German to express the struggle of one class with the other, he felt conscious of having overcome “French one-sidedness” and of representing, not true requirements, but the requirements of truth; not the interests of the proletariat, but the interests of Human Nature, of Man in general, who belongs to no class, has no reality, who exists only in the misty realm of philosophical fantasy.

This German Socialism, which took its schoolboy task so seriously and solemnly, and extolled its poor stock-in-trade in such mountebank fashion, meanwhile gradually lost its pedantic innocence.

The fight of the German, and especially, of the Prussian bourgeoisie, against feudal aristocracy and absolute monarchy, in other words, the liberal movement, became more earnest.

By this, the long-wished-for opportunity was offered to “True” Socialism of confronting the political movement with the Socialist demands, of hurling the traditional anathemas against liberalism, against representative government, against bourgeois competition, bourgeois freedom of the press, bourgeois legislation, bourgeois liberty and equality, and of preaching to the masses that they had nothing to gain, and everything to lose, by this bourgeois movement. German Socialism forgot, in the nick of time, that the French criticism, whose silly echo it was, presupposed the existence of modern bourgeois society, with its corresponding economic conditions of existence, and the political constitution adapted thereto, the very things whose attainment was the object of the pending struggle in Germany.

To the absolute governments, with their following of parsons, professors, country squires and officials, it served as a welcome scarecrow against the threatening bourgeoisie.

It was a sweet finish after the bitter pills of floggings and bullets with which these same governments, just at that time, dosed the German working-class risings.

While this “True” Socialism thus served the governments as a weapon for fighting the German bourgeoisie, it, at the same time, directly represented a reactionary interest, the interest of the German Philistines. In Germany, the petty-bourgeois class, a relic of the sixteenth century, and since then constantly cropping up again under various forms, is the real social basis of the existing state of things.

To preserve this class is to preserve the existing state of things in Germany. The industrial and political supremacy of the bourgeoisie threatens it with certain destruction; on the one hand, from the concentration of capital; on the other, from the rise of a revolutionary proletariat. “True” Socialism appeared to kill these two birds with one stone. It spread like an epidemic.

The robe of speculative cobwebs, embroidered with flowers of rhetoric, steeped in the dew of sickly sentiment, this transcendental robe in which the German Socialists wrapped their sorry “eternal truths,” all skin and bone, served to wonderfully increase the sale of their goods amongst such a public. And on its part, German Socialism recognized, more and more, its own calling as the bombastic representative of the petty-bourgeois Philistine.

It proclaimed the German nation to be the model nation, and the German petty Philistine to be the typical man. To every villainous meanness of this model man, it gave a hidden, higher, Socialistic interpretation, the exact contrary of its real character. It went to the extreme length of directly opposing the “brutally destructive” tendency of Communism, and of proclaiming its supreme and impartial contempt of all class struggles. With very few exceptions, all the so-called Socialist and Communist publications that now (1847) circulate in Germany belong to the domain of this foul and enervating literature.

2. Conservative, or Bourgeois, Socialism

A part of the bourgeoisie is desirous of redressing social grievances, in order to secure the continued existence of bourgeois society.

To this section belong economists, philanthropists, humanitarians, improvers of the condition of the working class, organizers of charity, members of societies for the prevention of cruelty to animals, temperance fanatics, hole-and-corner reformers of every imaginable kind. This form of Socialism has, moreover, been worked out into complete systems.

We may cite Proudhon's *Philosophie de la Misere* as an example of this form.

The Socialistic bourgeois want all the advantages of modern social conditions without the struggles and dangers necessarily resulting therefrom. They desire the existing state of society minus its

revolutionary and disintegrating elements. They wish for a bourgeoisie without a proletariat. The bourgeoisie naturally conceives the world in which it is supreme to be the best; and bourgeois Socialism develops this comfortable conception into various more or less complete systems. In requiring the proletariat to carry out such a system, and thereby to march straightway into the social New Jerusalem, it but requires in reality, that the proletariat should remain within the bounds of existing society, but should cast away all its hateful ideas concerning the bourgeoisie.

A second and more practical, but less systematic, form of this Socialism sought to depreciate every revolutionary movement in the eyes of the working class, by showing that no mere political reform, but only a change in the material conditions of existence, in economic relations, could be of any advantage to them. By changes in the material conditions of existence, this form of Socialism, however, by no means understands abolition of the bourgeois relations of production, an abolition that can be effected only by a revolution, but administrative reforms, based on the continued existence of these relations; reforms, therefore, that in no respect affect the relations between capital and labour, but, at the best, lessen the cost, and simplify the administrative work, of bourgeois government.

Bourgeois Socialism attains adequate expression, when, and only when, it becomes a mere figure of speech.

Free trade: for the benefit of the working class. Protective duties: for the benefit of the working class. Prison Reform: for the benefit of the working class. This is the last word and the only seriously meant word of bourgeois Socialism.

It is summed up in the phrase: the bourgeois is a bourgeois—for the benefit of the working class.

3. Critical-Utopian Socialism and Communism

We do not here refer to that literature which, in every great modern revolution, has always given voice to the demands of the proletariat, such as the writings of Babeuf and others.

The first direct attempts of the proletariat to attain its own ends, made in times of universal excitement, when feudal society was being overthrown, these attempts necessarily failed, owing to the then undeveloped state of the proletariat, as well as to the absence of the economic conditions for its emancipation, conditions that had yet to be produced and could be produced by the impending bourgeois epoch alone. The revolutionary literature that accompanied these first movements of the proletariat had necessarily a reactionary character. It inculcated universal asceticism and social levelling in its crudest form.

The Socialist and Communist systems properly so called, those of Saint-Simon, Fourier, Owen and others, spring into existence in the early undeveloped period, described above, of the struggle between proletariat and bourgeoisie (see Section 1. Bourgeois and Proletarians).

The founders of these systems see, indeed, the class antagonisms, as well as the action of the decomposing elements, in the prevailing form of society. But the proletariat, as yet in its infancy, offers to them the spectacle of a class without any historical initiative or any independent political movement.

Since the development of class antagonism keeps even pace with the development of industry, the economic situation, as they find it, does not as yet offer to them the material conditions for the emancipation of the proletariat. They, therefore, search after a new social science, after new social laws, that are to create these conditions.

Historical action is to yield to their personal inventive action, historically created conditions of emancipation to fantastic ones,

and the gradual, spontaneous class-organization of the proletariat to the organization of society specially contrived by these inventors. Future history resolves itself, in their eyes, into the propaganda and the practical carrying out of their social plans.

In the formation of their plans, they are conscious of caring chiefly for the interests of the working class, as being the most suffering class. Only from the point of view of being the most suffering class does the proletariat exist for them.

The undeveloped state of the class struggle, as well as their own surroundings, causes Socialists of this kind to consider themselves far superior to all class antagonisms. They want to improve the condition of every member of society, even that of the most favoured. Hence, they habitually appeal to society at large, without distinction of class; nay, by preference, to the ruling class. For how can people, when once they understand their system, fail to see in it the best possible plan of the best possible state of society?

Hence, they reject all political, and especially all revolutionary, action; they wish to attain their ends by peaceful means, and endeavour, by small experiments, necessarily doomed to failure, and by the force of example, to pave the way for the new social Gospel.

Such fantastic pictures of future society, painted at a time when the proletariat is still in a very undeveloped state and has but a fantastic conception of its own position correspond with the first instinctive yearnings of that class for a general reconstruction of society.

But these Socialist and Communist publications contain also a critical element. They attack every principle of existing society. Hence they are full of the most valuable materials for the enlightenment of the working class. The practical measures proposed in them—such as the abolition of the distinction between town and country, of the family, of the carrying on of industries for the account of private individuals, and of the wage system, the proclamation of social harmony, the conversion of the functions of the State into a mere superintendence of production, all these proposals, point solely to the disappearance

of class antagonisms which were, at that time, only just cropping up, and which, in these publications, are recognized in their earliest, indistinct and undefined forms only. These proposals, therefore, are of a purely Utopian character.

The significance of Critical-Utopian Socialism and Communism bears an inverse relation to historical development. In proportion as the modern class struggle develops and takes definite shape, this fantastic standing apart from the contest, these fantastic attacks on it, lose all practical value and all theoretical justification. Therefore, although the originators of these systems were, in many respects, revolutionary, their disciples have, in every case, formed mere reactionary sects. They hold fast by the original views of their masters, in opposition to the progressive historical development of the proletariat. They, therefore, endeavour, and that consistently, to deaden the class struggle and to reconcile the class antagonisms. They still dream of experimental realization of their social Utopias, of founding isolated “phalansteres,” of establishing “Home Colonies,” of setting up a “Little Icaria”—duodecimo editions of the New Jerusalem—and to realize all these castles in the air, they are compelled to appeal to the feelings and purses of the bourgeois. By degrees they sink into the category of the reactionary conservative Socialists depicted above, differing from these only by more systematic pedantry, and by their fanatical and superstitious belief in the miraculous effects of their social science.

They, therefore, violently oppose all political action on the part of the working class; such action, according to them, can only result from blind unbelief in the new Gospel.

The Owenites in England, and the Fourierists in France, respectively, oppose the Chartists and the Reformists.

IV. Position of the Communists in Relation to the Various Existing Opposition Parties

Section II has made clear the relations of the Communists to the existing working-class parties, such as the Chartists in England and the Agrarian Reformers in America.

The Communists fight for the attainment of the immediate aims, for the enforcement of the momentary interests of the working class; but in the movement of the present, they also represent and take care of the future of that movement. In France the Communists ally themselves with the Social-Democrats, against the conservative and radical bourgeoisie, reserving, however, the right to take up a critical position in regard to phrases and illusions traditionally handed down from the great Revolution.

In Switzerland, they support the Radicals, without losing sight of the fact that this party consists of antagonistic elements, partly of Democratic Socialists, in the French sense, partly of radical bourgeois.

In Poland, they support the party that insists on an agrarian revolution as the prime condition for national emancipation, that party which fomented the insurrection of Cracow in 1846.

In Germany, they fight with the bourgeoisie whenever it acts in a revolutionary way, against the absolute monarchy, the feudal squirearchy, and the petty bourgeoisie.

But they never cease, for a single instant, to instil into the working class the clearest possible recognition of the hostile antagonism

between bourgeoisie and proletariat, in order that the German workers may straightaway use, as so many weapons against the bourgeoisie, the social and political conditions that the bourgeoisie must necessarily introduce along with its supremacy, and in order that, after the fall of the reactionary classes in Germany, the fight against the bourgeoisie itself may immediately begin.

The Communists turn their attention chiefly to Germany because that country is on the eve of a bourgeois revolution that is bound to be carried out under more advanced conditions of European civilization, and with a much more developed proletariat than that of England was in the seventeenth, and of France in the eighteenth century, and because the bourgeois revolution in Germany will be but the prelude to an immediately following proletarian revolution.

In short, the Communists everywhere support every revolutionary movement against the existing social and political order of things.

In all these movements they bring to the front, as the leading question in each, the property question, no matter what its degree of development at the time.

Finally, they labour everywhere for the union and agreement of the democratic parties of all countries.

The Communists disdain to conceal their views and aims.

They openly declare that their ends can be attained only by the forcible overthrow of all existing social conditions.

Let the ruling classes tremble at a Communistic revolution.

The proletarians have nothing to lose but their chains.

They have a world to win.

WORKING MEN OF ALL COUNTRIES, UNITE!

FROM

NEWSPEAK **TO**

**CYBER
SPEAK**

A HISTORY OF SOVIET CYBERNETICS

SLAVA GEROVITCH

From Newspeak to Cyberspeak

A History of Soviet Cybernetics

Slava Gerovitch

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Introduction

Soviet Science and Politics through the Prism of Language

Communist society had indeed engendered a new universal language, detached from nature and subordinate to pure thought. . . . All situations, things, or men, can be reduced to a principle which transcends them.

—Françoise Thom, *Newspeak: The Language of Soviet Communism*¹

Cyberneticians argued that we were now at an historical conjuncture where machines were becoming sufficiently complex and the relationship between people and machines sufficiently intense that a new language was needed to span both: the language of cybernetics.

—Geoffrey Bowker, “How to Be Universal: Some Cybernetic Strategies”²

The history of Soviet cybernetics is a story of rebellion and conformity, of enchantment and disappointment, and of fascination with a new revolutionary language and frustration when this language was appropriated by the establishment. Soviet cybernetics was not simply an intellectual trend; it was a social movement for radical reform in science and in society in general. Cyberneticians came to believe in the possibility of a universal method of problem solving if only problems could be formulated in the right language. They viewed computer simulation as this universal method, and the language of cybernetics as a language of objectivity and truth. Soviet cybernetics challenged the existing order of things not only in the conceptual foundations of science but also in economics and politics.

Like cybernetics itself, which transcended the boundaries of any particular discipline, this book deals with social and political as well as intellectual developments in Soviet science from the late 1940s through the early 1970s. It looks at these developments through the prism of language because the aspiration to create a precise, universal scientific language was central to the cybernetic project and played a prominent role in the origins, the rise, and the fall of Soviet cybernetics.

Cybernetics is an unusual historical phenomenon. It is not a traditional scientific discipline, a specific engineering technique, or a philosophical doctrine, although it combines many elements of science, engineering, and philosophy. As presented in Norbert Wiener's classic 1948 book *Cybernetics, or Control and Communication in the Animal and the Machine*, cybernetics comprises an assortment of analogies between humans and self-regulating machines: human behavior is compared to the operation of a servomechanism; human communication is likened to the transmission of signals over telephone lines; the human brain is compared to computer hardware and the human mind to software; order is identified with life, certainty, and information; disorder is linked to death, uncertainty, and entropy.³ Cyberneticians view control as a form of communication, and communication as a form of control: both are characterized by purposeful action based on information exchange via feedback loops. Cybernetics unifies diverse mathematical models, explanatory frameworks, and appealing metaphors from various disciplines by means of a common language that I call *cyberspeak*. This language combines concepts from physiology (*homeostasis* and *reflex*), psychology (*behavior* and *goal*), control engineering (*control* and *feedback*), thermodynamics (*entropy* and *order*), and communication engineering (*information*, *signal*, and *noise*) and generalizes each of them to be equally applicable to living organisms, to self-regulating machines, and to human society.

In the West, cybernetic ideas have elicited a wide range of responses. Some view cybernetics as an embodiment of military patterns of command and control; others see it as an expression of liberal yearning for freedom of communication and grassroots participatory democracy. Some trace the origins of cybernetic ideas to wartime military projects in fire control and cryptology; others point to prewar traditions in control and communication engineering. Some portray cyberneticians' universalistic aspirations as a grant-generating ploy; others hail the cultural shift resulting from cybernetics' erasure of boundaries between organism and machine, between animate and inanimate, between mind and body, and between nature and culture.⁴

In the Soviet Union, cyberspeak acquired an entirely new set of connotations. In his memoirs, the prominent Soviet dissident Vladimir Bukovskii has chosen the following image to highlight the fundamental incompati-

bility between the rational thinking of some Soviet dissident scientists and the irrationality of the Soviet regime:

Just imagine for a moment that for some reason the KGB decides to arrest a computer. On the one hand, it is impossible to intimidate or trap a computer, to entice it to a compromise, to force a false confession, or even make it tell a lie, however small. On the other hand, a computer would not be able to understand the ambiguous language of inquiry, the language of the Soviet law. Its logical circuits would either give a binary yes-or-no answer, or—for an extended reply—would produce a long perforated tape with endless zeroes and ones. What will they do with it? Attach to a dossier?⁵

Here the logical rigor and mathematical precision of computer algorithms and calculations is opposed to the ideology-laden language of Soviet slogan-like public statements, which cannot be questioned or verified but are assumed to be true a priori. Since the publication of George Orwell's *Nineteen Eighty-Four*, this ideological language has often been referred to as *newspeak*.

Although Soviet ideological texts often appeared ambiguous and illogical, newspeak did have its own patterns and rules, which some observers, ironically, compared to computer algorithms. The cultural theorist Mikhail Epstein argued that "Soviet Marxist ideological language is by its very nature artificial; it would be possible to outline its structure using abstract formulas" and that "with specific formulas of functions and markers, a computer would be capable of composing Soviet ideological texts."⁶ William Griffith similarly suggested that the deciphering of Soviet "esoteric communications" could perhaps be "better and more easily done by an electronic computer."⁷ Then was there a fundamental conflict or an inherent concord between newspeak and cyberspeak?

This book focuses on the historical encounter between the language of cybernetics and the Soviet ideological language. It is a history of Soviet cybernetics as a discourse—an ensemble of practices of producing, articulating, communicating, and manipulating knowledge, a language phenomenon as much as a mathematical theory or an engineering technique. The analysis of cybernetics as a discursive phenomenon brings to the fore the rich political, ideological, and cultural connotations of the cybernetic notions of control and communication that reverberated in Soviet cybernetic texts. I argue, in fact, that cyberspeak was as much an ideological language as it was a language of science.

No other field of science or engineering underwent such frequent and profound changes of attitude in the Soviet Union as cybernetics. The historian Paul Josephson has recently called it “unimaginable that a culture so fascinated with the potential of science to build communism, a culture whose achievements in the 1950s included the hydrogen bomb, nuclear power, tokamaks, and Sputnik, could dismiss the promise of cybernetics.”⁸ As with many other unimaginable things, however, this was precisely what happened at first in the Soviet Union. In 1954 the *Short Philosophical Dictionary* defined cybernetics as a “reactionary pseudo-science” and “an ideological weapon of imperialist reaction.”⁹ By the late 1950s, cybernetics was recognized as an innocent victim of political oppression and “rehabilitated” along with some of the political prisoners of the Stalinist regime. Soviet cybernetics emerged as a movement for radical reform of the Stalinist system of science. It gained wide popularity, and in the early 1960s it was written into a new Party Program and hailed as a “science in the service of communism.” By the late 1960s, however, cybernetics began to lose intellectual content and turn into a fashionable trend. In the 1970s, disillusioned former cybernetics enthusiasts liked to tell a bitter joke: “They told us before that cybernetics was a reactionary pseudo-science. Now we are firmly convinced that it is just the opposite: cybernetics is not reactionary, not pseudo-, and not a science.”¹⁰ In the 1990s, the cybernetics boom was blamed for numerous shortcomings of Soviet science. “This doctrine, which called itself a science of control, chained the technological élan of a great nation,” wrote one commentator in a Russian on-line magazine. “Domestic science wasted immeasurable time and effort on the chimera of cybernetics, while the field of computer technology was deprived of full-scale funding.”¹¹

The fluctuating attitudes toward cybernetics represented something larger than mere rejection or acceptance of a particular scientific approach: they reflected profound changes in scientific language and research methodology across a wide range of disciplines, in the system of power relations within the scientific community, and in the political role of scientists and engineers in Soviet society. Through the lens of cybernetics, one can observe some salient features of Soviet science and engineering in the late Stalinist period and in the Khrushchev era.

The history of Soviet science and technology has often been seen as closely intertwined with political history. Adhering to a Cold War-inspired “totalitarian model” of Soviet history, some historians have portrayed

Soviet science as a victim of pervasive control by the Party/state apparatus; the late Stalinist period in particular has been described as a time of the “triumph of ideology” over science.¹² But this image of science suppressed by political interference is hard to reconcile with the impressive scientific achievements of the Stalinist era, which earned Soviet scientists a host of Nobel Prizes in physics and chemistry. In the postwar period, scientific and engineering institutions and large-scale industrial and construction projects aimed at fulfilling Stalin’s ambitious plan of the “great transformation of nature” mushroomed, and the Soviet Union celebrated an unprecedented “cult” of science and technology.¹³ It was during this period that Soviet scientists built their first atomic and hydrogen bombs. Paradoxically, Soviet science appeared to thrive under Stalin’s totalitarian rule better than in the relatively liberal climate of the Khrushchev regime. Indeed, the record of Soviet science seems to challenge the idea that democracy is vital to science. David Holloway has offered a solution to this paradox by describing Stalin’s defense laboratories as “islands of intellectual autonomy,” where political controls were relaxed and scientists could freely exchange ideas.¹⁴ Loren Graham has agreed that science sometimes provided a refuge from the harsh reality of Stalinism; however, after comparing the amount of funding and government support for science and technology under Stalin with later periods, he concluded provocatively that money appeared to be more important than freedom in producing breakthroughs.¹⁵

While adherents of the totalitarian model take for granted a fundamental conflict between the Soviet scientific community and the Party/state bureaucracy, other historians emphasize various forms of ideological accommodation, pragmatic cooperation, and even institutional integration between different groups of scientists and politicians. Graham has dispelled the popular myth of Soviet scientists’ being blinded by Marxist ideology and has shown how dialectical materialism, the official Soviet philosophy of science, was fruitfully integrated into the scientific outlook of many Soviet scholars.¹⁶ David Joravsky has radically revised the traditional view of the struggle between Soviet geneticists and the followers of the ignoramus Trofim Lysenko as an “ideological” conflict, explaining it instead as a competition between rival groups of scientists and their patrons in various branches of the government apparatus.¹⁷ Mark Adams has emphasized the role of “networks” of informal connections among scientists and between scientists and politicians, and described the relationships among the Party,

the government, and the scientists in terms of negotiation rather than direct administrative pressure.¹⁸ Nikolai Kremmentsov has further contended that “the control apparatus and the scientific community became fused not only in their overlapping organizational structures and networks, but also . . . in a common and quite peculiar set of shared images, rituals, and rhetoric,” which led to their “cultural unification.”¹⁹ As a result of the gradual diffusion of cultural norms from the Party life into science, Alexei Kojevnikov has argued, Soviet scientists began to play “games of intraparty democracy”: they reproduced public rituals of “criticism and self-criticism,” and they framed political denunciations as “creative discussions” of scholarly matters. In this case, politics affected science through the subtle mechanism of discursive domination rather than through the brute force of administrative control.²⁰ The closer we look, then, the more complex the picture of the relationship between Soviet scientists and Soviet politicians becomes. Instead of a simple binary opposition, we have a confusing Möbius strip: it is no longer entirely clear who is on which side.²¹

Adherents of the totalitarian model view the use of ideological language in science as an example of distorted reflection of reality and tend to dismiss it as “ideologization” of science; social historians, on the other hand, often describe the role of language as purely pragmatic and instrumental. In this study, I do not draw a sharp line between “ideological” and “scientific” elements of academic discourse. Instead of viewing this discourse as a tool of totalitarian oppression or an unproblematic instrument of negotiation, I interpret it as a cultural medium in which Soviet scientists lived and worked. I agree with Stephen Kotkin that Stalinism was a “specifically socialist civilization” with its own language (“Bolshevik”), which not only gave the speaker access to power but also powerfully shaped the speaker’s identity. “Stalinism was not just a political system, let alone the rule of an individual,” writes Kotkin. “It was a set of values, a social identity, a way of life.”²² In this book I examine how newspeak and later cyberspeak became central to the “way of life” of Soviet scientists under Stalin and under Khrushchev.

In chapter 1, “The Cold War in Code Words: The Newspeak of Soviet Science,” I explore the use of newspeak—the blending of scientific, philosophical, and ideological concepts—in political and academic discussions of the late Stalinist period. In particular, I examine how Soviet scientists used newspeak to balance the chief military and ideological priorities for

postwar Soviet science: to “overtake and surpass” science in the capitalist countries and at the same time to “criticize and destroy” Western scholarship for its alleged ideological flaws. Both the “ideologization” and the “de-ideologization” of science appear as variable discursive strategies, rather than inherent features of Soviet science. This chapter traces ideological disputes in three disciplines—mathematics, linguistics, and physiology—in the late Stalinist period. An analysis of “floating signifiers”—such terms as *formalism* and *idealism*, which crossed the boundaries between scientific disciplines and between science and politics—provides an insight into the discursive mechanism of “scientific newspeak.”

The emergence of cyberspeak in the “cybernetics circle” of American and European mathematicians, engineers, physiologists, sociologists, and philosophers in the 1940s is the subject of chapter 2, “Cyberspeak: A Universal Language for Men and Machines,” in which I examine the divergent research paths of Norbert Wiener and Andrei Kolmogorov. I also discuss the contributions of Julian Bigelow and Arturo Rosenblueth, Claude Shannon and Warren Weaver, Warren McCulloch and Walter Pitts, Alan Turing, John von Neumann, and Erwin Schrödinger. These scientists and engineers not only produced diverse theories, models, and techniques of control and communication; they also introduced an array of new concepts that lay the foundation of cyberspeak. I trace the evolution of cyberspeak from a set of narrowly defined technical concepts to a “universal language” for men and machines and finally to the vehicle of a cybernetics bandwagon.

In chapter 3, “Normal Pseudo-Science,” I examine the origins of the campaign against cybernetics in the Soviet press in the early 1950s in conjunction with the development of the first Soviet electronic digital computers. The first clash between newspeak and cyberspeak resulted in an ideological controversy over cybernetics’ man-machine analogies and in attempts to “de-ideologize” Soviet computers. Contrary to the common view of the anti-cybernetics campaign as having been directed by Party authorities, I interpret it as a chain of events spontaneously generated by self-perpetuating Cold War propaganda discourse. The myth that this campaign significantly delayed the development of Soviet computers has already been dispelled, but the negative ideological image of cybernetics in the late Stalinist period did seriously narrow the range of the first Soviet computer applications. Soviet computing was shaped by the tension between the practical goal of building major components of modern sophisticated weapons and the

ideological urge to combat alien influences. To shield themselves from potential ideological complications, computer specialists distanced computing from cybernetics and steered clear of computer modeling of biological and sociological phenomena. As a result, the digital computer was initially conceptualized in the Soviet Union as a giant calculator and stripped of all cybernetic metaphors.

In chapter 4, “Cybernetics in Rebellion,” I explore the transition from Stalinism to the Khrushchev period in Soviet science through the prism of changing Soviet attitudes toward computing and cybernetics. During Nikita Khrushchev’s political “thaw,” cyberspeak openly challenged newspeak. The computer came to symbolize a new spirit of rigorous thinking, logical clarity, and quantitative precision, contrasting sharply with the vague and manipulative language of Stalinist ideological discourse. This chapter examines the cybernetics movement as a vehicle of de-Stalinization in Soviet science. Soviet cyberneticians sought a new foundation of scientific objectivity in the rigor of mathematical formulas and computer algorithms and in the “precise” concepts of cybernetics. In contrast to “scientific newspeak,” they put forward a computer-based cybernetic criterion of objectivity as overtly non-ideological, non-philosophical, non-class-oriented, and non-Party-minded. The cyberneticians aspired to bring computer-based objectivity to the entire family of the life sciences and the social sciences by translating these sciences into cyberspeak.

In chapter 5, “The ‘Cybernetization’ of Soviet Science,” I examine the Soviet cyberneticians’ radical project of transforming a wide range of scientific disciplines along cybernetic lines. Soviet cybernetics emerged as an ambitious project for creating a single overarching conceptual framework, a general scientific methodology, a sort of substitute for the meta-scientific role of newspeak in academic discourse. Cybernetics began to serve as an institutional umbrella for various unorthodox research trends previously suppressed by dominant Stalinist schools. This chapter describes three such trends: “biological cybernetics” (genetics), “physiological cybernetics” (non-Pavlovian “physiology of activity”), and “cybernetic linguistics” (structural linguistics). In cybernetics’ “trading zone,” specialists from various disciplines attempted to communicate by means of a common language: cyberspeak. Initially, the flexibility of cyberspeak helped to establish the meta-scientific status of cybernetics and to bring a large number of disciplines under the umbrella of cybernetics. At the same time, this flexibility

resulted in the wide diversity of cybernetic discourses and in the suspicious elasticity of cybernetic concepts. The increasing ambiguity of the language of cybernetics threatened to undermine the entire project of bringing formal rigor and exact reasoning to Soviet science.

In chapter 6, “Cybernetics in the Service of Communism,” I document the climb of Soviet cybernetics to the height of official recognition and its concurrent fall to the depths of intellectual shallowness. After its inclusion in the 1961 Program of the Communist Party as one of the sciences crucial to the construction of communism, cybernetics became fully legitimized, officially recognized, and almost canonized. Inspired by their initial successes, Soviet cyberneticians set out to make cybernetics a “science of government.” With limited political options, they tried to work out a mathematical solution to the political problem of reforming the Soviet economy. This chapter examines their ambitious proposals for nationwide “optimal” planning and management, and discusses the roots of opposition to such plans both from traditional political economists and industry executives and from radical reformer economists who were calling for the introduction of market mechanisms. After Brezhnev replaced Khrushchev, the academic and political establishment began to appropriate cyberspeak and computer technology as means of conserving the existing administrative hierarchies and power structures. Party ideologues developed the concept of “scientific management of society” with the help of cybernetic models and methods of control to ensure the stability of the Soviet economy and Soviet society. By the early 1970s, cybernetics had been transformed from a vehicle of reform into a pillar of the status quo. As cybernetic concepts acquired the generality and universality characteristic of other categories of official philosophy, cyberspeak began to resemble newspeak. Former cybernetics enthusiasts began to distance themselves from this official discourse and even stopped calling themselves cyberneticians. Cyberspeak and newspeak seamlessly integrated into “CyberNewspeak.”

Emphasizing the close connections between language and theory, between knowledge and power, and between science and politics, I interpret newspeak and cyberspeak not as linguistic practices somehow imposed on Soviet scientists but as particular discursive strategies developed by scientists in their efforts to adapt to a specific political, ideological, and socioeconomic situation and to manipulate that situation to their advantage. I focus on the productive, rather than merely repressive, function of

ideological discourse in Soviet science.²³ At the same time, I question Soviet scientists' complete mastery of either newspeak or cyberspeak, emphasizing the subtle ways in which language controls its "masters."

It is hard for historians of science to escape the conceptual domination of their subject matter: historians of biology often speak of "the evolution of ideas" and "the growth of science," historians of geology talk about "the shifting of the argument" and "the shaping of scientific knowledge," and historians of the physical sciences explore "the construction of science" and "the diffusion of knowledge." Although I have resisted the temptation to conceptualize the history of cybernetics as a self-organizing process controlled by feedback loops with the political and social context of science, the encroachment of cyberspeak, now spoken almost universally, has been hard to stop. It is no longer possible to avoid such common terms as *information*, *control*, *communication*, *code*, *signal*, and *feedback*, which have acquired their broad meanings through cybernetics. At least we now know where they came from.

The Cold War in Code Words: The Newspeak of Soviet Science

We have created in our press a special language—the language of a socialist citizen for a socialist citizen only. . . . This language is unfamiliar and sometimes unintelligible to the masses, even to the leftist masses, in the capitalist countries.

—from a classified report by a Soviet propaganda expert, 1946¹

The leading Soviet specialist in mathematical logic—Sof'ia Ianovskaia, a professor at Moscow University—was a very experienced editor. A Bolshevik since 1918, in the 1930s she was trusted with an important political task: publishing the “mathematical manuscripts” of Marx. The manuscripts had not been published previously, for some mathematicians remained skeptical about their scholarly value. But Ianovskaia knew very well how to give the right ideological twist to a scholarly book and to disarm any potential critics. She discovered a great depth of dialectical thought in Marx’s marginal notes, and she underscored their immense philosophical significance. Her experience came in handy in 1947, when she edited the Russian translation of David Hilbert and Wilhelm Ackermann’s *Principles of Theoretical Logic*. To make the book more palatable to ideological watchdogs, she supplied a clever introduction, in which she distanced the book’s authors from the unnamed “idealists parasitizing on mathematical logic.” In addition, throughout the book she sprinkled dry logical formulas with a few lively examples of how mathematical logic could be relevant to socialism. Instead of using banalities like “all men are mortal,” Ianovskaia illustrated logical expressions with philosophical maxims and familiar political slogans. Thus, she explained the concept of logical implication by using the vocabulary of dialectical materialism, the official Soviet philosophy of science:

Let us consider the following premises:

1. The answer to the basic question of philosophy, which concerns the relationship between thinking and being, can be either materialistic or idealistic.

2. The materialistic answer is incompatible with the idealistic one.
3. If the followers of Mach are telling the truth, their answer to the basic question of philosophy is neither materialistic, nor idealistic.

It is immediately obvious that a logical implication of this is as follows:

4. The followers of Mach are not telling the truth.²

Ianovskaia further demonstrated the power of formal language by representing the declaration “All Soviet women have equal rights with Soviet men” with the following formula:

For any object (or “any person”) it is true that
if this object is a Soviet woman,
then this object has equal rights with Soviet men.³

Somehow this did not sound quite right. When formalized, revered philosophical dogmas and erstwhile sacred slogans turned hopelessly profane and sounded more like parodies. In 1950 the Russian edition of *Principles of Theoretical Logic* received a scathing review in the main Soviet philosophical journal *Voprosy filosofii* [*Problems of Philosophy*]. “Any mortal who is not intoxicated with formalism,” the reviewers wrote, “can see that these exercises . . . lead to nonsense.”⁴ The reviewers brushed aside Ianovskaia’s rhetoric and charged that Hilbert and Ackermann were marching “along the path of idealism and metaphysics, the path of further formalization of logic” and were “hiding their idealistic views behind specialized mathematical reasoning.” Ianovskaia’s efforts, in turn, were branded as a “*reconciliatory attitude toward idealism* in mathematics.”⁵

Soviet scientists, entangled in the terrifying web of Soviet politics in the early years of the Cold War, learned to play by the rules of the establishment. Soviet scientists’ close collaboration with the Party and with the government bureaucracy resulted in their gradual cultural adaptation to the behavioral patterns, modes of thought, and rhetorical styles of their patrons. Academic discussions were, more and more often, conducted in a language accessible to the bureaucracy: the value-laden ideological language of official Soviet discourse. This language constituted a pervasive ideological-linguistic medium in which Soviet scientists lived and worked.

Echoing George Orwell’s original insight, I call this language *newspeak*. In his 1949 novel *Nineteen Eighty-Four*, a sharp and gloomy caricature of Stalinist Russia, Orwell coined this term to denote a particular socialist language that embodied a totalitarian world view and the “mental habits” of Party members. Newspeak made all alternative views impossible, since

heretical thoughts would be “literally unthinkable.”⁶ After Orwell, *newspeak* became a popular analytical category in studies of Soviet public discourse. Newspeak reappeared time and again under the names *ideological language*, *official speech*, *Stalinese*, *totalitarian language*, *Communist speech*, *the Soviet language*, *authoritarian Russian*, and *langue de bois* [wooden language].⁷ Several dictionaries of “Communist jargon” came out.⁸ Soviet discursive practices, or specific ways of speaking, writing, and otherwise producing meaning, were diverse, and they evolved considerably over time. Without assuming the existence of a single, uniform, universally accepted language for the entire public discourse under Stalinism, I use *newspeak* in a narrow sense that refers specifically to the blending of scientific, philosophical, and ideological concepts in political and academic discourses of the late Stalinist period.

This chapter analyzes the dynamics of postwar Soviet science through the lens of the language of scientific debates. Instead of viewing scientific discourse as a mere servant of the state or a victim of totalitarian oppression, I focus on the productive, rather than repressive, function of newspeak in Soviet science. This chapter explores various discursive strategies developed by scientists themselves in their efforts to adapt to the current political, socioeconomic, and cultural situation, and to manipulate ideological discourse at the same time.

In the postwar period, Soviet scientists struggled to balance the chief military and ideological priorities for Cold War science: to “overtake and surpass” science in the capitalist countries and to “criticize and destroy” Western scholarship for its alleged ideological flaws. One popular strategy was to draw a boundary between the “objective content” of scientific knowledge and its philosophical meaning. Thus, I view the ideological disputes of the early 1950s not only as a clash of competing philosophical and ideological interpretations of particular theories but also as a contest over the exact position of the boundary between science and ideology.

Following Mark Adams, I interpret Soviet ideology not as an essential set of beliefs somehow imposed on the scientific community from above but as a flexible language of negotiation involving the Party, the government, and the scientists.⁹ Using the tetrad model of the Soviet “ideolanguage” developed by Mikhail Epstein, I attempt to elucidate the discursive mechanisms of newspeak.¹⁰ In contrast to Orwell, I emphasize the flexibility rather than the rigidity of this ideological language. Newspeak did not

simply express some pre-established orthodoxy; quite the opposite, the skillful use of newspeak allowed one to manipulate ideology and, to some extent, to define what was permitted in a Soviet context. A manifestly dogmatic and at the same time strikingly flexible language, newspeak did not embody the truth; it provided mechanisms for negotiating the truth.

The ideological language of the postwar period had two major components, two rich repertoires, which complemented each another: the political language of the Cold War and the philosophical language of dialectical materialism. Philosophical terminology served as a bridge between the scientific and the political. It was used not only by the professional philosophers who served as ideological watchdogs of the Soviet academic community, but also by scientists themselves. By learning newspeak, scientists acquired necessary rhetorical tools for refashioning scientific ideas, legitimizing their own position and discrediting their opponents.

The interplay of various discursive strategies of newspeak is illustrated by case studies of ideological disputes that occurred in three scientific disciplines—mathematics, linguistics, and physiology—in the late Stalinist period. Despite the vast intellectual differences among these disciplines, in all three cases Soviet scientists employed very similar strategies to translate their intellectual disagreements and institutional squabbles into principled ideological conflicts. Newspeak became a common language for Soviet mathematicians, linguists, physiologists, philosophers, and politicians.

Balancing Military and Ideological Priorities for Cold War Science

In March of 1954 researchers of the Mathematical Institute of the Soviet Academy of Sciences in Moscow were preparing a comprehensive book-length survey of the entire mathematical discipline, *Mathematics: Its Content, Methods, and Significance*. An Institute-wide “philosophy seminar,” whose mission was to instill the right ideological principles into the researchers’ minds, held a special session devoted to the discussion of a draft introduction to the book. One of the discussants displayed heightened ideological vigilance and proposed to de-emphasize in the introduction the contributions of American mathematicians. “There is no progressive science or progressive music in America now,” he argued. “They have lured in a number of scientists from all over the world, and now barely manage to maintain their military potential. We will not promote American math-

ematics.”¹¹ Another participant voiced a different opinion: “There are some progressive-minded people [in the United States], and one should not lump them together with the Wall Street,” he said. “A great majority of American mathematicians have come out of the ordinary folk and do not represent the American monopolistic bourgeoisie.”¹² The question how to treat science produced by a Cold War enemy—as a value-neutral body of knowledge or as an ideological Trojan horse—acquired central importance in Soviet public discourse on Western science in the early years of the Cold War.

In the murky waters of Cold War politics, Soviet scientists and engineers were caught between the Scylla of the national defense and the Charybdis of ideological purity. In February of 1946 Stalin personally formulated as the chief priority for Soviet science “not only to overtake but to surpass in the near future the achievements of science beyond the borders of our country.”¹³ Soviet scientists and engineers were instructed to catch up with the West in the shortest possible time, particularly in nuclear physics and rocketry. In the cases of the first Soviet nuclear bomb and the first rocket, the Soviet leaders, distrustful of domestic scientists’ abilities, chose the path of directly copying Western exemplars. At the same time, paradoxically, Party ideologues urged Soviet scientists to treat Western scholarship as “idealistic and reactionary.” The Cold War was fought not only on the international scene but also on the “home front”—against any real or imaginary opposition to the chosen political course of international confrontation. After brief period of ideological relaxation and active cultural contacts among the Allies during World War II, Soviet authorities began to tighten ideological screws, cutting off international contacts and trying to bring intellectuals in line with the Party policy. In April of 1947, Agitprop—the Department of Agitation and Propaganda of the Party Central Committee—issued a secret “Plan of Measures for the Propagation of Soviet Patriotism among the Population.” In August of 1947 *Literaturnaia gazeta* demanded that several Soviet scientists be brought to “public court” for their “lack of patriotism.” A few days later *Pravda*, the main Party mouthpiece, criticized several leading scientists for “unpatriotic acts” and “servility to the West”—crimes that consisted largely of publishing articles in foreign periodicals. In March of 1949 the Politburo set the ideological priorities for the second edition of the *Great Soviet Encyclopedia*, urging the authors to “criticize from the Party position modern bourgeois trends in science and technology.”¹⁴ In May of 1952 the Academy of Sciences

Institute of Philosophy promised “to criticize and destroy all reactionary philosophical trends that appear in bourgeois countries under new, modish names and spread the propaganda of a new war.”¹⁵

Soviet scientists and engineers faced a fundamental dilemma. Soviet scientific progress was measured against Western science, though the same Western science was also branded in public discourse as a source of alien ideology. Both priorities—“overtake and surpass” and “criticize and destroy”—featured prominently in Soviet public discourse. This created a permanent unresolved tension; it also created considerable confusion. One could hardly obey both orders at once, following the Western path while moving in the opposite direction. To “overtake and surpass” Western science, Soviet scientists needed to borrow Western knowledge; if they borrowed, they could easily be accused of “kowtowing” before the West. However, completely disregarding the latest trends in Western science could be seen as a deliberate attempt to slow down Soviet science and to put it behind its Western counterpart. The historian David Joravsky has pointed out the absurdity of this situation:

The need to overtake and surpass the West coexisted with the need to stop kowtowing to the West, each inflaming the other. However inconsistently, the Stalinist mentality laid both demands on scientists, to be true to their own Russian knowledge and to surpass Westerners in universal knowledge; to take a rock-solid stand in the native monolith and to be as disputatiously creative as scientists in the West.¹⁶

Soviet discourse on Western science thus became saturated with paradoxes. Soviet authors often dismissed a Western-born scientific theory as bourgeois nonsense and in the same breath claimed national priority in elaborating the very same theory. Soviet leaders, distrustful of Soviet scientists, tended to support scientific and technological innovations only if they were recognized in the West. At the same time, Western reports were often regarded as a source of false information intended to mislead Soviet scientists and put them on the wrong track; it was assumed that a really worthy idea would not be advertised in the open press.

In the fragmented and contradictory ideological discourse of the late Stalinist period, there was no general rule for politically correct behavior. In every particular situation, Soviet scientists had to choose between two alternative slogans: “Criticize and Destroy!” and “Overtake and Surpass!” This was not simply a choice of rhetoric; it implied a radical change in their vision of science, and it had profound epistemological implications. The

followers of the “criticize and destroy” approach viewed knowledge as ideological through and through, and therefore they regarded any scientific theory born in a capitalist society as an expression of “reactionary, imperialist ideology” by definition. In contrast, the champions of the “overtake and surpass” strategy did not regard Western science as an ideological threat. They assumed that scientific knowledge was “objective,” value-neutral, and universal across political borders.

The slogan “Criticize and Destroy!” was taken up by the supporters of Trofim Lysenko, who discarded much of contemporary Western knowledge and attempted to build a distinct, ideologically superior socialist science. In July 1948 Lysenko delivered his infamous address “On the Situation in Biological Science,” in which he contrasted two “opposing and antagonistic” trends in biology. He labeled the first trend (Western-born “Weismannism-Mendelism-Morganism,” which lay at the basis of modern genetics) unscientific, idealistic, metaphysical, reactionary, scholastic, feeble, and sterile. As a healthy alternative, Lysenko put forward his own doctrine, a variation on the Lamarckian idea of the inheritance of acquired characteristics, which he praised as truly scientific, materialistic, creative, productive, and progressive. He patriotically called this doctrine “creative Michurinist Darwinism,” tracing its roots to a Russian “founding father”: the agronomist Ivan Michurin. Caught in a fierce battle for the control over Soviet biological research and educational institutions, Lysenko painted his opponents—Soviet geneticists and evolutionary biologists—as enemies of Soviet ideology. Emphasizing the Western origins of his opponents’ work, he dubbed them “Morganists,” attacked them for their alleged philosophical and ideological errors, and even attached political labels to them.¹⁷

Stalin himself edited Lysenko’s address, bringing its rhetoric in line with the ideological priorities of the unfolding Cold War. Initially, Lysenko relied on the criterion of class to divide science into the “Soviet” and “bourgeois” kinds; this principle, popular among Soviet Marxists in the 1920s and the 1930s, was now completely obsolete. During the war, in the spirit of cooperation between the Allies, it had been supplanted by proclamation of the unity of international science. The dawning of the Cold War invalidated this thesis too. Stalin went over Lysenko’s manuscript scrupulously and replaced the obsolete references to “bourgeois” scientific theories with the words *idealistic* and *reactionary*; he also substituted *scientific biology* for *Soviet biology*.¹⁸ Stalin’s revisions signaled a discursive turn from class-based analysis

of science to the concept of “two worlds—two ideologies in science”—an idea that was much better suited to the tasks of Cold War propaganda.

An opposite attitude toward Western science—one based on the Party slogan “Overtake and Surpass!”—was expressed by Soviet defense physicists concerned with closing the “nuclear gap.” The atomic bomb, then the most potent symbol of political and military power, effectively rendered Western physics legitimate in the eyes of Soviet officials. This may have played a decisive role in resolving an ideological controversy over quantum mechanics and relativity theory—a conflict that was stimulated by the institutional rivalry between physicists from the Academy of Sciences and physicists from Moscow University. In 1949 the Academy physicists reportedly prevented an ideological pogrom in physics by claiming the importance of these Western-born theories for the construction of nuclear weapons. The University physicists, who chose to trumpet the “Criticize and Destroy!” slogan, found that their ideological arguments had little weight against the bomb.¹⁹ The relationship between science and ideology in the postwar period was not fixed; it varied from discipline to discipline, and it was often hotly contested.

Shifting Boundaries between Knowledge and Ideology

The followers of the two opposite approaches—“to criticize and destroy” and “to overtake and surpass”—developed distinct discursive strategies to legitimize their views of the relationship between knowledge and ideology. Both groups drew on Marxist theory; however, playing on the complexity and the inconsistency of the Soviet interpretation of Marxist philosophy of science, they evoked different aspects of it. This interpretation paradoxically combined social constructivism (science as a product of socioeconomic and political forces) with scientific realism (science as objective truth about nature).²⁰

The first group, best represented by Lysenko and his followers, attempted to “ideologize” science by translating scientific theories into an explicitly ideological language. They rendered their own theories into Marxist philosophical and political terms. The theories of their opponents, on the other hand, were labeled as philosophical and political deviations from Marxism. The “ideologizers” thoroughly traced their opponents’ views to Western-born scientific theories—an embodiment of alien ideology—and thus made

them an easy target of ideological attacks under the slogan “Criticize and Destroy!” It was precisely this type of discourse that prompted historians to speak of the essential “ideologization” of Soviet science in the late Stalinist period.

In the late 1940s and the early 1950s, a series of vociferous ideological campaigns dominated by the “criticize and destroy” trend swept Soviet science. Lysenko’s proclamation of the uncompromising struggle between socialist and capitalist trends in biology set an example for the rest of Soviet science. In August through December of 1948, meetings discussing the new line were held in numerous research and educational institutions across all disciplines. Mathematicians, physicists, geologists, and astronomers all were now supposed to expose “idealistic and reactionary” elements in their academic fields. In early 1949 a widespread campaign against “cosmopolitanism” (an ideological label for anti-patriotism and “kowtowing before the West”) began. That campaign was accompanied by vicious anti-Semitic attacks and by the expulsion of Jews (labeled “rootless cosmopolitans”) from many cultural and academic institutions.²¹ These campaigns destroyed careers and effectively banned whole areas of research; in a number of disciplines, the most dogmatic trends prevailed, imposing narrow conceptual frameworks and stifling creative thought.²²

In postwar ideological campaigns, the boundary between academic debate and political dispute was completely erased. Philosophical categories that denoted various deviations from dialectical materialism (the official Soviet philosophy of science) mixed with political clichés of Cold War propaganda. Discussions of genetics, relativity theory, quantum mechanics, and various Western-born theories in economics, chemistry, physiology, linguistics, and mathematical logic became saturated with the pejorative labels *idealism*, *mechanicism*, *metaphysics*, *formalism*, and *cosmopolitanism*.²³ Campaign activists often reduced Western scientific knowledge to philosophical errors and further to ideological and political mistakes. Soviet scientists who saw in Western science a rational, “objective” kernel opposed this trend. Insisting on the value-neutral, impartial character of scientific knowledge, they elaborated an alternative strategy of “de-ideologization” of science. They ingeniously split Western scientific theories into two presumably independent parts: the ideologically neutral and objective “core” and the ideology-laden philosophical “shell.” The “de-ideologizers” tried to rescue what they saw as the “essential” elements of Western theories

while sacrificing only the “dispensable” ones. They freely “criticized and destroyed” the latter while safely adopting and further developing the former. The “de-ideologization” strategy gradually shaped a popular image of science as a centaur with a solid body of scientific facts and a manifestly political face, socialist or capitalist.

In their defense of quantum mechanics and relativity theory from ideological critique, the Academy physicists often resorted to the “de-ideologization” strategy. For example, they insisted on the mathematical correctness of the uncertainty principle, but they cautiously distanced themselves from the controversial Copenhagen interpretation. At the same time, they worked hard to elaborate an acceptable philosophical interpretation of quantum mechanics so as to bring it into harmony with dialectical materialism.²⁴ The “de-ideologization” strategy also profoundly influenced contemporary Soviet writings on the history of science and technology.²⁵

Different authors not only offered competing philosophical and ideological interpretations but also disputed the exact location of the boundary between scientific knowledge and ideology. For example, the localization of the physical principle of complementarity was hotly contested. Militant philosophers claimed that this principle was a part of the “philosophical interpretation” and therefore belonged to their professional domain; they argued that this principle was “idealistic” and therefore false. The Academy physicists, on the other hand, tried to present it as a part of the core theory and offered an alternative materialistic interpretation. The boundary between the scientific “core” and the ideological “shell” was constantly shifting back and forth, depending on who was drawing it.²⁶

Remapping science—drawing a boundary between knowledge and ideology—was not only an epistemological task; it was also political activity. It effectively delineated the spheres of authority between scientists and non-scientists and between politicians and professional ideologues. Throughout Soviet history, these spheres were redefined many times, and epistemological boundaries were redrawn accordingly. The relative intellectual autonomy of scientists in the early Soviet period was followed by the increasing involvement of politicians, government officials, and philosophers in resolving scientific disputes during the Stalin era.²⁷ Depending on the position of their Party and government patrons, competing groups of scientists constantly shifted the knowledge/ideology boundary back and forth, trying either to invite or to prevent the authorities’ intervention. Because of the

tensions inherent in postwar politics, this boundary could never be fixed. Perhaps all sides had a stake in maintaining this discursive flexibility, for it allowed them substantial room to maneuver.

The postwar ideological campaigns affected all Soviet scientists, and political rhetoric became truly pervasive. Both the “ideologizers” and the “de-ideologizers” spoke the Stalinist ideological language. The first group attempted to translate entire scientific theories into this language, while the second limited its uses to the discussion of “philosophical interpretations” of those theories. Fluency in this language, which I will call *newspeak*, became one of the prerequisites for scientific work. Newspeak provided potent discursive strategies for linking science to philosophy to politics, and it allowed a skillful speaker to attach political labels to any scientific target.

Newspeak: The Fundamentals

Newspeak allowed a seamless transition from science to philosophy to ideology, for its vocabulary was composed largely of “empty” or “floating” signifiers—words that easily passed between different realms and were (re)filled with different meanings.²⁸ For example, as the political scientist Rachel Walker has argued, *Marxism-Leninism* could appear in different contexts as “an ideology, a science, a morality, a theory, a philosophy, a political practice, a sociology, an economics, a party line, and so on.”²⁹ This term could acquire wide currency and become truly pervasive precisely because of its capacity to accommodate a broad range of meanings:

“Marxism-Leninism” is an empty signifier: a word which means everything and nothing, a word which has no intrinsic meaning, no ontological and epistemological entailments, no necessary and essential relationship with any of the ostensible objects with which it has been conventionally associated—whether these be a general “body of ideas,” a social value system, or the theoretical output of two historical individuals named Marx and Lenin. This is not to argue that the word is entirely incapable of meaning, but rather to suggest that any meanings predicated on it are contingent—they shift and change according to context and usage.³⁰

Words of newspeak easily crossed contextual boundaries, but they did not walk value-free. These words blended description with evaluation: they did not simply name things, but passed a strong positive or negative value judgment about them. For example, *Marxism-Leninism*, whatever it meant, always carried a positive evaluation, while such words as *idealism*, *metaphysics*, and *formalism* could refer to a wide range of scientific and

philosophical doctrines but always labeled them negatively. The cultural theorist Mikhail Epstein has called such words “ideologemes”:

Ideologemes, being the elementary particles of ideological thinking, are not simply words, but concealed judgments that take the form of words. Usually a judgment is developed in an entire sentence, where it is divided into a subject and predicate. This kind of judgment is open to discussion because the link between subject and predicate is explicitly relative. . . . The factual meaning of the ideologeme usually serves as the subject of the judgment, the evaluative meaning, as the predicate. . . . An ideologeme is nothing other than an idea that is hidden in one word (or, sometimes, in one indivisible phrase or idiom). In this way it can be inserted into the listener’s consciousness without the possibility of argumentation or objection. One cannot quarrel with a single word.³¹

For example, newspeak reduced the explicit assertion “Striving for peace in this situation is wrong” to the simple label “conciliatoriness,” which already embodied a negative judgment. Ideologemes resembled what Orwell called “words which had been deliberately constructed for political purposes” and “a sort of verbal shorthand, often packing whole ranges of ideas into a few syllables.”³²

Since ideologemes have dual (descriptive and evaluative) meanings, the relations between them are more complex than relations between ordinary words. Instead of simple synonymy and antonymy, ideologemes can relate to one another in four different ways.

The first way, full antonymy, denotes the opposition of both the descriptive and the evaluative meanings. Such binary oppositions as socialism vs. capitalism, proletariat vs. the bourgeoisie, freedom vs. slavery (or oppression), internationalism vs. nationalism (or chauvinism), and collectivism vs. individualism were characteristic of early Marxism. They embodied a Manichean vision of the world deeply divided into two antagonistic and irreconcilable camps.³³

Relationships of the second type involve pairs of words with synonymous descriptive meanings but opposite evaluations. The same concept is expressed in different words, depending on whether one is talking about a friend or an enemy. For example, the word *spy* could refer only to the enemy’s intelligence officers, and never to our own agents.³⁴ Different vocabularies are used to describe similar actions of socialist and capitalist governments and to characterize comparable activities of the “revolutionary avant-garde” and its “class enemies.” What in our case is called “peacefulness” is in their case labeled “conciliatoriness”; what in one context is

hailed as “internationalism” is branded “cosmopolitanism” in another; our “loyalty to principles” translates into “dogmatism.” As concepts travel from one ideological context to another, their ideological coloration changes accordingly: positive evaluative meanings are turned into negative ones, and vice versa.

Pairs of ideologemes with similar (either strictly positive or strictly negative) evaluations but opposite descriptive meanings belong to the third type. For example, both members of such pairs as internationalism and patriotism, peacefulness and irreconcilability, and tradition and innovation have equally positive connotations. By contrast, both parts of such oppositions as idealism vs. mechanicism, subjectivism vs. objectivism, and formalism vs. naturalism bear equally negative evaluation. By placing strong emphasis on evaluation, newspeak helps mask contradictions in ideological discourse. Thus, one can claim to be at once an internationalist and a patriot, or one can accuse an opponent of deviating from dialectical materialism in two opposite directions (e.g., toward idealism and toward mechanicism) at the same time.

The fourth type of relationship connects fully synonymous ideologemes, which can be freely substituted for one another. For example, discipline is identified with organization and consciousness, while permissiveness is associated with lack of control and anarchy. By constructing series of synonymous ideologemes, newspeak provided political “supplements” to philosophical semantics and philosophical “supplements” to political semantics.³⁵ Thus, one could easily jump from science to politics by translating scientific, philosophical, and political concepts into synonymous ideologemes. In this way, Lysenko skillfully connected his “agrobiology” with dialectical materialism and with socialism and at the same time linked genetics to idealism to capitalism. Lysenko’s doctrine was thus effectively portrayed as a socialist science, while genetics was labeled as a product of capitalism. By linking scientific, philosophical, and political concepts into chains of ideological synonymy, one could blow up a smallest flaw into a major political mistake, and turn an intellectual opponent into a principled political enemy. For example, in 1940 the militant Marxist philosopher Ernest Kolman stepped into the dispute between Lysenko and the prominent Soviet mathematician Kolmogorov over the use of statistical methods in biology. Kolman skillfully built a chain of ideological associations—from Kolmogorov to the German mathematician Richard von Mises (because

both of them worked on probability theory), then from von Mises to the Austrian philosopher Ernst Mach (because von Mises cited Mach in his work), and finally from Mach to “subjective idealism” (which Lenin himself had branded as a major deviation from dialectical materialism).³⁶ Kolmogorov thus ended up in a subjective idealistic hole, and this smelled of a grave political error. The construction of ideological synonymy served not only as a powerful rhetorical tool of criticism, but also as a convenient propaganda device. As the historian Stephen Kotkin has observed, one could appeal to various groups by appropriately shifting the emphasis from one “mobilizing phrase” (such as *Soviet*, *socialism*, *Bolshevism*, or *Marxism-Leninism*) to another, since all of them functioned as ideological synonyms.³⁷

Ideologemes related by synonymy or antonymy of their descriptive or evaluative meanings could be organized into tetrads (figure 1.1).³⁸ In every tetrad, both words in the left column have positive evaluations, even though their descriptive meanings are opposed (internationalism and patriotism). The descriptive meanings of both words in the right column are also opposed, but this time they have negative evaluations (cosmopoli-

internationalism	—	cosmopolitanism
patriotism	—————	nationalism

freedom	————	anarchy
discipline	—	repression

peacefulness	————	appeasement
irreconcilability	—	aggressiveness

innovation	————	avant-gardism
loyalty to tradition	—	backwardness

pragmatism	————	bendiness
loyalty to principles	—	dogmatism

Figure 1.1

tanism and nationalism). Both words in each row have identical descriptive meanings but opposite evaluations. For example, an equal love for all nations can be either ideologically approved (internationalism) or disapproved (cosmopolitanism); similarly, an exclusive love for one's own nation can be evaluated either positively (patriotism) or negatively (nationalism or chauvinism).

Manipulations with ideologemes within each tetrad give newspeak enormous flexibility. Through a series of substitutions, contrasts, and conversions, one can easily provide an ideological justification or refutation of any position that can be formulated in tetrad terms. Any tetrad concept can be turned into its opposite, as Orwell so vividly illustrated in the famous slogans inscribed on the façade of the Ministry of Truth³⁹:

War is Peace

Freedom is Slavery

Ignorance is Strength

For example, one could start with the ideologeme "freedom," substitute its evaluative synonym "revolutionary discipline," then equate that with "revolutionary violence," convert that into "repression" (different evaluations, but similar denotations), and finally replace "repression" with "slavery."⁴⁰

Newspeak turned elements of various political slogans, which were introduced in different historical periods and reflected tortuous paths of Bolshevism, into "floating" or "empty" signifiers. They could be filled with any content, descriptive or evaluative, depending on the intentions of the speaker. For example, the binary opposition "internationalism vs. nationalism," characteristic of early Marxist ideology, was widely employed in early Soviet years to support the dream of a "world revolution." During World War II, however, the ideological emphasis was switched to "patriotism," which in postwar years formed a stable binary set with its ideological opposite, "cosmopolitanism." The political culture of late Stalinism did not eliminate the first opposition but instead combined it with the second to form a flexible ideological tetrad. If political expediency demanded that postwar cooperation with the Western Allies be ideologically condemned, the proponents of such cooperation could be labeled "cosmopolites"; if the opposite view was expedient, the opponents of such cooperation could easily be branded as "chauvinists." A cunning newspeak adept like Stalin could thus use "leftist slogans to defeat the right, rightist slogans to defeat the left."⁴¹

Thanks to the flexibility of newspeak, virtually any position could be rendered ideologically flawed, and anyone could be painted as an ideological enemy. As Epstein argues, the “speaker who controls the tetrad does not so much participate in conflicts as he uses them, playing upon their contradictions.”⁴² As Walker has argued, there was no safe ideological position in Soviet politics:

The result is a “Catch-22” which keeps the whole community to which [the term *Marxism-Leninism*] is addressed in a state of perpetual tension since this community can never predict exactly what behavior, what “interpretation” of “Marxism-Leninism,” will be correct at any point in time. To defend the “purity” of Marxism-Leninism is to risk being labeled a “pendant” or a “dogmatist.” To creatively develop it is to risk being labeled a “revisionist” or some sort of “deviationist.” To be seen doing neither is to risk the accusation of not being a Marxist-Leninist at all.⁴³

The inherent uncertainty of the Soviet ideological discourse both created room for maneuver and made any maneuvers potentially dangerous: one had to repeat dogmas without sounding dogmatic, and to show originality without introducing too much novelty. It is not difficult to find some similarities between these techniques and the rhetorical devices used by scientists, who often paradoxically combine the claim of “organized skepticism” with a firm commitment to the dominant paradigm.⁴⁴ Although “science” is often perceived as an opposite of “ideology,” in the Soviet case the language of science proved inextricably linked with newspeak.

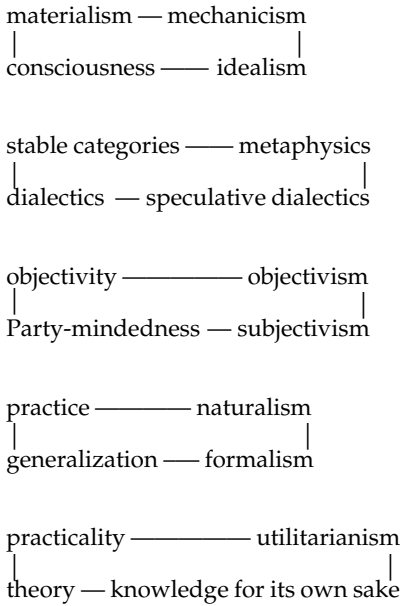
Scientific Newspeak

Newspeak was not just the language of Party bureaucracy; it permeated the entire public discourse of Stalinist society, and it became a pervasive medium in which Soviet scientists lived and worked. As the role of Soviet science in national defense, in industry, and in agriculture gradually grew, Party and government officials began to serve as referees of scientific disputes. Soviet philosophers, who since the 1930s had been involved in the ideological policing of the scientific community, often served as mediators between scientists and politicians, translating scientific theories into ideological language for the politicians and transforming political slogans into research agendas for the scientists. Newspeak emerged at the nexus of scientific, philosophical, and political discussions, tying them inextricably together. Newspeak became a language of negotiation for Soviet scientists,

philosophers, politicians, and government officials, providing a means for seamlessly translating science into philosophy into ideology into politics and vice versa.

Under Stalin, dialectical materialism, once a thriving and productive field of philosophical scholarship, turned into an official canon, gradually “calified,” and began to serve as a philosophical cudgel.⁴⁵ In the ideological battles of the 1930s and the 1940s, the basic principles of dialectical materialism (realism; nonreductionism; the view of the material world as an infinitely complex, interconnected, and evolving whole; belief in the relative nature of human knowledge) were often subjected to the most dogmatic interpretations.⁴⁶ Doctrinaire thinkers identified “the material world” with entities directly observable with contemporary scientific instruments; everything else, including genes, was labeled as the product of “idealistic” speculation. At the same time, the categories of dialectical materialism and the three laws of dialectics (the transition of quantity into quality, the unity and struggle of opposites, and the negation of the negation) acquired great “elasticity,” which allowed one to bend philosophical argument in any direction. After making numerous twists while attempting to catch up with the winding Party line, Soviet philosophy of science finally degenerated into a complex rhetorical system that combined ostentatious adherence to the canonical formulas of Marx and Lenin with extremely flexible techniques of cunningly reinterpreting those formulas to make them fit the political agenda of the day. Such philosophical oppositions as “materialism vs. idealism,” “dialectics vs. metaphysics,” and “practice vs. formalism” acquired strong political connotations in the Soviet academic discourse in the late Stalinist period. The second (negative) term in each pair did not simply denote alleged philosophical flaws in the opponent’s argument; it also labeled the opponent as an ideological deviant and a potential enemy. As figure 1.2 illustrates, dialectical materialism became a part of newspeak, a convenient rhetorical tool for translating scholarly debates into ideological and ultimately political conflicts.

Both the scientists and the Party and state officials showed genuine creativity in using the discursive techniques of “quotation-mongering” (fetching a supporting quote from the canonical writings of Marxist “classics”) and “label-sticking” (painting the opponent as a political or philosophical deviationist).⁴⁷ Stalin’s technique of accusing his political opponents of deviating from the (constantly changing) Party line echoed in the disputes

**Figure 1.2**

between the Lysenkoites and geneticists over the interpretation of Darwinism and in Pavlov's disciples' competing claims to his legacy. The debates over who was a "true Darwinist" or a "true Pavlovian" were modeled after the struggle for the right to be called a "true Leninist." In a truly postmodern fashion, both sides in such disputes jiggled "classical" quotations to prove their exclusive connection with the sacred source of scientific/political authority. Some scholars even kept elaborate catalogues of Marxist quotations for any occasion.⁴⁸ The philosophical/political labels scientists were trying to stick to their opponents were also parts of the regular Party lexicon.

The emergence of scientific newspeak as a fusion of scientific and political discourses was a part of the general "cultural unification" between scientists and their Party and government patrons in the late Stalinist period.⁴⁹ As a new elite group, Soviet scientists began to play "games of intraparty democracy," to model scientific conferences after Party meetings, to reproduce the public rituals of "criticism and self-criticism," and to frame political denunciations as "creative discussions" of scholarly matters.⁵⁰ Ideological language was not just an instrument of scientists' interaction with the state; to a large extent, newspeak was the product of this interaction.

Both professional ideologists and scientists contributed to the expansion and the sophistication of newspeak's uses. Not only did the scientists borrow terms and rhetorical tools from the Party lexicon, but the Party and state officials were also striving to speak "scientifically." Officially approved scientific doctrines, such as Lysenkoism and Pavlovianism, entered the newspeak canon and provided models and terminology for public discourse far beyond their disciplines. Images and metaphors circulated back and forth between the realms of politics and science.

In the late Tsarist and early Soviet periods, Communist ideology drew its legitimacy in part from its claim of being "scientific," thus in fact placing science on an even higher pedestal than itself. It was somewhat ironic, but historically quite logical, that the roles reversed when Communist ideology was firmly installed in its supreme position: now science had to draw its own legitimacy from the claim of its complete compatibility with this ideology. Both the Marxist ideologists and the scientists seemed quite content with this circular argument: Marxism is correct because it is scientific; Soviet science is correct because it is Marxist. Ideologists interpreted this formula as the affirmation that Marxist ideology was the supreme judge of scientific truth; they made it the foundation of the so-called principle of "Party-mindedness" of science. Scientists, by contrast, often construed it as the assertion of the inherent affinity between science and Marxism, as they were both based on the same rational foundation. Sergei Vavilov, president of the Soviet Academy of Sciences, came up with a cunning formula: The Party-mindedness of science is "the expression of its correctness."⁵¹ As the historian Alexei Kojevnikov has observed, this statement could be read either as an affirmation that Party-mindedness was vital for attaining correct knowledge or as an assertion that correct scientific theories would certainly be useful for socialist construction and should therefore be regarded as "Party-minded."⁵² Scientists who preferred to elude ideological watchdogs and to use their own criteria of scientific validity, of course, favored the latter interpretation.

Like many other national languages in the twentieth century, the Russian language dramatically expanded its vocabulary with new scientific and technical terminology. As the literary critic Andrei Sinyavsky argued, newspeak turned these terms into political symbols:

Everyday speech is full of these "beautiful" words, such as *mechanization, industrialization, electrification, melioration, chemicalization, aviation, radio, antenna,*

accumulator, commutator, automat, cadres, et cetera. These terms [imply] some greater meaning. Behind the word *cadres*, one hears Stalin's famous dictum "The cadres decide everything"; and behind the word *electrification*, the Leninist formula "Socialism equals Soviet power plus the electrification of the entire country."⁵³

Newspeak borrowed from the language of science not only florid terminology but also a whole set of rhetorical devices that helped create the impression of objectivity. Both languages favored an impersonal style, avoided verbs, used an excessive number of nouns, and were well suited for generalizations and decontextualized, timeless proclamations.⁵⁴ Newspeak, for example, deliberately excluded the first person pronoun *I*, replacing it with the collectivist *we*, in much the same way as scientists often let "facts speak for themselves" instead of providing a personal account. As in scientific discourse, which is marked by an elaborate classification of entities, newspeak piled up nouns on top of one another, essentializing ideological constructs and increasing the social distance between the author and the listener.⁵⁵ In short, newspeak aspired to what the Soviet dissident Aleksander Zinov'ev has dubbed the "scientoid" style.⁵⁶

Clever manipulation of ideologemes allowed one to translate science into philosophy into politics in many different ways. Rival groups of scientists translated the same scientific theories into opposite ideologemes. They rendered their own scientific theories into positive ideologemes, such as "objectivity," "practicality," "dialectics," and "materialism." At the same time, they translated their opponents' theories into negative ideologemes, such as "objectivism," "utilitarianism," "idealism," "mechanicism," and "metaphysics." The outcomes of scientific disputes often hinged on the disputants' mastery of newspeak and their ability to perform an ideological translation. The fundamental indeterminacy of ideological translation, however, often made it difficult to predict which translation would work better and which side in an ideological debate would win.⁵⁷

Three case studies of ideological disputes in mathematics, linguistics, and physiology are discussed below. Despite the vast differences in the intellectual contents of these disciplines, Soviet scientists resorted to very similar discursive strategies in their attempts to translate science into politics in all three cases. Rival groups transformed their intellectual disagreements and institutional squabbles into principled ideological conflicts. Academic discussions in various disciplines were conducted within a unified philosophical/ideological discourse that revolved around

such concepts as “idealism” and “formalism.” Mathematicians, linguists, physiologists, and politicians found a common language: newspeak.

“Formalism” as a Floating Signifier

The “floating signifiers” of newspeak not only changed meanings in different contexts but also carried “traces” of meaning from one realm to another. They easily crossed boundaries between scientific disciplines and various spheres of culture. If a term acquired a strong positive or negative ideological evaluation in one field, it carried this value into other domains. *Formalism* became a particularly pervasive word, encompassing a wide range of negatively evaluated phenomena in various fields of science and culture.

When Soviet critics castigated Hilbert and Ackermann’s book *Principles of Theoretical Logic* for formalism, they purposefully attached an ideological label to a technical term. Among the mathematicians, Hilbert’s school of mathematical thought had been internationally known as formalism. This school emerged in the early twentieth century as one of the three major approaches to the foundation of mathematics. Hilbert’s formalism aspired to apply the axiomatic method of Euclid’s geometry to other branches of mathematics, treating mathematical theories as totalities of formulas written in accordance with arbitrarily defined formal rules without any reference to the meaning of these formulas. Formalists thus derived the validity of mathematical theories from the consistency of the formal language of mathematics. Two alternative approaches to the foundation of mathematics were intuitionism and logicism. Intuitionism, introduced by the Dutch mathematician L. E. J. Brouwer, contended that the basic objects of mathematics were mental constructions governed by self-evident laws. Intuitionists thus accepted only those mathematical axioms and rules of inference that they intuitively perceived as true, dismissing the arbitrarily defined formulas of the formalists as mathematically meaningless. Logicism was developed by the German mathematician Gottlob Frege and the British mathematician Bertrand Russell. They attempted to deduce mathematics from purely logical principles without using any specifically mathematical concepts, such as set or number. Back in the 1930s, Soviet mathematicians discussed formalism, intuitionism, and logicism as important mathematical trends without attaching to them any particular ideological labels.⁵⁸

The word *formalism* acquired a much more ominous meaning during the militant debate over the interpretation of Marxist philosophy in 1930. One group of Marxist philosophers accused another of “formalistic views and errors,” which consisted in “the repetition of abstract formulas instead of solving concrete problems posed by life itself.”⁵⁹ The accused charged that their opponents were themselves guilty of “formalistic deviations,” and qualified formalism as an “idealistic” departure from dialectical materialism. “The formalistic perversion of materialistic dialectics,” they wrote, “is in essence an idealistic revision of Marxism.”⁶⁰

Formalism, now clearly a derogatory label, became widespread in the Soviet public discourse in the context of a vociferous anti-formalist campaign in 1936, which was opened by two editorials in *Pravda* condemning “formalist perversions” in the music of Shostakovich.⁶¹ His experiments with complex musical language, often unfamiliar and alien to a wide audience, were publicly castigated for their “anti-popular” character. “A formalist” became a common label for anyone detached from life and the needs of the people.

The new meaning of *formalism* loomed large in the attacks Trofim Lysenko and his followers launched against Soviet geneticists in the late 1930s. Accusing geneticists of reducing their work to “formal” statistical analysis, the Lysenkoites put into circulation the label “formal genetics.” In 1936–37 several articles harshly criticized “formalism” in Soviet genetics.⁶² Soon this term began penetrating official discourse. The Presidium of the Soviet Academy of Sciences used the term *formal genetics* in its official resolution in May of 1938.⁶³ In 1939 Stalin’s chief of secret police, Lavrentii Beria, wrote to the foreign minister, Viacheslav Molotov, about “the bourgeois school of so-called formal geneticists.”⁶⁴ In 1940, in an attempt to discredit the use of mathematical methods in biology, the philosopher Ernest Kolman unearthed some “incriminating evidence” of the direct connection between “formal genetics” and Hilbert’s mathematical formalism.⁶⁵ In a 1930 article, Hilbert drew parallels between genetics and geometry and argued that “the laws of heredity appear as an application of the linear axiom of congruence of the elementary geometrical propositions about plotting the intercepts.”⁶⁶ Quoting this passage, Kolman charged that Hilbert had “utterly simplified” and reduced to “meaninglessness” the biological concept of heredity.⁶⁷ The ideological label *formalism*, like a floating signifier, passed from mathematics to philosophy to music to biology and

back to mathematics, and from academic disputes to Party documents and back to academic disputes.

Formalism resurfaced in official discourse soon after the war. The February 1948 resolution of the Party Central Committee “On the Opera *Great Friendship* by V. Muradeli” condemned “formalist perversions” in Soviet music. The chief ideologist, Andrei Zhdanov, defined formalism as “a rejection of the classical heritage under the banner of innovation, a rejection of the idea of the popular origin of music, and of service to the people, in order to gratify the individualistic emotions of a small group of select aesthetes.” He condemned “the formalist trend” in music as “ugly and false, permeated with idealist sentiment, alien to the broad masses of the people, and created not for the millions of Soviet people.”⁶⁸

Lysenko’s infamous triumph over Soviet geneticists at the July-August 1948 session of the Academy of Agricultural Sciences further popularized the label *formal genetics*. A vociferous campaign against “reactionary” scientific theories followed, quickly spreading Lysenko’s style of argument to other disciplines. By analogy with *formal genetics*, the opponents of the structural method in linguistics labeled this approach *formal linguistics*.⁶⁹ In structural chemistry, the charge of formalism was leveled against the proponents of the resonance theory.⁷⁰ Critics accused some astronomers of formalism and “astronomical idealism” for engaging in research far removed from the practical needs of industrial construction, and a prominent ethnographer was labeled a formalist for placing more emphasis on cultural forms than on social content.⁷¹ *Formalism* became a traveling label, a floating signifier, that could be stuck wherever logical or mathematical reasoning appeared in the sciences. *Formalism* not only traveled across disciplines; most important, it tied together different layers of discourse (scientific, philosophical, and political), effectively erasing boundaries between them. By speaking of formalism, one could easily link mathematical reasoning in science with philosophical mistakes and eventually with political errors. The charge of formalism became one of the basic discursive patterns, or motifs, of newspeak.

From Formulas to “Formalism” in Mathematics

Stalin evidently did not consider mathematics a particularly appropriate subject for ideological critique. When he received a draft of Trofim

Lysenko's 1948 speech, which stated that "any science is class-oriented by its very nature" (i.e., that science must be either "bourgeois" or "proletarian"), Stalin underlined that phrase and wrote in the margin: "Ha-Ha-Ha! And what about mathematics? And what about Darwinism?"⁷² The sarcastic tone of this comment suggests that, in Stalin's view, the idea of applying class-based analysis to mathematics was laughable. Yet Soviet mathematicians had little reason to laugh. Stalin's marginal remark remained private, while the vociferous campaigns against "idealism" and "formalism" went on publicly across all scientific disciplines. Some campaign activists managed to find ideological heresy where Stalin himself could not find it: in mathematics.

Taking advantage of the ongoing ideological campaigns, some mathematicians quickly translated their intellectual and institutional conflicts into newspeak. For example, in the summer of 1948 Konstantin Ermilin, a professor of mathematics at Leningrad State University, sent a letter of denunciation to the Party Central Committee. Accusing Soviet specialists in set-theory topology and the algebra of associative systems of using "fancy terminology," of giving "arbitrary definitions" to mathematical objects, and of studying them with a "formal apparatus," Ermilin labeled this a "decadent trend." He argued that such theories emphasized only the "surface, formal side" of mathematics, did not reflect the material world, and were therefore "idealistic." "Pure knowledge," Ermilin declared, could only lead to "total ignorance." "Mathematics is not an entertainment for the devotees of logical constructions," he told Party officials. "One cannot easily accept the fact that the mental energy of highly qualified people is wasted on leisurely play with concepts, while this country is awaiting effective practical help from these people." In conclusion, Ermilin called for mathematical research institutions and university curricula to be purged of the "decadent influences."⁷³

Party authorities trusted an investigation to the Leningrad University Party organization, which announced a public discussion of the ideological situation in mathematics. The prominent Leningrad mathematician Aleksandr Aleksandrov realized that Ermilin's letter was motivated by personal intrigues rather than by deep conceptual disagreement. According to the rules of the game, however, the charge formulated in newspeak had to be answered in newspeak. Therefore, Aleksandrov, like Ermilin, assumed the role of an uncompromising fighter against "formalism" and "idealism"

in mathematics, but he located these evils elsewhere. Later Aleksandrov recalled: "The next morning I brought to the Party bureau my theses on formalism in mathematics. I seized the stand first and thereby set the direction for the whole discussion. We harshly criticized formalism, but found no formalists in Soviet mathematics, not to speak of our university mathematics."⁷⁴ Contrary to Ermilin's expectations, the affair did not result in any disciplinary action. Aleksandrov's rendition of "formalism in mathematics" as a phenomenon characteristic of Western and not of Soviet mathematics successfully diverted the discussion away from personal attacks on his colleagues.

Aleksandrov achieved his goal by applying the familiar "de-ideologization" strategy, which he further developed in a series of articles in the popular magazine *Priroda* [*Nature*] in the early 1950s. In these publications, he drew a sharp line between the "objective content" and the "philosophical interpretation" of scientific theories. He also claimed that the content of scientific knowledge was "independent from the social system or ideology"; it was only the general cultural meaning of scientific ideas that bore an "imprint of society's ideology."⁷⁵ The same mathematical theory, Aleksandrov argued, would receive different philosophical interpretations in different ideological contexts. He admitted that mathematics was in a deep ideological crisis in bourgeois societies. In particular, he condemned Hilbert's formalism and Brouwer's intuitionism as philosophical speculations based on "the separation of mathematics from material reality, from practice."⁷⁶ He explained that, in their search for the foundations of mathematics, the formalists relied on logical consistency and the intuitionists on the mathematician's personal intuition. Soviet mathematicians, in contrast, were armed with the postulates of dialectical materialism, verified mathematical truths with practice, and were thus protected from philosophical errors. Therefore, while "idealistic perversions" led to the crisis of mathematics in bourgeois societies, Soviet mathematics was ideologically safe. "In a socialist society, a crisis of science is impossible," Aleksandrov declared, "since Marxism—the ideology of socialism—is a scientific ideology and therefore, by its own nature, must be in harmony with the objective content of science."⁷⁷ He concluded that, despite the idealistic philosophical interpretations of mathematical logic in the West, Soviet mathematicians should further develop the "objective content" of formal mathematical logic and formal calculi, since without formal logical

consistency mathematical theories would lose their power as instruments of science.⁷⁸

In his 1950 review of the philosopher Leonid Maistrov's dissertation "The Struggle of Materialism against Idealism in the Theory of Probabilities," the mathematician Aleksei Liapunov employed a similar discursive technique. Imitating Lysenko, Maistrov discovered two opposite ideological trends in probability theory: a materialistic trend and an idealistic one. He labeled Émile Borel, Richard von Mises, Karl Pearson, and Henri Poincaré "bourgeois scientists." In particular, he argued that, in the statistical interpretation of probability offered by von Mises, probabilities resulted "not from the objective internal characteristics of phenomena but from our experimentation." Maistrov also condemned the attempts to apply probability theory to sociology as a dangerous effort to substitute probabilistic laws for the "real regularities" of social development. "By perverting the theory of probabilities in an idealistic fashion, the educated lackeys of the capital are trying to vindicate the incurable vices of the capitalist society and justify reactionary political theories," he argued.⁷⁹ In his review, Liapunov attempted to rescue Western work on probability theory by separating this work from the philosophical views of Western scientists. He covered his ideological bases by declaring his support for Maistrov's negative evaluation of the philosophical and political views of the aforementioned "bourgeois scientists." "I believe, however," Liapunov added, effectively neutralizing the ideological critique, "that it is necessary to emphasize that the main scientific activity of Poincaré, Borel, and von Mises in essence contradicts their own philosophical views."⁸⁰

Various authors drew the line between "scientific activity" and "philosophical views" differently. In her introduction to the Russian edition of Hilbert and Ackermann's *Principles of Theoretical Logic*, Sof'ia Ianovskaia attempted to distance mathematical logic per se from the "idealists parasitizing on mathematical logic." Her critics, led by Maistrov, refused to draw a line between knowledge and ideology, and argued that "the reactionary philosophical views of the founders of bourgeois mathematical logic have perverted the content of this discipline."⁸¹

By blurring the line between scientific and philosophical terminology, newspeak provided rhetorical resources for both sides in this debate. Militant critics identified mathematical formalism with the philosophical error of "formalism." Drawing heavily on the Marxist principle of practice as the

criterion of truth, they associated abstract mathematical reasoning with the “detachment from practice.” Their opponents identified abstract thought not with formalism but with generalization (a positive member of the tetrad) and cited Lenin to argue that practice, which always deals with the concrete, can never serve as the final criterion for verifying general knowledge.⁸² When particular mathematical theories came under ideological attack, their defenders resorted to the same flexible techniques of newspeak as the critics, shifting the imaginary boundary between knowledge and ideology and providing alternative philosophical interpretations of controversial theories.

From Literary Form to “Formalism” in Linguistics

Linguistics, like any other scientific discipline, was subjected to ideological scrutiny in the search for “formalists” and “idealists” in the course of the ideological campaigns of late Stalinism. The situation was exacerbated by the fact that in linguistics, as in mathematics, *formalism* had a technical meaning, and a number of linguists and literary critics called themselves “the Formalists,” thereby directly inviting ideological criticism.

The Formalist movement comprised two distinct groups: the Society for the Study of Poetic Language (Opojaz), founded in 1916 in Petrograd (later Leningrad and now St. Petersburg), and the Moscow Linguistic Circle, founded in 1915.⁸³ The Opojaz members were primarily literary critics who modeled literary analysis on linguistic studies; the Muscovites were largely professional linguists interested in expanding the scope of linguistic analysis to literary texts. Both groups adopted the view of language as a formal system of signs developed by the Swiss linguist Ferdinand de Saussure. Opposed both to symbolist analysis and to sociological criticism, the Formalists aspired to create “scientific” methods of literary analysis. Their analysis focused on formal artistic devices and textual techniques; they viewed the ideological, psychological, and sociological dimensions of literature as secondary. “A literary work is pure form,” declared Viktor Shklovskii, the leader of Opojaz. “It is not a thing or an object; it is a relationship between objects.”⁸⁴ “If literary history wants to become a science,” proclaimed Roman Jakobson (one of the founders of the Moscow Linguistic Circle), “it must recognize the artistic device as its only concern.”⁸⁵

In the early Soviet years, the radical stance taken by the Formalist movement met sharp criticism from various sides. Its former supporter Viktor

Zhirmunskii, for example, attacked the Formalists for their exclusive reliance on a single analytical method. “For some adherents of this new trend,” he wrote, “the Formalist method becomes the only legitimate scientific theory, not merely a method, but a full-fledged world view, which I would prefer to call formalistic rather than formal.”⁸⁶ The Formalists’ opposition to sociological analysis also prompted a reprimand from Marxist scholars; Leon Trotsky, in particular, condemned “the superficiality and reactionary character” of the Formalist theory. This criticism, however, remained largely academic; even Trotsky admitted that Formalists’ work, despite their errors, was still “useful.”⁸⁷

In the 1920s the linguistic branch of the Formalist movement produced a major innovation in phonological studies. Jakobson and the prominent linguist Prince Nikolai Trubetskoi emigrated from the Soviet Union to Czechoslovakia, where they became the core of the Prague Linguistic Circle. The Prague school defined phonemes both functionally (as sound elements which differentiate between morphemes) and structurally (as elements in a phonological system). The sound side of language was thus presented as a structural whole rather than a chaotic assembly of individual phonemes. The organization of phonemes into a system with a strictly defined set of relations led Jakobson to the idea of implicational laws (rules) of phonology, which made it possible to predict unknown structural elements on the basis of knowledge of the available elements.⁸⁸

Jakobson aspired to extend the structural approach beyond phonology to the whole of linguistics and eventually to other disciplines. In his view, structural linguistics was part of a larger structuralist trend dominating contemporary science:

Were we to comprise the leading idea of present-day science in its most various manifestations, we could hardly find a more appropriate designation than *structuralism*. Any set of phenomena examined by contemporary science is treated not as a mechanical agglomeration but as a structural whole, and the basic task is to reveal the inner, whether static or developmental, laws of this system.⁸⁹

After moving to the United States in the 1940s, Jakobson co-founded the New York Linguistic Circle, which further propagated structuralist ideas. With the rise of the Cold War hysteria, however, he became a victim of an anti-Communist “witch hunt.” Some of Jakobson’s colleagues resented his active contacts with scholars from Eastern Europe and accused him of Communist sympathies. One activist even demanded that Jakobson be

“registered as a foreign agent because of the sympathies for the Marxian ideology in his writings.”⁹⁰ On the other side of the Atlantic, ironically, the same Cold War discourse made him an equally suspect figure in the eyes of ideological watchdogs among Soviet linguists.

In the course of the postwar campaigns against “cosmopolitanism,” “kowtowing before the West,” and “reactionary and idealistic” Western science, the Formalists and the structural linguists became targets of severe ideological criticism. Jakobson’s “faults”—his emigration to the West and his “cosmopolitan” (i.e., Jewish) origins—threw a long dark shadow over his work on structural linguistics, which became almost synonymous with “bourgeois linguistics.” The Formalist movement was now closely identified with the philosophical and ideological errors of “formalism” and “idealism.” One critic accused Jakobson of reducing “the essence of [linguistic] phenomena to a system of formal oppositions” and branded his views “idealistic.”⁹¹ This critic further argued that the Moscow Linguistic Circle had represented a “pillar of formalism, a reactionary bourgeois trend, which attempted to poison the mind of the Soviet intelligentsia.” “It is characteristic,” this critic continued, “that Jakobson has currently retreated to the USA—a bastion of bourgeois reaction, the principal source of imperialist aggression.”⁹² The Formalist movement in literary criticism and the structural approach in linguistics were painted as mortal enemies of Marxism, their intellectual flaws closely intertwined with political errors:

Structuralism is one of the most reactionary trends in modern bourgeois linguistics. This militant formalist doctrine is mobilized completely for the struggle against Marxism in linguistic science and embodies the basic features of bourgeois cosmopolitanism. . . . The theoretical and organizational formation of structuralism in linguistics is inextricably connected with the names of the typical bourgeois cosmopolitan scholars Roman Jakobson and Prince Trubetskoi, people who relinquished their motherland.⁹³

As a healthy alternative to this “relic of bourgeois ideology,” the critic praised the “new doctrine of language” of the late Marxist linguist Nikolai Marr and extolled the concept of the “dialectical unity of language and thought.” This “unity,” of course, was not amenable to any formal representation.⁹⁴

Jumping on the bandwagon of the campaign against “reactionary and idealistic” Western science, ideological vigilantes among the linguists rushed to expose in public the “alien” trends in linguistics. Marr’s followers, in particular, launched a frontal attack on their main opponents, specialists in

comparative historical linguistics and structuralists, carefully couching their critique in ideological terms.⁹⁵ Closely adhering to Lysenko's rhetorical style at the July-August 1948 session of the Academy of Agricultural Sciences, campaign activists put Marr on a pedestal as the Russian founding father of genuine linguistics and contrasted his "materialistic" views with "idealistic" and "metaphysical" linguistic theories. Their desire to imitate Lysenko, who had referred to "two trends in biology," was so strong that the leading Marrist critic Fedot Filin entitled a 1948 talk "On Two Trends in Linguistics," even though Marrism and its two main "enemies," comparative historical linguistics and structuralism, made up *three* trends.⁹⁶

Referring to structural linguistic studies in the West, the Marrists quickly labeled them "bourgeois phonology" and accused Soviet structuralists of pursuing "a general formalistic approach to language." Since structuralism and comparative historical linguistics were lumped together as a single "idealistic" trend in linguistics, the charge of formalism was soon extended from one to the other. The Marrists branded the efforts to trace the origins of Indo-European language to the parent Proto-Indo-European language in comparative historical linguistics a "formal-genetic" approach to the study of language, echoing Lysenko's favorite label "formal genetics." The Marrists then could not decide whether it would be better to accuse their opponents of "the identification of thought with language" or "the detachment of thought from language"; in the end, they brought both accusations and concluded that either error led to "pure formalism." To make it clear that ideological errors had to be corrected with administrative measures, Filin listed the most important positions occupied by his opponents, implying that those positions would be better filled with Marrists.⁹⁷ As a result of these attacks, several prominent specialists in comparative historical linguistics lost their jobs, and one died from a heart attack after two weeks of incessant public castigation.⁹⁸

In June of 1950, with yet another turn of the ideological screw, Stalin suddenly distanced himself from the "excesses" of policies he had previously promoted.⁹⁹ In a lengthy article on linguistics in *Pravda*, he branded the Marrist "new doctrine of language" as "truly idealistic" and restored the legitimacy of comparative historical linguistics. Moreover, he condemned the practice of administrative pressure (the "Arakcheev regime"¹⁰⁰) in science, rejected the idea that one school could hold a monopoly on truth, and called for "freedom of criticism." Stalin's authoritarian call for free-

dom, however, produced precisely the opposite: comparative historical linguistics immediately assumed monopoly status, the comparativists were again elevated to high positions, and large-scale administrative persecution of the former Marrists began.

Distributed in millions of copies and prescribed for obligatory “discussion” in all academic institutions, “Stalin’s doctrine of language” was quickly canonized, along with all its accidental blunders (including Stalin’s claim that the hitherto unknown “Kursko-Orlovskii dialect” was the basis of the Russian language¹⁰¹). To say that Stalin’s sudden about-face came as a shock to the Marrists would be an understatement. Two of them reportedly went mad.¹⁰² Others quickly “reeducated themselves” and began to write prolifically on the imaginary Kursko-Orlovskii dialect. The endless public praise of Stalin’s “linguistic genius” was mocked in an underground song:

Comrade Stalin, you’re a truly great savant,
By your knowledge of linguistics I am stunned.
I’m a humble Soviet inmate with no roots
And no comrades but gray wolves in the woods.¹⁰³

Stalin evidently enjoyed his role as a great scholar, for he even took time to answer some of the questions sent to him by his readers. One reader asked the “linguistic genius” to clarify the meaning of the traveling label *formalism*, indiscriminately used by the Marrists: “Many linguists consider *formalism* one of the main reasons for stagnation in Soviet linguistics. It would be very desirable to know your opinion, what is formalism in linguistics and how to overcome it?”¹⁰⁴ Stalin, a true expert in newspeak, patiently explained the political function of this floating signifier in linguistics:

N. Ia. Marr and his “disciples” accuse of “formalism” all linguists who do not support the “new doctrine” of N. Ia. Marr. This is, of course, preposterous and mindless. N. Ia. Marr considered grammar a simple “formality” and called “formalists” those who viewed the grammatical structure as the foundation of language. This is altogether silly. I think that “formalism” was invented by the authors of the “new doctrine” to facilitate the struggle against their opponents in linguistics.¹⁰⁵

By that time, the label *formalism* had become so pervasive in Soviet academic discourse that even Stalin’s explicit critique could not put an end to the wanderings of this empty signifier. Traditional historical comparative linguists, who had taken the place of Marrists as the officially approved linguistic school, often similarly translated their intellectual disagreement with the structuralists into ideological terms. Traditional linguists studied

phonemes in isolation from one another, while structuralists attempted to arrange the phonemes into a unified system and measure their characteristics. Comparative linguists branded such efforts formalistic. “The structural approach is not a method of linguistic research,” one critic argued, “but merely a technique of formal mathematical representation of the results of research, and this form of representation does not add any new knowledge of language.”¹⁰⁶ In politically charged academic debates, the use of mathematical formulas in the social sciences and the life sciences almost invariably provoked the allegation of formalism.

The Specter of “Idealism” in Physiology

Under the banner of the struggle against “formalism,” the Lysenkoites expelled mathematics from Soviet biology, while the Marrists and the comparativists barred it from Soviet linguistics. Physiology remained one of the very few disciplines in which one could still expect mathematical formalisms to play a legitimate role. After all, it was Ivan Pavlov himself, the icon of Soviet physiology, who proclaimed in the early 1910s that “the edge of physiological knowledge, its true goal is to express the infinitely complex relationship between the organism and its environment in a precise mathematical formula.” “This,” Pavlov continued, “is the ultimate goal of physiology, this is its frontier.”¹⁰⁷ Pavlovian experiments on conditioning always involved precise quantitative measurements. In his programmatic 1902 address “Natural Science and the Brain,” Pavlov expressed confidence that even human thought would eventually lend itself to mathematical description:

All life—from the most elementary to the most complex organisms, including, of course, the human beings—is a long chain of more and more complex, up the highest degree, balances with the environment. The time will come—maybe in the distant future—when mathematical analysis based on natural-scientific research will embrace with magnificent formulas of equations all these balances, eventually including itself.¹⁰⁸

Ivan Petrovich Pavlov (1849–1936) occupied an exceptional place in Soviet science, both as a man and as a symbol.¹⁰⁹ Named the “leading physiologist of the world” at the Fifteenth International Psychological Congress in 1935, he enjoyed great international fame. Before World War II, he was the only Nobel Laureate among Soviet scientists, and thus he was a valu-

able national asset in both scientific and propaganda terms. To advance his ambitious agenda of experimental and theoretical research, Pavlov made use of his administrative authority as the head of several laboratories and his personal access to Soviet leaders as an expert advisor to the government. After his death in 1936, his theory of conditional reflexes became a canonical conceptual framework for Soviet physiology.

For Pavlov, modern technology was an emblem of sophistication and he aspired to “elevate” his theory of nervous activity to an equally high level of complexity. Accordingly, his laboratory at the Institute of Experimental Medicine in St. Petersburg was organized as a giant “physiology factory,” with rigid division of labor, strict discipline, uniform experimental procedures, and checking of knowledge claims to make sure that they conformed to the Pavlovian theoretical framework.¹¹⁰ Pavlov strove to uncover precise quantitative laws governing physiological processes. He insisted on the “complete exclusion of psychic influence” from his experiments, so that machine-like regularities underlying the functioning of physiological mechanisms would not be obscured. Viewing the organism as a complex machine, Pavlov modeled his ideal of scientific investigation after engineering practice:

Man is definitely a system (more crudely speaking, a machine), which, like any other system in nature, obeys the inescapable and uniform laws of all nature; but this system, in the horizon of our contemporary scientific view, is unique in its highest degree of self-regulation. Among the products of man’s hands, we are already sufficiently familiar with machines that regulate themselves in various ways. From this point of view, the method of studying the man-system is the same as for any other system: decomposing into parts, studying the role of each part, studying the connections among the parts, studying the relations with the environment, and finally, based on all this, understanding the overall functioning of this system and, if within human capacity, controlling it.¹¹¹

In his writings, Pavlov constantly borrowed metaphors from contemporary technology; for example, he called the digestive system a “chemical factory.”¹¹² His central metaphor—the human nervous system as a central telephone switchboard—illuminated the crucial distinction between unconditional and conditional reflexes.¹¹³ According to Pavlov, inborn unconditional reflexes fixedly tied a particular stimulus (e.g., the viewing of food) to a specific response (e.g., salivation). Pavlov compared such reflexes to a set of direct, permanent telephone lines. Conditional reflexes (e.g., salivation in response to the ringing of a bell, previously associated with the viewing

of food) could be acquired, lost, and reestablished. Pavlov likened conditional reflexes to flexible temporary connections between telephone users through a switchboard. “In technology, as in our daily life,” he wrote, “the principle of [temporary] connection is applied so often that it would be odd not to expect the implementation of the same principle in the mechanism of the higher nervous system, which establishes most complex, subtle connections.”¹¹⁴ In the same way as the central telephone station solved the problem of communication for a large number of users, the mechanism of conditional reflexes, in Pavlov’s view, solved the problem of the organism’s reaction to diverse stimuli.

In the early Soviet years, new machines, such as tractors and electric generators, were hailed as signs of progress and emblems of the bright communist future. Man-machine metaphors permeated public discourse. Poetry and popular songs propagated images of “iron men” with “a flaming engine in place of heart.” The Soviet fascination with Taylorism and Fordism produced a popular movement for the “scientific organization of labor” led by Aleksei Gastev, a visionary who saw increasing mechanization and standardization of workers’ movements, language, and even thoughts as means of improving the efficiency of labor. In 1920 Gastev organized the Central Institute of Labor in Moscow and launched a wide range of scientific studies and training courses.¹¹⁵ In 1922 Gastev hired the physiologist Nikolai Aleksandrovich Bernshtein (1896–1966), who would later play the leading role in Soviet “physiological cybernetics.”

Throughout his career, Bernshtein spoke openly and consistently about his disagreement with Pavlov’s doctrine of conditional reflexes, even when it became an official dogma.¹¹⁶ Pavlov’s experiments focused principally on the correlation between a dog’s salivary gland activity and various stimuli; he dealt neither with locomotion nor with human experimental subjects. Bernshtein studied precisely what Pavlov left out: human motor activity. In the early 1920s, Bernshtein conducted a series of experiments at the Institute of Labor, measuring the trajectories and speeds of human limbs, while his subjects performed such tasks as hammering, typing, or playing piano. He further developed and refined experimental techniques of the European science of “biomechanics,” which conceptualized the human body as a mechanical system of muscular masses and forces.¹¹⁷ Classical neurophysiology assumed that each muscle had a singular representation in the corresponding area of the brain; a central impulse from a given area

was thought to direct a determined movement of the corresponding muscle. Bernshtein called this “the push-button control-board model of the cortex, similar in plan to an organ keyboard.”¹¹⁸ In contrast, his experiments showed that the same labor task was performed differently—with varying tensions of different muscles—at various times. Muscular movements were “constructed” anew, so to speak, each time the task was performed.¹¹⁹ Bernshtein argued that reflex theory could not explain muscular movements, and he interpreted locomotion not as a sequence of pre-determined actions (as in the Pavlovian reflex theory) but as a cycle of actions and corrections. As early as 1934, Bernshtein proposed to replace the classical Pavlovian concept of the “reflex arc” with a “reflex circle.”¹²⁰

After the establishment of Pavlov’s reflex theory as an unquestionable physiological canon, all alternatives to the Pavlovian doctrine quickly disappeared from academic discourse. When the “patriotic” campaign of the late 1940s prompted the search for Russian-born “founding fathers” in all scientific disciplines, physiologists did not have to search very hard. Pavlov was already recognized as the leader of their discipline, and the only question was who would benefit from the closest association with Pavlov’s name. Like many other canons, the Pavlovian legacy proved amenable to diverse interpretations. Some of Pavlov’s former students laid exclusive claims to his legacy and began to pull Soviet physiology in different directions. An institutional conflict between rival groups within the physiological community took the form of a debate over the correct interpretation of Pavlov’s teaching.

In accordance with the political style of the day, this debate was conducted in newspeak. Some of the “Pavlovians” cleverly linked Pavlov’s teaching to the newly canonized doctrine, Lysenko’s theory of the inheritance of acquired characteristics. They disregarded the well-known fact that Pavlov had supported Soviet geneticists in the 1930s and had even ordered that a monument to Gregor Mendel be erected in front of his laboratory. Instead, they “creatively” reinterpreted the scientific legacy of their late mentor and claimed its full compatibility with Lysenkoism. They revived Pavlov’s early hypothesis (which he later rejected) about the inheritance of conditional reflexes and their transformation into unconditional ones, and they portrayed this hypothesis as the central dogma of the Pavlovian doctrine. In June and July of 1950, with the blessing of Party authorities, they convened a special joint session of the Academy of Sciences and the Academy of

Medical Sciences in Moscow to solidify their reinterpretation of Pavlov's teaching and to purge their opponents.¹²¹ Several prominent physiologists who resisted this dogmatic interpretation of Pavlov's teaching were accused of the "perversion of Pavlov's line" and dismissed from their jobs. Pavlov's most talented disciple, Leon Orbeli, who did not support the myth of "Pavlov the Michurinist," was relieved of his administrative positions in Soviet physiology, and this opened wonderful career opportunities for orthodox Pavlovians. The name Pavlov now stood for Party-blessed dogma in science. The rejection of such a dogma amounted to a political mistake.

Bernshtein, who disagreed with Pavlov conceptually and did not even attempt to portray himself as an orthodox Pavlovian, became a prominent target of ideological criticism. The onset of the anti-Semitic campaign against "cosmopolitanism" made Bernshtein, a Jew, particularly vulnerable. Despite his highly acclaimed wartime work on the restoration of damaged motor functions at the Moscow Scientific Research Institute of Prosthetics and the 1947 awarding of the prestigious Stalin Prize for his book *On the Construction of Movements*, Bernshtein was subjected to severe public criticism. His critics pointed out that he had mentioned Pavlov's name in his book only once and accused him of attempting to "belittle" Pavlov's significance. Furthermore, since Bernshtein cited foreign authors, he was charged with "kowtowing before foreign scientists" and "anti-patriotism."¹²² The critics also attached to Bernshtein's doctrine the usual labels: *idealism* (for using mathematical analysis) and *mechanicism* (for regarding the human body as a self-regulating mechanism). They even accused him of holding onto the "false theory of mutations" (i.e., genetics). At the 1950 "Pavlov session," critics alleged that he knew "neither the letter nor the spirit of Pavlov's teachings." Such criticism amounted to political denunciation, and Bernshtein was soon forced out of his job and lost any opportunity for research and publication.¹²³

Instead of exploring the intellectual differences between Pavlov's doctrine and Bernshtein's approach, the critics simply followed the clichés of newspeak and portrayed the conflict between Pavlov and Bernshtein as a clash between dialectical materialism and an alien philosophy of science. Ironically, they "discovered" ideological heresy in precisely those aspects of Bernshtein's work that were, if anything, dialectical and materialist. If Bernshtein's extensive use of mathematical analysis was "idealistic," why was Pavlov's call for "magnificent formulas of equations" not idealistic? If

Bernshtein's flexible circular mechanism of physiological regulation was mechanistic, what should Pavlov's crude telephone-switchboard model of the nervous system be called?

Under the banner of the campaign against "reactionary and idealistic science," the critics systematically translated Pavlov's teaching into positive ideologemes and Bernshtein's doctrine into negative ones. The critics did not even notice that they had accused Bernshtein of deviating from dialectical materialism in two opposite directions—idealistic and mechanistic—at the same time. The charges of "idealism" and of "mechanicism" belonged to the same discursive strategy of "criticizing and destroying" the scientific opponent as an ideological enemy. As ideological synonyms, idealism and mechanicism were interchangeable, despite their opposite descriptive meanings.

Like any ideological language, scientific newspeak did not recognize inner contradictions. The same theory could be translated into newspeak in many different ways, and Soviet scientists took full advantage of this opportunity to produce alternative translations. The main ideological question in postwar Soviet science was not whether or not to be a dialectical materialist. The question was whose claim to being a dialectical materialist would be officially recognized.

In this chapter I have attempted to analyze postwar academic discourse not as a container of a particular ideology but as a mechanism for formulating and advancing political and research agendas in the language of newspeak. Instead of postulating a cleavage between "science" and "ideology," I have explored the common language spoken by both scientists and ideologues. Newspeak did not express any pre-existing ideological system; instead, it functioned as a surrogate ideology. By fusing description with evaluation, newspeak fulfilled the main function of ideology: it provided a framework for value judgments.¹²⁴ One can hardly derive diverse and eclectic newspeak practices from some fixed ideological principles. Quite the opposite: Soviet ideology itself may be more productively viewed as the result of conscious attempts to explicate and rationalize assorted discursive strategies, or mechanisms, of newspeak, in much the same way as grammatical rules are invented to describe diverse linguistic practices.¹²⁵

While the meaning of "Soviet ideology" varied widely, the Soviet ideological discourse seems to have followed very specific patterns and rules.

No individual controlled it: as we have seen, the Stalinist view of mathematics as an ideology-laden field was not necessarily Stalin's own view. Stalin could condemn a particular use of the label *formalism* in linguistics, but he could not stop its spread into other fields. Discursive inertia carried "floating signifiers" from one area to another. Newspeak had a life of its own. Both the rulers and the ruled were similarly caught up, as Mikhail Heller has put it, "in the magical circle" of the Soviet language: "There are no augurs in the Soviet logocracy who, using the Word as a weapon, can protect themselves from its influence. Everybody lives and works within the bounds of the Soviet vocabulary and of the Soviet patterns of thought."¹²⁶

When Sof'ia Ianovskaia faced ideological criticism of her role in the publication of the Russian translation of Hilbert and Ackermann's *Principles of Theoretical Logic*, she immediately performed the ritual of "self-criticism" and published a repentant letter, admitting "idealistic confusion" in her work. To vindicate herself, she now had to show some effort at "criticizing and destroying" ideological heresy in her field. A learned student of Marxism, she quickly dug up a passage in *Anti-Dühring* in which Engels ridiculed the analogy between the calculating machine and the human mind. Fitting Engels into the ongoing campaign against idealism in science, Ianovskaia wrote that his critique had been "aimed against the reactionary ambitions of idealism . . . to substitute [human] reasoning with computing." Having thoroughly "criticized and destroyed" this idealistic error, Ianovskaia successfully fulfilled her ideological obligations. She then turned quietly to rescuing the "scientific meaning" of Hilbert and Ackermann's work by emphasizing its importance for "technicians working on the construction of calculating-solving machines and automata."¹²⁷ Using the "de-ideologization" strategy, she drew a sharp line between the objective content of mathematical logic and its "idealistic" philosophical interpretation.

While the "overtake and surpass" attitude dominated in Soviet physics and the "criticize and destroy" trend triumphed in biology, the field of computing, or "machine mathematics," was an intriguing borderline case. The first electronic digital computers were built in Britain and the United States in the 1940s and the early 1950s as direct products of military-sponsored research and became vital components of weapons systems. The popular perception of computers in the West was largely shaped by cybernetics' man-machine analogies: computers were seen as "giant brains," while

human behavior was often interpreted within the framework of calculation, manipulation, and control. Together, the military importance of computing and the ideological ramifications of cybernetic metaphors created an ambiguous and complex situation for Soviet computer specialists.

I will discuss Soviet attitudes toward American cybernetics and explore the interplay of the “overtake and surpass” and “criticize and destroy” motifs in the early history of Soviet computing in chapter 3. Before examining the Soviet case, however, it is appropriate to explicate the fundamentals of cybernetics. In chapter 2, I will discuss major cybernetic ideas developed by American and Western European scientists and engineers in the 1940s, viewing the diverse and eclectic analogies between the human brain and the computer, between human communication and information exchange, and between negative entropy and biological and social order through the prism of a shared metaphorical language that I call *cyberspeak*.

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Cyberspeak: A Universal Language for Men and Machines

Recently, two facts have entered to change the situation. On the one hand, devices have been built that (so it is said) are more like men than the old machines were. Modern computers can be programmed to act unpredictably and adaptively in complex situations. That is, they are intelligent. On the other hand, men have behaved in ways that (so it is said) correspond rather well to our old ideas about mechanisms. They can be manipulated, “brain washed” and apparently controlled without limit. With this sharp increase in the number of properties that men and machines seem to have in common, the analogy between them becomes more compelling.

—Ulric Neisser, “Computers as Tools and Metaphors” (1966)¹

In November of 1959 the magazine *Science* published a jocular article titled “Report of the Special Committee.” Its author lamented the fact that special advisory committees on important scientific and technological problems were spending too much time and effort in preparation of their final reports, and suggested a radical simplification of this whole process. He observed that such reports always used the same arguments and ended with the same conclusion: an urgent request for more funding. Therefore, it would be expedient to abolish such committees altogether and instead utilize a ready-made “standard, universal form” already including all the clichés that distinguished an efficient, well-thought-out report:

Summary Report of Special Committee On X

1. This is a scientific field of critical importance, with obvious and widely ramified interconnection with national defense and with the health of our national economy. The intellectual and esthetic importance of deepening our knowledge in this area cannot be overemphasized.
2. This field has been meagerly supported in the past, and there is every reason to expect that modest but suitable financial support (say, roughly 20 times the present level) could lead promptly to results of the highest significance.
3. . . . the present moment [is] a particularly fortunate and promising one for undertaking an energetic attack.

4. . . . [the experts] have developed a momentum which is a great present asset, but which might decay rapidly if encouragement is not promptly supplied.
5. Your Committee deeply deplores—indeed condemns—international rivalries in science. But we nevertheless feel compelled to point out that the Russians appear to be, in this field X, well ahead of us.
6. Your Committee thus recommends the immediate creation of a National Institute on X . . . together with a broad program of research funds, fellowships, travel grants, and so on.²

One only needed to replace X with “a word or phrase suitable to the special case in hand” (for example, “the design and construction of a much larger, faster, and more flexible computer”), and a perfect committee report would be completely ready.

American scientists returning to civilian research after working on wartime defense projects sought ways to apply new techniques they had developed during the war to a wide range of problems, and, incidentally, to extend the generous funding they had received during the war to the post-war period. They argued that methods they had developed for solving specific military problems (for example, tracking airplanes and enciphering messages) had general theoretical significance and could solve numerous communication and control problems far beyond the military realm. This argument became a recurrent theme in innumerable proposals showered on funding agencies after the war by American scientists and engineers. The jocular “universal form” merely reflected this trend of unrestrained expansion of the new methods and theoretical models “in all suitable institutions,” “throughout the waters of the oceans of this planet,” “within the deep core of the earth,” “in the arctic and antarctic areas,” “throughout the troposphere,” and “in space.”³

One field of study seemed particularly well suited for this “universal form,” for it had truly universal aspirations to explain “all human behavior.” From wartime efforts to build computers and feedback devices for control and communication, a set of techniques of formal logical representation, mathematical modeling, and computer simulation of purposeful behavior had emerged. A number of American and British scientists and engineers from various fields had compared the human brain to the electronic digital computer both structurally and functionally, and had drawn parallels between thinking and computation, between human memory and computer storage, and between the all-or-none principle of neuron “firing” and the binary mathematics of digital computers. More generally, they had

come to view negative feedback (used in such devices as servomechanisms and computers) as the fundamental mechanism of self-regulation in human physiology and in society. Patterns of military control and communication became models for a wide range of human activities. The American mathematician Norbert Wiener introduced these ideas to a broad audience in his 1948 book *Cybernetics, or Control and Communication in the Animal and the Machine*.⁴ A new “universal” language, which I call *cyberspeak*, tied together a diverse set of man-machine metaphors. Living organisms, control and communication devices, and human society were now described in the same terms: *information*, *feedback*, and *control*.

It was hardly a coincidence that the “Summary Report of Special Committee On X” was written by an active promoter of cybernetic ideas. The author was Warren Weaver, director of the Alfred P. Sloan Foundation and of the Natural Science Division of the Rockefeller Foundation, chairman of the Basic Research Group of the Department of Defense, trustee of the Sloan-Kettering Institute, and member of National Science Foundation—an outstanding science administrator widely regarded as “the dean of American Science.”⁵ It was at the request of Weaver, then the chief of the Applied Mathematics Panel of the National Defense Research Committee (NDRC), that the mathematician John von Neumann drafted his famous report on the first electronic digital computers, in the spring of 1945. In the autumn of 1940 Weaver (then the head of NDRC section D-2, a fire-control division) let a contract for “General Mathematical Theory of Prediction and Applications,” carried out by Wiener and the electrical engineer Julian Bigelow. A year later, Weaver let another contract for Wiener to write up his theoretical results in a report, titled *The Extrapolation, Interpolation, and Smoothing of Stationary Time Series*, that lay a mathematical foundation for cybernetics. After the war, Weaver, through the Rockefeller Foundation, actively supported Wiener’s early studies in cybernetics. In 1940 Weaver also let a contract to the communications engineer Claude Shannon for “Mathematical Studies Relating to Fire Control,” under which Shannon studied the problem of improving the smoothness of tracking. In parallel, Shannon worked at the Bell Telephone Laboratories on designing a speech-encipherment system under a top-secret cryptological project whose code name, “Project X,” was later echoed in Weaver’s jocular “Summary Report on X.” This project contributed much to Shannon’s “mathematical theory of communication,” later generalized as “information theory”—one of the

central components of cybernetics. And it was Weaver who in 1949 publicly promoted information theory as a suitable explanatory framework for “all human behavior.”

Cybernetics was a very peculiar historical phenomenon. It could not be classified as a conventional scientific discipline, as an engineering approach, or as a philosophical doctrine, though it resembled each somewhat. The cybernetic mixing of humans and machines produced very diverse socio-political and cultural ramifications, and attitudes toward cybernetics were equally varied. A number of recent historical studies have traced the origins of major cybernetic concepts to Wiener’s work on anti-aircraft gun control during World War II and have argued that cybernetics embodied distinct military patterns of control and communication. In particular, Peter Galison has suggested that cybernetics, operations research, and game theory formed a set of “Manichean sciences” that made military conflict a model for human interaction with the world.⁶ Andy Pickering has traced the role of the World War II “regime” in the emergence of a whole range of “cyborg objects” (such as computers and anti-aircraft gunnery, with its “coupling of men and artillery”) and “cyborg sciences” (including cybernetics and operations research).⁷ Paul Edwards has argued that American cybernetics bore the distinctive cultural imprint of the Cold War and that the language of cybernetics became a vehicle of Cold War discourse. The cultural imagery of computers, or the “cyborg discourse,” embodied in integrated human-machine systems and artificial intelligence devices, was closely linked to the “closed-world discourse,” which reflected the ideological stereotypes of the Cold War. Reified in military command-and-control systems, the two discourses intertwined to form a vision of the political and social world as a closed, computable system subject to manipulation and control.⁸

David Mindell, in contrast, emphasizes that Wiener began working on the general principles of cybernetics only after his anti-aircraft predictor project had been canceled, and that cybernetics became for Wiener a civilian enterprise in the midst of World War II. Mindell argues that “cybernetics recast military control in a civilian mold”⁹ and sees the roots of cybernetics in the prewar traditions of control engineering and communication engineering, which widely employed the metaphor of intelligent behavior for servomechanisms and the analogy between the nervous system and a telephone network.

Different faces of cybernetics are reflected in the personalities of two major contributors to cybernetics: Wiener and von Neumann. Steve Heims has sharply contrasted Wiener, a “New Englander with a social conscience,” with von Neumann, a “hard-boiled, weapons-minded ‘realist.’”¹⁰ David Noble has similarly emphasized that the two men had very different views of the social implications of cybernetic ideas:

Unlike John von Neumann, whose mathematical axiomatic approach reflected his affinity for military authority and control, Wiener insisted upon the indeterminacy of systems and a statistical, probabilistic understanding of their functioning. Any technical part of social systems, he stressed, should be designed to complement, to be compatible with, and therefore to sustain and enhance human life. Overly determined systems, in delimiting the full range of human thought and action, would tend toward instability and breakdown because they narrowed down the range of negative feedback: self-adjusting, self-correcting action. . . . Wiener wanted to ensure that advances in technology would benefit labor, not industrial corporations.¹¹

Wiener’s liberal social outlook and pacifist ideas paradoxically combined in the cybernetic discourse with von Neumann’s cold rationalism and militaristic attitude. This discourse thus became permeated with tension and ambiguity and opened up to various interpretations.

Cybernetics seems difficult to pin down to a single political or social trend. While some view it as an extension of military patterns of thinking and behavior into the civilian realm, others see it as a liberating movement that destroys traditional cultural divisions and stereotypes. For example, Donna Haraway has argued that humans are in fact becoming “cyborgs”—cybernetic organisms, or hybrids of machine and organism—and that cybernetics effectively undermines the rigid “dichotomies between mind and body, animal and human, organism and machine, public and private, nature and culture, men and women, [and] primitive and civilized.”¹²

In this chapter, instead of looking for a single interpretation of cybernetic ideas, I focus on the emergence and cultural ramifications of the language of cybernetics. In my view, the diversity of meanings ascribed to cybernetic ideas owes much to the flexibility of this “universal” language. Cybernetics is viewed here as a product of juxtaposition of diverse disciplinary and cultural discourses characteristic of postwar American science. To place American cybernetics in a larger international context, I start by comparing the research trajectories of two great mathematicians, Norbert Wiener and Andrei Kolmogorov—trajectories that initially closely paralleled each other but suddenly diverged precisely at the birth of cybernetics.

Norbert Wiener and Andrei Kolmogorov: Two Mathematicians Tackle Biology

Norbert Wiener was a prodigy. He read at age 3, enrolled in high school at 9, entered Tufts College at 11, and graduated at 14.¹³ Andrei Kolmogorov also showed his talents very early. At age 5 he edited the mathematical section of a homemade magazine, inventing mathematical problems and curiosities for each issue. At 14 he taught himself higher mathematics by reading an encyclopedia and, upon entering Moscow University, quickly passed exams for the freshman year and moved up to sophomore status, thus obtaining a food ration privilege—vitally important in the post-revolution years.¹⁴

Kolmogorov started as a mathematics student but soon developed a strong interest in the humanities. At Moscow University he studied with the prominent historian Sergei Bakhrushin and, at age 17, wrote a meticulously researched paper based on a statistical analysis of tax records in medieval Russia. Bakhrushin praised the paper but did not recommend it for publication. “You have found only one proof,” he explained to young Kolmogorov. “That is very little for a historian. You need at least five proofs.”¹⁵ At that moment, as Kolmogorov would later tell his students, he decided to concentrate in mathematics, where one proof would suffice.

Wiener also majored in mathematics, but he too showed diverse interests in other fields, trying first biology and then philosophy. He enrolled in the doctoral program in philosophy at Harvard University and, at age 18, completed a dissertation in mathematical logic. He then went to Cambridge University to study symbolic logic with Bertrand Russell. Kolmogorov too became interested in mathematical logic—particularly the contemporary controversy over intuitionism, a logical theory that challenged the transfinite applications of the *tertium non datur* principle (the law of the excluded middle) in classical logic. In 1925 he published a seminal article in which he “embedded” classical logic in intuitionist logic, thus showing that all finite conclusions of classical logic could be proved without transfinite application of the *tertium non datur* principle.

Kolmogorov quickly attained international fame among mathematicians. In 1923, at age 19, while still an undergraduate at Moscow University, he made a fundamental contribution to the theory of functions. He constructed an odd, counterintuitive mathematical object—a summable function whose



Figure 2.1

Norbert Wiener in Moscow, 1960. Courtesy of Natal'ia Liapunova.

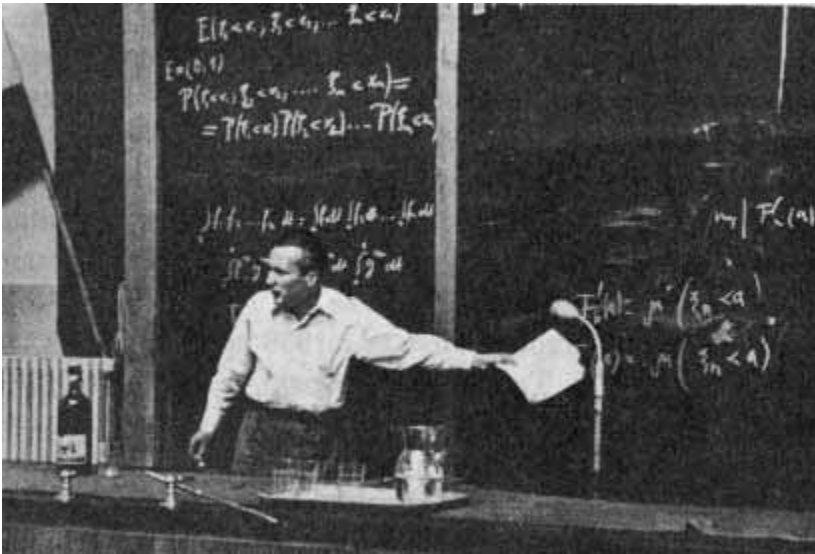


Figure 2.2

Andrei Kolmogorov at a Mathematics Congress in Hungary, 1950. From Shiriaev, ed., *Kolmogorov v vospominaniakh*.

Fourier series diverged almost everywhere. In 1931 Kolmogorov was appointed professor of mathematics at Moscow University, where he taught until his death in 1987. Wiener, who taught mathematics at the Massachusetts Institute of Technology from 1919 until his retirement in 1960, first became famous for his 1920–1923 studies of Brownian motion. Each Brownian particle moves in an unpredictable zigzagging trajectory and the space of such trajectories has an infinite number of dimensions; Wiener provided a statistical description of this process by assigning probabilities to subsets of such trajectories. This measure of probabilities for infinite-dimensional spaces became known as *Wiener measure*. The theory of stochastic processes emerged from Wiener’s work, and Kolmogorov became one of the leading researchers in that field. In 1933 he co-authored an important article on Brownian motion. Soon after that, Kolmogorov reformulated the foundations of the theory of probability in terms of measure theory, producing his famous system of axioms for probability theory (a system now considered standard).

Kolmogorov and Wiener were intensely aware of each other’s work. Wiener later recalled: “Khintchine and Kolmogoroff, the two chief Russian exponents of the theory of probability, have long been involved in the same field in which I was working. For more than twenty years, we have been on one another’s heels; either they had proved a theorem which I was about to prove, or I had been ahead of them by the narrowest of margins.”¹⁶ While visiting Moscow in the summer of 1960, Wiener told an interviewer: “When I read works of Academician Kolmogorov, I feel that these are my thoughts as well, this is what I wanted to say. And I know that Academician Kolmogorov has the same feeling when reading my works.”¹⁷

The race between Kolmogorov and Wiener took a particularly dramatic turn during their parallel work on prediction theory. In a series of articles published in 1939, 1940, and 1941, Kolmogorov developed a theory of interpolation and extrapolation of stationary sequences.¹⁸ Wiener reportedly learned of Kolmogorov’s work in late 1941, when he was working on similar problems.¹⁹ In February of 1942 Wiener submitted a classified report for the military on extrapolation, interpolation, and smoothing of stationary time series.²⁰ Kolmogorov derived his theory from purely theoretical studies of the geometry of Hilbert spaces, whereas Wiener arrived at this topic while working on the problem of predicting the trajectories of enemy aircraft, but their results, by Wiener’s admission, “turned out to be

equivalent.”²¹ While Kolmogorov was more rigorous in his theoretical formulations, Wiener developed an application-oriented theory of filtration, which provided a convenient basis for solving engineering problems.²²

In the 1930s, both Wiener and Kolmogorov developed an acute interest in the applications of mathematics to biology. Wiener took an active part in the Philosophy of Science Club, an interdisciplinary seminar on scientific method led by the prominent physiologist Walter Cannon at the Harvard Medical School. This seminar, attended largely by medical scientists, focused on the attempts to solve the mind-brain relationship with newest neurological methods, including brain wave encephalography. Kolmogorov applied various mathematical methods to the problems of evolutionary biology and genetics. In particular, he studied traveling waves associated with the spread of an advantageous gene in a linear habitat and concluded, almost simultaneously with and independently from R. A. Fischer, that the range inhabited by individuals with favorable characteristics would expand with an asymptotically constant velocity. This model laid the foundation of a general mathematical theory of reaction-diffusion that would later serve to describe the spread of epidemics, rumors, advertising, and cultural innovations. Kolmogorov’s involvement with biological problems, however, proved short-lived.

In 1940 Kolmogorov entered a scholarly dispute over the use of statistical methods in genetics, which quickly turned into an ideological controversy. In 1939 the Soviet journal *Vernalization*, a mouthpiece of the supporters of the notorious hack scientist Trofim Lysenko, published a paper by a graduate student named Ermolaeva, who claimed to have empirically refuted the famous Law of Segregation of Gregor Mendel’s classical genetics. She had replicated the basic monohybrid cross experiment, which, theoretically, was to result in the 3:1 split of traits in the second generation, with one-fourth of the offspring displaying the recessive trait and three-fourths displaying the dominant trait. Ermolaeva found certain deviations from the 3:1 ratio and concluded that Mendelian genetics did not withstand the test of statistical experiment. Soviet geneticists approached Kolmogorov, and he wrote a brief rebuttal, explaining that the small size of Ermolaeva’s statistical sample accounted for this discrepancy and that her data in fact amounted to “a new brilliant confirmation of Mendel’s laws.”²³

The Lysenkoites immediately redirected their ideological attacks at Kolmogorov. If statistics did not become their ally, it had to be treated as

an enemy. In his published reply to Kolmogorov, Lysenko equated the use of probability theory and statistics in biology with the submission to “blind chance”: “We biologists . . . do not want to submit to blind chance, even though this chance is mathematically admissible,” he wrote. “As a biologist, I am not interested in the question whether or not Mendel was a good mathematician. As to my opinion of Mendel’s statistics,” he continued, “I claim that they have no bearing whatever on biology. . . . [Kolmogorov’s] article, too, has no bearing whatever on biological science.”²⁴ In short, if statistics did not confirm the dogmas of Lysenkoism, it was not applicable to biology. The philosopher Ernest Kolman labeled Kolmogorov’s mathematical treatment of genetics a philosophical heresy. Kolman argued that, by deriving biological conclusions from his mathematical analysis, Kolmogorov had committed a serious methodological error:

Unfortunately, . . . instead of confining himself to the purely mathematical aspect of the matter, [Kolmogorov] draws conclusions which go beyond the limits of mathematics, and declares himself in favor of Mendel and Morgan’s genetics. . . . If Mendel’s law is a biological law, it cannot be proved or disproved by mathematical or statistical means. Only on the basis of biology . . . can Mendel’s law be proved or disproved as a universal biological law. . . . Mendel’s law of segregating factors deduced from a definite group of cases of inheritance is nothing but a statistical rule, and not a universal biological law. . . . Statistics as applied to biology must occupy a subordinate place.²⁵

Threatened with the prospect of involvement in a dangerous philosophical and ideological debate, Kolmogorov was forced into a defensive position and had to reassure the critics that he had spoken only of the statistical “confirmation” of Mendel’s law, not of its “proof.”²⁶

This incident put an abrupt end to Kolmogorov’s inroads into biology. To avoid ideological complications, for many years he limited the applications of his mathematical techniques to physical problems and stayed away from the life sciences and the social sciences, urging his students to do the same.²⁷

During World War II, both Wiener and Kolmogorov joined the war effort. Under a contract with the NDRC, Wiener worked on the problems of anti-aircraft gun control. From the results of his prediction theory, Wiener elaborated a statistical model that maximized the probability of a successful hit. With the engineer Julian Bigelow, he designed and built an “anti-aircraft predictor,” a device for predicting the trajectory of an enemy plane. Wiener suspected that Kolmogorov was following a similar path; he even asked Vannevar Bush, director of the US Office of Scientific Research

and Development, for permission to collaborate with Kolmogorov. Even though the Soviet Union was then viewed as an ally of the United States, this permission evidently never came.²⁸

Kolmogorov was indeed following a parallel path, working on problems of ballistics and also using a statistical approach, although he focused on ground artillery rather than anti-aircraft fire and applied a different mathematical apparatus. He elaborated the general principles of evaluating the effectiveness of artillery fire on the basis of target-destruction probability, and he estimated the conditions under which artificially induced dispersion of combat shells would be efficient.²⁹ Years later these principles would be employed by the designers of Soviet anti-aircraft and anti-missile defense systems.³⁰ For his wartime work, Kolmogorov was twice awarded the Order of Lenin, the highest mark of honor in the Soviet Union. Wiener's results in prediction theory would later also prove very important for weapons design; however, his anti-aircraft predictor did not work very well, and in January of 1943 his wartime project was terminated.³¹ At this point, the paths of Wiener and Kolmogorov began to diverge.

Control via Feedback: The Body as a Servomechanism

While Kolmogorov made mathematical applications in the life sciences a taboo, Wiener chose precisely this subject for his new research project. Upon returning to civilian research, Wiener switched his interests to neurophysiology. He decided to use some of the insights gained during his wartime work to explain neurophysiological mechanisms of human behavior.

Wiener's work on the anti-aircraft predictor led him to a far-reaching analogy between the operation of servomechanisms (feedback-based control devices) and human purposeful behavior. This predictor was designed to forecast the future trajectory of an enemy airplane and to help point a gun at the position where the airplane and a shell would meet. This function was usually performed by human gun pointers and gun trainers, and Wiener's device would therefore "usurp a specifically human function." Wiener described the actions of human gunners in mathematical terms "in order to incorporate them mathematically into the machines they control," and he came to view his anti-aircraft predictor as an engineering model of human behavior in this situation. In order to test the predictor, Wiener and Bigelow had to generate a set of trajectories of the enemy aircraft, and they

used servomechanisms to simulate these trajectories. They assumed that “an aviator under the strain of combat conditions is scarcely in a mood to engage in any very complicated and untrammelled voluntary behavior.”³² The enemy pilot’s evasion technique, they concluded, would follow the same feedback principle that was implemented in servomechanisms. Wiener and Bigelow then extended this analogy to the anti-aircraft gunner and began thinking of the role of feedback in a broader range of human activities.³³

After the termination of the anti-aircraft predictor project, Wiener and Bigelow teamed up with the Mexican physiologist Arturo Rosenblueth, whom Wiener had met at a session of the Philosophy of Science Club, to explore the implications of the servomechanism analogy. They concluded that human voluntary activity was based on “circular processes, emerging from the nervous system into the muscles, and re-entering the nervous system through the sense organs.”³⁴ They attributed various pathological conditions in which a simple voluntary act such as picking up a pencil would lead to uncontrollable oscillations to breaking of this circular mechanism. Wiener, Rosenblueth, and Bigelow came to view this circularity as a universal principle of purposeful behavior. They called it *feedback*, borrowing the term from control engineering. They interpreted any goal-directed action—from pointing a gun to picking up a pencil—as a negative feedback process in which the attempts to approach the goal were constantly corrected by return signals from the goal that indicated the current distance from its (anticipated) position, or “the amount by which we have failed to pick up the pencil at each instant.”³⁵

In 1943 Wiener, Rosenblueth, and Bigelow published a joint article, in which they suggested that purposeful human behavior was governed by the same feedback mechanism that was employed in servomechanisms.³⁶ Combining terms from control engineering (*feedback*), psychology (*purpose*), philosophy (*teleology*), and mathematics (*extrapolation*), they constructed a classificatory scheme of behavior that was equally applicable to human action and machine operation (figure 2.3).

Both human activity and machine operation could be purposeful, if the feedback principle was involved. By showing how organisms and machines could achieve their goals by means of the deterministic feedback mechanism, Wiener, Rosenblueth, and Bigelow undermined the philosophical oppositions between teleology and determinism, between voluntary acts and mechanical actions, and ultimately between men and machines. They

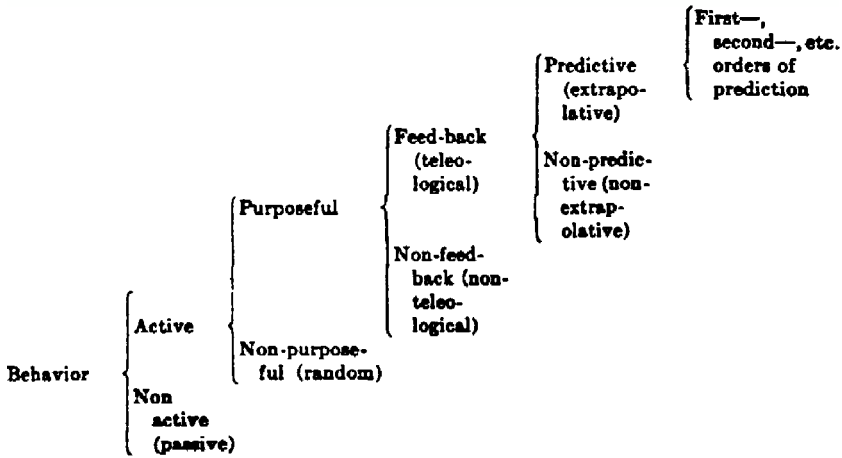


Figure 2.3

The classification of behavior. From Rosenblueth, Wiener, and Bigelow, "Behavior, Purpose and Teleology."

argued that "the broad classes of behavior are the same in machines and in living organisms."³⁷ Therefore, they argued, organisms and machines could be described in similar terms and studied with the same methods.

Several years later, Wiener attempted to show that the feedback mechanism would explain such widely recognized neurophysiological phenomena as Pavlovian conditional reflexes. He introduced the notion of *affective tone*, which, "arranged on some sort of scale from negative 'pain' to positive 'pleasure,'" would make it possible to reinforce favorable reactions:

An increase in affective tone favors all processes in a nervous system that are under way at the time and gives them a secondary power to increase affective tone; and . . . a decrease in affective tone tends to inhibit all processes under way at the time and gives them a secondary ability to decrease affective tone. . . . Note that the mechanism of affective tone is itself a feedback mechanism.³⁸

Wiener then drew a direct analogy between reflexes and anti-aircraft fire control. In both cases, he argued, "anticipatory feedback" was involved, which included an effector with a lagging characteristic and a compensator that acted as a predictor. "Feedbacks of this general type are certainly found in human and animal reflexes," he wrote. "Any system of anti-aircraft fire control must meet the same problem."³⁹

Wiener's formulation of the problem of prediction and control—in living organisms as well as in machines—in the mathematical terms of statistical

mechanics raised some fundamental physical and philosophical questions. One of these questions concerned the paradoxical ability of living organisms to develop and sustain a high degree of organization, the ability that stood in apparent contradiction to some basic physical laws.

The Order of Life: The Organism as an Entropy-Reducing Machine

The second law of thermodynamics—“nature’s tendency to degrade the organized and to destroy the meaningful,” as Wiener put it⁴⁰—posed a serious obstacle for any attempt to provide a statistical description of living processes. The Austrian theoretical physicist Erwin Schrödinger directly confronted the problem of paradoxical stability and even increasing complexity of living organisms, which apparently disobeyed the “natural tendency of things to approach the chaotic state”—“the same tendency,” he remarked, “that the books of a library or the piles of papers and manuscripts on a writing desk display.”⁴¹ Schrödinger’s 1944 book *What Is Life?* served as an important source for Wiener’s attempts to find a universal basis for order and control in living and non-living nature.

The topic of Schrödinger’s book was somewhat unusual for a physicist.⁴² In the introduction, Schrödinger referred to some “difficulties of language” he had experienced. Although he may have meant that his English needed polishing, he also faced linguistic difficulties of a different kind. Schrödinger aspired to give a physical account of biological phenomena—that is, to explain them by physical and chemical laws. He decided to abandon the language of mathematics, the “physicist’s most dreaded weapon,” because his subject seemed “much too involved to be fully accessible to mathematics.”⁴³ Instead, Schrödinger offered a systematic reinterpretation of biological phenomena in physical terms. In other words, he attempted to translate life into the language of physics.

Schrödinger defined life as “orderly and lawful behavior of matter”; however, as he quickly discovered, living matter seemed to violate the accepted physical order and behaved rather unlawfully. He suggested that the thermodynamic concept of entropy be used as a quantitative measure of organization and order in living organisms, and he immediately faced a physical paradox. The living organism—a complex, highly organized system—seemed to disobey the second law of thermodynamics, which prohibited entropy decrease in closed systems. Instead of plunging into entropy/chaos, living

organisms demonstrated “admirable regularity and orderliness” and were able to keep up the existing order. Schrödinger formulated the chief problem of life as a strictly physical problem: How were living organisms able to obviate this inescapable tendency toward a “thermal death”? To solve this paradox, he set out to find a physical basis of order in the living cell.

Schrödinger suggested that the stability of life was maintained by a specific mechanism of “order from order.” Citing the contemporary chromosome theory of heredity, he found the internal source of order in the organism in extremely stable genetic structures—the chromosome fibers—which he translated into the physical language as “aperiodic crystals.” Schrödinger argued that these “incredibly small groups of atoms, much too small to display exact statistical laws, [played] a dominating role in the very orderly and lawful events within a living organism”⁴⁴; they controlled the development of the organism’s features and provided for the transmission of traits from generation to generation. The stability of chromosomes, in turn, was supposedly maintained by the constant influx of order in the form of highly ordered organic molecules contained in food. Schrödinger interpreted the organism as a thermodynamic system that counteracted the inevitable increase of entropy by absorbing “negative entropy,” or “drinking orderliness,” from the environment.

Paradoxically, the root of life was found in the clockwork-type, mechanical principle of “order from order.” “We seem to arrive at the ridiculous conclusion that the clue to the understanding of life is that it is based on a pure mechanism,”⁴⁵ Schrödinger admitted. The fundamental divide thus lay not between living organisms and machines but between order and chaos. Both organisms and machines fought chaos with order, and by similar means: “order from order.”

ORGANISMS AND MACHINES	vs.	NATURE
order	vs.	chaos
control	vs.	statistical laws
influx of negative entropy	vs.	increasing entropy
organization	vs.	disorganization

Figure 2.4

Adapted from Schrödinger, *What Is Life?*

The emerging parallels between genetics and quantum physics fascinated Schrödinger, one of the founders of quantum mechanics. “The significant fact [of genetic mutations] is the discontinuity,” he wrote. “It reminds a physicist of quantum theory—no intermediate energies occurring between two neighboring energy levels. He would be inclined to call [Hugo] de Vries’s mutation theory, figuratively, the quantum theory of biology. We shall see later that this is much more than figurative. The mutations are actually due to quantum jumps in the gene molecule.”⁴⁶ Schrödinger borrowed this interpretation of genetic mutations as physical events from the 1935 joint article of the Russian geneticist Nikolai Timoféeff-Ressovsky, the German biologist Karl Zimmer, and the German (later American) physicist Max Delbrück. They calculated the rate of x-ray-induced genetic mutations as a function of the dose, and they estimated the target area for mutation-inducing ionization. Schrödinger, unaware at the time that many of the claims made in this article had later been revised, assumed that this estimate indicated “the size of the gene.” The biochemist Max Perutz later soberly concluded that “what was true in [Schrödinger’s] book was not original, and most of what was original was known not to be true even when the book was written.”⁴⁷

Despite its technical blunders, Schrödinger’s book proved extremely significant in its conceptual impact: it rendered the biological problem of life as a physical problem of molecular order that could be addressed with the help of physical methods, theories, and instruments. This book played a major role in what Evelyn Fox Keller has called a “political and cognitive synergy” of physics and biology—a synergy that resulted in the emergence of molecular biology.⁴⁸ Molecular biology became an attractive alternative for those physicists who were dissatisfied with the military engagement of nuclear physics. Under the influence of Schrödinger’s book, a number of young scientists—including James Watson, Francis Crick, Maurice Wilkins, and Seymour Belzer—began applying physical methods to biology. Schrödinger’s translation of biological concepts into the language of physics allowed molecular biologists to borrow their research agenda, their attitudes, their techniques, and eventually their social authority from physics. At the same time, the physical-biological language Schrödinger forged made it possible to channel biological analogies into the realm of machines and vice versa. His terminology became one of the main components of the language of cybernetics.

Schrödinger introduced into scientific discourse the central metaphor of molecular biology: the chromosome as a message written in code.⁴⁹ He called the chromosome fiber a “hereditary code-script,” an “architect’s plan” for a future organism. Using a social metaphor, he also compared chromosomes to “stations of local government dispersed through the body, communicating with each other with great ease, thanks to the code that is common to all of them.”⁵⁰ The code metaphor seemed particularly appropriate in a wartime cultural context filled with allusions to code making and code breaking. At that very moment, across the Atlantic, the communication engineer Claude Shannon was working on cryptological problems and applying the same code metaphor to human communication. Like Schrödinger, Shannon borrowed the notion of entropy from thermodynamics and used it in a broader sense. Schrödinger’s conceptual framework differed from Shannon’s in two important respects: Schrödinger interpreted codes as instruments of control, while Shannon viewed them as tools of communication; and Schrödinger interpreted entropy as a measure of disorder, while Shannon used it as a measure of uncertainty. On the basis of shared metaphors, however, biophysicists and communication engineers began to find a common language: cyberspeak.

Human Communication as an Engineering Problem: Man as an “Information Source”

One of the crucial steps in the formation of cybernetics was made when major problems in control engineering and in communication engineering were reformulated in the mathematical terms of statistical mechanics and reinterpreted as problems of filtering and prediction. During World War II, Wiener worked simultaneously on two projects: the construction of an anti-aircraft predictor and the design of wave filters to eliminate background noise in electrical networks. He quickly realized that smoothing a rough curve to predict the future aircraft trajectory was akin to filtering out high-frequency oscillations from electrical signals. It proved impossible, however, to construct an ideal prediction mechanism. An apparatus designed to achieve high precision on very smooth curves turned out to be very sensitive and was sent into oscillation by the smallest departure from smoothness. To balance the contradictory demands of smoothness and stability, Wiener elaborated statistical means of analysis of the signal and the noise, thus laying a

mathematical foundation for the design of both the anti-aircraft predictor and the wave filters. Wiener and his collaborators became convinced that “the problems of control engineering and of communication engineering were inseparable.”⁵¹ The same conclusion was reached at about the same time by a group of control and communication engineers at the Bell Laboratories. Working under defense contracts from the NDRC, they compared the central problem of anti-aircraft fire control—tracking the trajectory of an enemy airplane—with the standard communication engineering problem of filtering out noise. They found “an obvious analogy between the problem of smoothing the data to eliminate or reduce the effect of tracking errors and the problem of separating a signal from interfering noise in communications systems.”⁵² Out of this juncture of control engineering and communication engineering emerged the idea of unity of control and communication—one of the theoretical underpinnings of cybernetics.

Bell Labs engineers formulated the mathematical problems of anti-aircraft fire control—data smoothing and prediction—in terms of communication engineering as “a special case of the transmission, manipulation, and utilization of intelligence.”⁵³ Engineers used such terms as *intelligence*, *information*, and *communication* in narrow technical senses. Ironically, these strictly technical terms would later be used widely to describe in “precise terms” the cultural phenomena to which they initially referred only in a metaphorical sense.

In 1924 Harry Nyquist, an engineer working for the American Telephone and Telegraph Corporation, referred to the rate of sending of a signal as “the speed of transmission of intelligence.”⁵⁴ He proposed to measure this speed as the logarithm of the number of possible “values” to be transmitted (the number of different characters, in case of telegraphy). In 1928 the Bell Labs engineer Ralph Hartley replaced the anthropomorphic term *intelligence* with the more neutral *information*, which he proposed to measure as the logarithm of the number of possible messages, or symbol sequences of a specified length.⁵⁵ In his article “Transmission of Information,” Hartley suggested this logarithmic function as a common “quantitative measure whereby the capacities of various systems to transmit information may be compared.”⁵⁶ By introducing bandwidth as a common measure of information capacity, Nyquist and Hartley together provided a unified framework for analyzing transmission of various types of signals: radio waves, telegraph messages, telephone voice signals, television images. This uni-

versal mathematical description of diverse communication technologies laid the foundation for a unified theory of communication. As David Mindell has argued, this theoretical synthesis reflected corporate goals: AT&T capitalized on communication theory to advance its claim of a natural monopoly over all media. AT&T portrayed its efforts to become the unified communications company controlling telephony, telegraphy, radio broadcasting, and television as a necessary means for providing “substantially instantaneous transfer of intelligence” throughout the United States.⁵⁷

Man-machine metaphors are double-edged: while describing machines in anthropomorphic terms, one often conceptualizes humans in machine-like terms. In 1935 the Bell Labs engineer Homer Dudley patented the Vocoder, a device that extracted the amplitudes of a speech signal at ten different frequencies, transmitted them separately, and then reconstructed the speech signal at the receiving end. Dudley viewed his device as a model of human communication. “Communication by speech consists in sending by one mind and the receiving by another of a succession of phonetic symbols with some emotional content added,” he wrote. Dudley compared human speech to “a radio wave in that information is transmitted,” and effectively blurred the distinction between human communication and transmission of radio signals—both were merely regarded as transfer of information.⁵⁸ During World War II, on the basis of the Vocoder, Bell Labs engineers designed the “X-system,” which added a digital “key” to each speech component and served to provide secure top-level communication between Washington and London.⁵⁹ This project, known as “Project X,” helped establish a crucial link between communication engineering and cryptology.

Claude Shannon, a communication engineer at Bell Labs, participated both in the fire-control project and in “Project X,” and his work tied the problems of communication engineering, control engineering, and cryptology together. Later he recalled: “The work on both the mathematical theory of communications and the cryptology went forward concurrently from about 1941. I worked on both of them together and I had some of the ideas while working on the other. I wouldn’t say one came before the other—they were so close together you couldn’t separate them.”⁶⁰ Shannon’s conceptualization of the central problems in communication engineering (the accurate and efficient transmission of signals) and in cryptology (the accurate and efficient coding and decoding of messages) began to converge and resulted in the emergence of information theory.⁶¹

Like Wiener, who faced the contradictory demands of smoothness and stability for his anti-aircraft predictor, Shannon tried to balance two contradictory requirements of accuracy and efficiency of communication. For greater efficiency, signals had to be shorter; for greater accuracy and elimination of possible errors resulting from noise, signals had to be “redundant” (that is, longer). To balance the two criteria, Shannon introduced a quantitative measure that allowed him to evaluate both the degree of distortion by noise and the efficiency of encoding. Following Hartley, Shannon called this measure *information*. Shannon proposed to measure information in *bits* (binary digits), a term first suggested by John W. Tukey, a Princeton mathematician who also worked on fire-control problems during the war and who joined Bell Labs in 1945. One bit stood for an elementary choice between two equally probable alternatives; the choice from among four possibilities could be split into two elementary binary choices and thus required two bits of information; and so on. As Hartley had shown, the information required to choose one message from a set of equally probable messages amounted to the logarithm of the number of messages. Shannon considered a more general case in which different messages could occur with different probabilities. Shannon’s formula for information in this case proved identical to that of entropy in statistical mechanics, and he suggested calling it the “entropy of the information source,” measured in bits per symbol. Shannon proved a “fundamental theorem” which asserted that, by efficient encoding, one could transmit information at a rate arbitrarily close to the channel capacity when measured in bits per second. In his view, the notions of information, entropy, and certainty were closely intertwined: information measured the final certainty resulting from the decoding of a message, while entropy measured (the equal amount of) initial uncertainty before decoding the message. Shannon used the terms *entropy* and *uncertainty* interchangeably.⁶²

After the war, Shannon published his results on communication engineering and on cryptology in two separate articles, using essentially the same conceptual framework.⁶³ In the first article, he subsumed various forms of communication under the universal scheme of a “general communication system,” which used the same generic names for both human and machine components:

- (1) the *information source*, which produces a message or sequence or messages to be communicated to the receiving terminal;
- (2) the *transmitter*, which operates on the message in some way to produce a signal suitable for transmission over the channel;

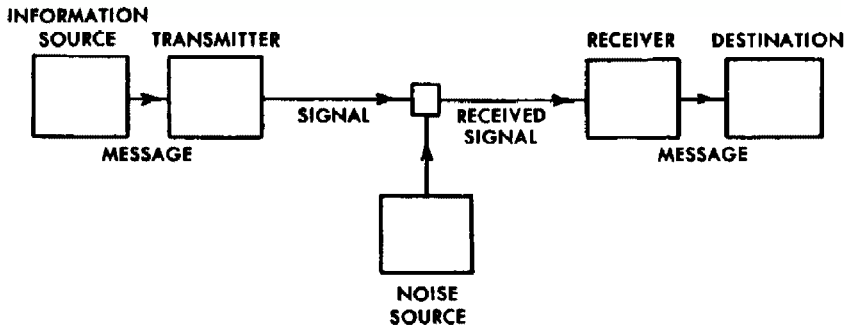


Figure 2.5

Schematic diagram of a general communication system. From Shannon and Weaver, *The Mathematical Theory of Communication*.

- (3) the *channel* is merely the medium used to transmit the signal from the transmitter to receiver;
- (4) the *receiver* ordinarily performs the inverse operation of that done by the transmitter, reconstructing the message from the signal;
- (5) the *destination* is the person (or thing) for whom the message is intended.⁶⁴

Other essential elements of the scheme were (6) the *message* itself (written or spoken words, pictures, etc.) selected out of a set of possible messages; (7) the *information*, “a measure of one’s freedom of choice when one selects a message”; (8) the *signal*, a sequence of discrete symbols or a continuous function sent over the communication channel; and (9) the *coding process*, which changed the message into the signal.

In “Communication Theory of Secrecy Systems,” Shannon translated this scheme into the cryptological language, calling the transmitter the *encoder* and the receiver the *decoder*. In both cases, he interpreted human communication as an exchange of encoded messages. (See figure 2.6.)

Viewing man as an “information source,” Shannon not only interpreted human communication as transmission of messages generated with certain probabilities; he also viewed human language itself as a stochastic process. In other words, he interpreted texts in natural language as sequences of individual letters generated with certain probabilities. This was how cryptanalysts had traditionally viewed natural language. One well-known cryptanalytic technique was based on the comparison between the average frequencies of letters in natural-language texts and the observed frequencies of symbols in a given encoded message; in the simplest code, the most frequent symbols would likely correspond to the most frequent letters.

HUMAN COMMUNICATION	COMMUNICATION SYSTEM / SECRECY SYSTEM
addresser	information source
sender	transmitter/encoder
medium	channel
uncertainty	entropy/noise
meaning	information
recipient	receiver/decoder
addressee	destination

Figure 2.6

Adapted from Shannon and Weaver, *The Mathematical Theory of Communication*, and Shannon, "Communication Theory of Secrecy Systems."

Shannon proposed a series of simple artificial languages as approximations to English and argued that "a sufficiently complex stochastic process [would] give a satisfactory representation" of natural language as a discrete source of information.⁶⁵

Shannon's interest in man-machine analogies dated back to his work at MIT in the second half of the 1930s. In his master's thesis, Shannon employed Boolean algebra (a symbolic description of the "laws of thought") to express the functioning of relay switching circuits, thus laying the foundation for the seminal analogy between human thinking and digital computing.

The Computer and the Mind as Universal Logical Machines

The transition from analog to digital computers in the 1940s involved changing not only the technical basis of computing but also its disciplinary affiliation and its cultural connotations.⁶⁶ Mathematicians and logicians began to play an important role in the design and programming of computers. Computing could now be described in purely logical terms, independent of its physical implementation. Various mathematical, engineering, and psychological concepts were gradually translated into logical terms,

and a new general language of computation emerged. Both the digital computer and the human mind were conceptualized as logical machines. In the next step, the mind came to be seen as a computer.

One of the most powerful analog computers of the day, the Differential Analyzer, was developed at MIT under the direction of Vannevar Bush, and both Norbert Wiener and Claude Shannon had close contacts with Bush's group. While a graduate student, Shannon worked on improving the controls of the Differential Analyzer, which involved a complex digital circuit built of electromechanical relays. In his 1937 master's thesis, Shannon represented on-off relay switches with true-or-false logical propositions of Boolean algebra, thus turning relay circuit design into a logical problem that could be solved by formal means. As the official history of Bell Labs proclaimed, Shannon effectively transformed relay circuit design from "a somewhat esoteric art to . . . a science."⁶⁷ From Shannon's work, it became clear that one could build a digital calculating machine using only interconnected relay switches. An important advantage of this approach was that electromechanical relays operated much faster than the cogged wheels traditionally used in mechanical calculating machines, such as Bush's Differential Analyzer. In 1937–38 Bush circulated a proposal to build a "rapid arithmetical machine" based on electronic switching, which eventually reached Howard Aiken, the designer of the first electromechanical digital computer, Mark I.⁶⁸

In the summer of 1940 Wiener came up with his own proposal for building a digital computing machine for the solution of partial differential equations. In a memorandum submitted to Bush, he formulated several main requirements for such a machine: the construction of a numerical central apparatus for addition and multiplication, the use of electronic tubes instead of gears or mechanical relays as switching devices, the adoption of the binary numerical system, the laying out of the entire sequence of logical and mathematical operations on the machine itself, and the construction of an erasable and reusable storage of data. Although Wiener himself did not pursue this computer project, he later realized that these ideas were "of interest in connection with the study of the nervous system."⁶⁹

The concept of a logical machine played a crucial role not only in the first proposals for digital computers but also in the theoretical conceptualization of computation by the British mathematician Alan Turing. He

attempted to solve the problem whether mathematics was decidable, that is, whether there existed a definite method to determine if any given mathematical assertion was true. This problem, called the Entscheidungsproblem, was proposed by the German mathematician David Hilbert at the Mathematical Congress in 1928. In his 1936 article “On Computable Numbers,” Turing suggested a machine-like logical procedure for imitating any computation performed by a human mathematician. He postulated that “the behavior of the [human] computer at any moment is determined by the symbols which he is observing, and his ‘state of mind’ at that moment.”⁷⁰ He compared “a man in the process of computing a real number to a machine which is only capable of a finite number of conditions.”⁷¹ Instead of reading off and writing on two-dimensional paper, this abstract logical machine would “scan” a symbol from a one-dimensional tape and then make a “move” along that tape, perhaps erasing this symbol and writing a different one. Turing represented each “state of mind” of the human computer with a particular “configuration” of the machine, which determined which “move” would be made and how the configuration changes if a particular symbol is scanned. Together, the operations performed by this machine and the changes in its configuration constituted its “behavior.” Turing further generalized his model by introducing “the universal computing machine,” which could read off the tape the description of any other computing machine of this type and then carry out the computations of that machine. Any possible human computation, Turing concluded, could be performed by this “universal computing machine,” and therefore the limitations of this machine would indicate the limitations of mathematics in general. In that article, Turing showed that some definable numbers could not be computed by the “universal machine” and thus proved that the Entscheidungsproblem had no solution. One of the steps leading to this negative conclusion eventually proved more culturally significant than the end result: the concept of the “universal computing machine” and the proof of its existence became a powerful positive result of Turing’s work.

Shannon’s description of the digital computer as a logical machine and Turing’s conceptualization of human computation as a logical procedure converged in the unified vision of the computer and the mind as logical machines. In 1943, while visiting Bell Labs, Turing met with Shannon, and they had long conversations. According to Turing’s biographer Andrew

Hodges: “Shannon had always been fascinated with the idea that a machine should be able to imitate the brain; he had studied neurology as well as mathematics and logic, and had seen his work on the differential analyzer as a first step towards a thinking machine. [Turing and Shannon] found their outlook to be the same: there was nothing sacred about the brain, and that if a machine could do as well as a brain, then it *would* be thinking.”⁷² This outlook was soon shared by Wiener, who began speaking of “the computing machine, and consequently the brain, as a logical machine” and calling for logic to be reduced to “the study of the logical machine, whether nervous or mechanical, with all its non-removable limitations and imperfections.”⁷³

The Logic of the Brain: The Nervous System as a Turing Machine

While Shannon and Turing were making functional comparisons between human thinking and logical computing, the neurophysiologist Warren McCulloch and the logician Walter Pitts were drawing structural analogies between the nervous system and a network of logical elements.

In May of 1942 the Josiah Macy Jr. Foundation sponsored a conference (held in New York) on cerebral inhibition. At this meeting, which focused on hypnosis and on the physiology of the conditional reflex, Arturo Rosenblueth presented the results of his joint studies with Norbert Wiener and Julian Bigelow. Conceptualizing purposeful behavior in engineering terms, they suggested that the human body be viewed as a “black box,” and they compared it to a feedback-controlled servomechanism. Among those present was Warren McCulloch (of the University of Illinois Medical School), who was interested precisely in the content of this “black box”—that is, in the neurophysiological mechanisms of human behavior. In collaboration with Walter Pitts (of the University of Chicago), he elaborated a logical model of the brain functioning, which turned out to be remarkably similar to Turing’s logical machine.

Trained initially as a philosopher, McCulloch liked big questions.⁷⁴ In particular, he aspired to answer the most fundamental epistemological question: “How do we know anything about the world?”⁷⁵ In search of the answer, he had turned from philosophy to psychology and then to neurophysiology. His approach, which he called *experimental epistemology*, was to find “a satisfactory explanation of *how we know what we know*, stated

in terms of the physics and chemistry, the anatomy and physiology, of the biological system.”⁷⁶ McCulloch started with the search for “what an atom was to chemistry, or a gene to genetics”—the “simplest psychic act,” the most fundamental element, the ultimate building block of mental activity:

My object, as a psychologist, was to invent a kind of least psychic event, or “psychon,” that would have the following properties: First, it was to be so simple an event that is either happened or else it did not happen. Second, it was to happen only if its bound case had happened . . . that is, it was to imply its temporal antecedent. Third, it was to propose this to subsequent psychons. Fourth, these were to be compounded to produce the equivalents of more complicated propositions concerning their antecedents.

In 1929 it dawned on me that these events might be regarded as the all-or-none impulses of neurons, combined by convergence upon the next neuron to yield complexes of propositional events.⁷⁷

Taking the activity of a single neuron as the sought-after “psychon,” McCulloch attempted to build a model of mental activity from the bottom up. In 1941, having taken a job in the Department of Psychiatry at the University of Illinois, he came into contact with a small group of researchers in “mathematical biophysics” at the University of Chicago, led by the Russian-born physicist Nicolas Rashevsky. Rashevsky’s group aspired to create a systematic mathematical biology modeled after mathematical physics: it would relate to experimental biology as mathematical physics related to experimental physics. Rashevsky focused his research on mathematical modeling of nerve excitation and conduction, but he viewed those merely as electrical phenomena rather than as processes of communication. As Wiener later put it, for Rashevsky’s group “the engineering of the body [was] a branch of power engineering.”⁷⁸ In this group, McCulloch found a kindred soul in Pitts, a young, very talented, self-educated logician. Together they elaborated an entirely different vision of neurophysiological processes. Instead of viewing the brain as an electrical circuit, they conceptualized it as a logical machine.

Inspired by Turing’s 1936 paper “On Computable Numbers,” McCulloch and Pitts took a purely logical approach to the description of mental activity. In McCulloch’s words, they “abstracted from real neurons everything irrelevant”⁷⁹ and represented all-or-none neuron impulses as true or false logical propositions. Then they linked these “formal neurons” into networks, which could be represented as functions of propositional logic. “What we thought we were doing (and I think we succeeded fairly

well),” McCulloch later recalled, “was treating the brain as a Turing machine.”⁸⁰

In 1943 McCulloch and Pitts published a highly influential paper, “A Logical Calculus of the Ideas Immanent in Nervous Activity,” in the *Bulletin of Mathematical Biophysics*, which was edited by Rashevsky. Postulating “all-or-none” as a “law of nervous activity,” they explored the possibilities it offered. McCulloch and Pitts showed that, under certain assumptions, any network of formal neurons could be represented by a logical function, and that, conversely, for any logical function one could construct a neural net that would produce that function. “The behavior of every net can be described in these [logical] terms, with the addition of more complicated logical means for nets containing circles,” they wrote, “[and] for any logical expression satisfying certain conditions, one can find a net behaving in the fashion it describes.”⁸¹

For McCulloch and Pitts, the construction of formal neural networks served only as an auxiliary device in their ultimate quest for “the nervous theory of knowledge.”⁸² They were concerned with the epistemological implications of their model, rather than its neurophysiological validity. From their point of view, the most significant result of their paper was proving that any particular logical function could be implemented in formal neural networks in many different ways. From that it followed that one could not determine the structure of the neural network unambiguously by observing its output, since “inference from any sample of overt behavior to nervous nets is not unique.”⁸³ Solely by observing the outputs of the “black box,” one could not deduce what mental activity was going on inside it.

McCulloch and Pitts interpreted the limitations of their formal model of the brain in philosophical terms as the fundamental limitations on our knowledge of the world. Without knowledge of the structure of the neural network, they argued, it would not be possible to deduce the input to the network (the “facts” about the “external world”) from its output (our “perception”), insofar as “if our nets are undefined, our facts are undefined.”⁸⁴ Furthermore, even if one knew the structure of the network and the current states of all the neurons, one could compute only the succeeding states. The presence of disjunctive relations and/or circular paths in the net would make it impossible to determine all the preceding states. In other words, from the perceptions retrieved from one’s memory, it was not possible to

deduce the “facts” that caused those perceptions. “Thus our knowledge of the world, including ourselves, is incomplete as to space and indefinite as to time,” McCulloch and Pitts concluded.⁸⁵ They suggested that our knowledge of the world was fundamentally limited in two ways: (1) From our present activities, we could not gain knowledge of ourselves (i.e., determine our neural net structure). (2) Although we could sometimes predict the future, we could not determine the past.

As a by-product of their analysis of formal neural networks, McCulloch and Pitts proved that such networks could compute any number computable by the Turing machine. This result had no significance for “experimental epistemology”; it could be useful, they thought, only to psychologists. McCulloch and Pitts argued that their analysis of formal networks provided “a psychological justification” for Turing’s definition of computability: “If any number can be computed by an organism, it is computable by these definitions, and conversely.”⁸⁶ Blurring the distinction between the formal neural network and the actual mental activity, McCulloch later argued that “a nervous system can compute any computable number.”⁸⁷ Even if we cannot know the world, we can at least compute as many numbers as a universal logical machine.

While an undergraduate at Haverford College, McCulloch had been asked about his plans in life. “I have no idea,” he replied, “but there is one question I would like to answer: What is a number, that a man may know it, and a man, that he may know a number?”⁸⁸ It seems that for McCulloch the question “How do we know anything about the world?” eventually turned into “How do we know a number?” Summing up his 1947 article with Pitts, “How Do We Know Universals,” McCulloch wrote that this article “cannot be wrong in its all-important proof that for a man to know such universals as shape regardless of size or chord regardless of key it would be sufficient for his brain to compute enough averages.”⁸⁹ Having reduced the knowledge of the world to the computability of numbers, McCulloch and Pitts successfully found a common denominator for the epistemology of humans and that of machines. “We had proved, in substance,” McCulloch wrote later with evident pride, “the equivalence of all general Turing machines—man-made or begotten.”⁹⁰

In 1943, after publishing his article with McCulloch, Pitts went to MIT to study with Wiener, who introduced him to Shannon’s work on the logical analysis of switching circuits.⁹¹ At the intersection of McCulloch and

Pitts's logical model of the brain, Turing's logical model of the mind, and Shannon's logical model of digital computing, an integrated vision of the mind and the brain as logical computing machines began to emerge. This vision was most forcefully expressed by the prominent mathematician John von Neumann.

The Computer as a Brain and the Brain as a Computer

It is hard to imagine two people as different as Norbert Wiener and John von Neumann.⁹² Wiener was a myopic, absent-minded professor, an arm-chair scientist, a liberal intellectual with strong ethical convictions, and a vehement pacifist who refused to participate in any US government projects after the bombing of Hiroshima. Von Neumann was an energetic and practical man, a cold-blooded rationalist, an indefatigable traveling consultant, an active contributor to the Manhattan Project, and a member of the Atomic Energy Commission; he knew his way around the Pentagon and the corridors of power. Yet they were very much interested in each other's work, they maintained a "professional friendship,"⁹³ and they had a productive working relationship that resulted, in particular, in the crystallization, formulation, and mathematical analysis of diverse cybernetic ideas. Although the conceptual trajectories of their thought went in opposite directions (Wiener was "humanizing" machines whereas von Neumann was "rationalizing" humans), their destinations, ironically, were the same. Their views eventually converged on a common set of man-machine metaphors that formed the basis of cyberspeak.

Various uses of formal logical language—by Shannon to describe relay circuits, by Turing to define a universal logical machine for computation, and by McCulloch and Pitts to build a model of the nervous system—linked electrical engineering, computing, and neurophysiology together and provided an opportunity for sharing models, tools, and metaphors among these fields. By describing the operations of the first electronic digital computers in similar logical terms, and comparing them to the human brain, von Neumann added a crucial link to this nexus.

Von Neumann, who had major responsibility for the complex mathematical calculations needed for the Manhattan Project, collaborated closely with the designers of the first electronic digital computer (ENIAC) and its stored-program successor (EDVAC). In the spring of 1945 Warren Weaver,

the head of the NDRC Applied Mathematics Panel, asked him to write a report on the current state and the future prospects of computing machines.⁹⁴ This report clearly outlined the concept of the stored program and explicated what became known as the “von Neumann architecture” for digital computers. Von Neumann did not describe the stored-program computer in terms of the electromechanical relays or the vacuum tubes used in the computers of his day, however; he employed McCulloch and Pitts’s language of idealized neurons. Observing that both the formal neurons and the switching and relay components of computing machines worked on the all-or-none principle (being either on or off, either excited or quiescent), he argued that any digital computing device, natural or artificial, would consist of such logical “elements.”

Von Neumann conceptualized the formal neural networks of McCulloch and Pitts as a general model of computation, applicable both to the man-made computers and to the human brain. While McCulloch and Pitts employed a logical formalism to illuminate neurophysiological questions, von Neumann took their neurophysiology to illuminate the logical design of computers. The logical language served to describe both the functioning of vacuum tubes and the human neural activity, and through logic the computer and the brain became inextricably connected. The “neuron analogy,” which for von Neumann initially served to separate the principal logical design of computers from its implementation in specific engineering components, would provide a powerful metaphorical framework for computing for years to come.

Von Neumann radically reinterpreted McCulloch and Pitts’s 1943 article. He decided that their logical description of “formal neurons” captured the “essential traits” of actual nervous cells. McCulloch and Pitts had merely argued that a formal network of simple all-or-none elements could produce some complex patterns of behavior. Von Neumann concluded that all complex behavior (“anything that you can describe in words”) could be ultimately reduced to computation in a formal neuron network:

It certainly follows that anything that you can describe in words can also be done with the neuron method. And it follows that the nerves need not be supernaturally clever or complicated. In fact, they needn’t be quite as clever and complicated as they are in reality, because an object which is a considerably amputated and emasculated neuron, which has many fewer attributes and responds in a much more schematic manner than a neuron, already can do everything you can think up.⁹⁵

Von Neumann argued that “the neurons of the higher animals are definitely elements in the above sense”⁹⁶ and proceeded to consider living organisms “as if they were purely digital automata.”⁹⁷ In other words, von Neumann reinterpreted the logical model of McCulloch and Pitts as an essential description of neural activity and then projected this description onto computers, suggesting a “neural model” for computing.

In von Neumann’s interpretation, the correspondence between the formal neural network and the abstract logical computer (the Turing machine), which McCulloch and Pitts established as a “psychological” by-product of their epistemological study, became the crux of the whole matter. Von Neumann argued that the main result of McCulloch and Pitts’s article was that “the generality of neural systems is exactly the same as the generality of logics.”⁹⁸ Linking their work with a similar result obtained by Turing for abstract computing machines, von Neumann built a chain of direct analogies from the actual brain to the formal neuron network to logical functions to the Turing machine to the computer, and aspired to create a general theory encompassing this whole range of artificial and living computers.

In the late 1940s, von Neumann began working on a general theory of “self-reproducing automata,” the term he applied both to the living organisms and to complex logical-mathematical objects. Whether the brain was a computer was longer a question for him; the question was what type of computer—analogue or digital—it was. Von Neumann admitted that a purely digital model for neurophysiology was too simplistic, but suggested that the answer had to combine elements of analogue and digital computing: “In the lingo of computing machinery one would say that [the nerve cell] is an analogue device that can do vastly more than transmit or not transmit a pulse.” There was nothing unique in the non-digital aspects of the nervous system, he argued, for “vacuum tubes, electromechanical relays, etc. are not switching devices either, since they have continuous properties.” If the operations of the brain and the digital computer were not entirely logical, that only brought them closer, not apart. In the end, von Neumann concluded that some “trans-continuous alternation between digital and analogue mechanisms” took place both in the computers and in the nervous system, and that this was “probably characteristic of every field.”⁹⁹

While Schrödinger looked at living organisms as a physicist and identified the “physical aspect” of life with the thermodynamic order, von

Neumann looked at the nervous system “from the mathematician’s point of view”¹⁰⁰ and identified the nervous system with a logical automaton. “The nervous system, when viewed as an automaton, must definitely have an arithmetical as well as a logical part,” he argued. “This means that we are again dealing with a computing machine in the proper sense and that a discussion in terms of the concepts familiar in computing machine theory is in order.”¹⁰¹ Using these concepts, von Neumann compared the computer and the brain in terms of size, speed, energy dissipation, efficiency, precision, the relations between parts, and the methods of dealing with errors. Viewed as logical-mathematical automata, the brain and the computer could be described in the same terms and quantitatively compared.

Von Neumann suggested the language of self-reproducing automata as a common language for describing both the computer and the human brain. This language, composed of terms borrowed from diverse scientific and engineering disciplines, effectively blurred the boundary between

BRAIN	AUTOMATON	COMPUTER
neuron	switching/relay organ	electromechanical relay/vacuum tube
impulse	stimulus	electric pulse
excitation	on	1
quiescence	off	0
all-or-none	on/off	binary system
nervous system	formal neural network	trigger circuit
synaptic delay	reaction time	delay time
afferent neurons	input organ	input device
motor/efferent neurons	output organ	output device
associative neurons	arithmetical, control, and memory organs	arithmetical, control, and memory units

Figure 2.7
Adapted from von Neumann, “The General and Logical Theory of Automata” and “Theory and Organization of Complicated Automata.”

organisms and machines. Taken out of their narrow technical context, these terms now circulated freely between the domains of neurophysiology and computing:

The terminology used in the following is taken from several fields of science; neurology, electrical engineering, and mathematics furnish most of the words. No attempt is made to be systematic in the application of terms, but it is hoped that the meaning will be clear in every case. It must be kept in mind that few of the terms are being used in the technical sense which is given to them in their own scientific field. Thus, in speaking of a neuron, we do not mean the animal organ, but rather one of the basic components of our network which resembles an animal neuron only superficially, and which might equally well have been called an electrical relay.¹⁰²

When von Neumann wanted to distinguish between living and non-living automata, he now had to refer specifically to “(artificial) computing machines” and “(natural) organisms.”¹⁰³ One could just as easily speak about “natural automata” or “artificial organisms,” since they were described in the same logical language. The powerful dual metaphor—the brain as a computer and the computer as a brain—became an essential component of the emerging universal language for men and machines.

The Making of Cyberspeak and the Emergence of Cybernetics

Wartime projects often forced specialists from diverse scientific and engineering fields to cross the conventional boundaries of their disciplines and to work together. Extensive networks of collaboration and information exchange emerged, and one of them formed around Norbert Wiener, whose assorted interests in mathematical logic, philosophy, neurophysiology, control and communication engineering, and computing brought him into close contact with specialists from all these fields. Starting in 1946, the ideas circulating in this network were discussed at a series of conferences on Circular Causal and Feedback Mechanisms in Biological and Social Systems, sponsored by the Josiah Macy Jr. Foundation. At Wiener’s suggestion, conference participants decided to call “the entire field of control and communication theory, whether in the machine or in the animal, by the name *Cybernetics*,”¹⁰⁴ derived from a Greek word for *steersman*, and changed the series title to Conference on Cybernetics. In his recollections, Wiener described the emergence of cybernetics as a purely intellectual process based on “the essential unity of the set of problems centering about communication, control, and statistical mechanics, whether in the machine

or in living tissue.”¹⁰⁵ Viewed as a social process, however, the formation of cybernetics hardly seemed natural or smooth; it involved some determined and difficult efforts to bring together a very diverse group of researchers and to create a language that all group members could share. The intellectual construction of cybernetics as an interdisciplinary field did not happen by itself; it was closely tied to the formation of a new social network, the “cybernetics group.”¹⁰⁶

In the summer of 1943 Wiener, who with Rosenblueth and Bigelow had just published “Behavior, Purpose, and Teleology,” received a copy of McCulloch and Pitts’s article “A Logical Calculus of the Ideas Immanent in Nervous Activity.” Wiener quickly discovered significant parallels between the work the Cambridge group and that of the Chicago group, and he immediately invited Pitts to MIT and introduced him to electronic computing. “He was very much interested when I showed him examples of modern vacuum tubes and explained to him that these were ideal means for realizing in the metal the equivalents of his neuron circuits and systems,” Wiener later recalled. As a result of these discussions, Wiener and his colleagues concluded that “the problem of interpreting the nature and varieties of memory in the animal has its parallel in the problem of constructing artificial memories for the machine.”¹⁰⁷ They quickly shared these ideas with von Neumann and with the computer designers Howard Aiken and Herman Goldstine: “Everywhere,” he recalled, “we met with sympathetic hearing, and the vocabulary of engineers soon became contaminated with the terms of the neurophysiologist and the psychologist.”¹⁰⁸

In early 1944 von Neumann and Wiener held an interdisciplinary meeting at Princeton at which several computer engineers, physiologists, and mathematicians discussed the analogy between the computer and the nervous system. Wiener evaluated the results of the meeting with great enthusiasm: “At the end of the meeting, it had become clear to all that there was a substantial common basis of ideas between the workers in the different fields, that people in each group could already use notions which had been better developed by the others, and that some attempt should be made to achieve a common vocabulary.”¹⁰⁹

The pronounced need for a common vocabulary highlighted a major problem of communication facing the emerging “cybernetics group.” After the proliferation of wartime projects, often isolated from one another by secrecy, the terminology in each of the constituent fields of cybernetics was

in flux. In control engineering, for example, standardization of terminology became a major concern: "Almost every early postwar paper made some reference to a 'new language,' to 'problems with terminology,' to the need to 'translate' the jargon of one or another group."¹¹⁰

The 1943 Rosenblueth-Wiener-Bigelow article drew heavy criticism for mixing philosophical, biological, and engineering terms.¹¹¹ Responding to critics, Wiener and Rosenblueth explained that the proposed assorted vocabulary was just a research tool that allowed them to study humans and machines with the same methods:

We also wish to explain why we use the humanistic terms purpose and teleology in the description of the behavior of some machines. . . . We believe that men and other animals are like machines from the scientific standpoint because we believe that the only fruitful methods for the study of human and animal behavior are the methods applicable to the behavior of mechanical objects as well. Thus, our main reason for selecting the terms in question was to emphasize that, as objects of scientific inquiry, humans do not differ from machines.¹¹²

Wiener and Rosenblueth made it clear that, rather than offer a ready-made overarching theory for humans and machines, they were suggesting some new tools for research in this direction. Cybernetics did not emerge as the finished product of a "natural" synthesis of diverse scientific and engineering fields; on the contrary, cybernetics was proposed as an ambitious research program aimed at forging such a synthesis. "It is the purpose of Cybernetics," Wiener proclaimed, "to develop a *language* and techniques that will enable us indeed to attack the problem of control and communication in general."¹¹³ Since some new techniques had already been developed in control and communication engineering and in neurophysiology, the creation of a common language was a crucial step that would allow a transfer of these techniques from one field to another. This common language emerged gradually over the course of ten meetings sponsored by the Macy Foundation in 1946–1953. Among the active participants of these meetings were mathematicians (Wiener, von Neumann, Pitts), engineers (Bigelow, Shannon, Heinz von Förster), philosophers (Filmer Northrop), neurophysiologists (Rosenblueth, Ralph Gerard, Rafael Lorente de Nó), psychiatrists (McCulloch, Lawrence Kubie, Henry Brosnin), psychologists (Heinrich Klüver, Kurt Lewin, Alex Bavelas, Joseph Licklider), biologists (W. Ross Ashby, Henry Quastler), linguists (Roman Jakobson, Charles Morris, Dorothy Lee), and social scientists (Gregory Bateson, Lawrence Frank, Paul Lazarsfeld, Margaret Mead).¹¹⁴

The participants discussed a whole parade of man-machine analogies: the body as a feedback-operated servomechanism, life as an entropy-reducing device, man as an “information source,” human communication as transmission of encoded messages, the human brain as a network of logical elements, and the human mind as a digital computer. This assortment of mathematical models, explanatory frameworks, and appealing metaphors presented a rather chaotic and eclectic picture. What held it together was a set of interdisciplinary connections: control was viewed as an aspect of communication and vice versa; information was defined as “negative entropy” and interpreted as a measure of order, organization, and certainty, while entropy was associated with chaos, noise, and uncertainty; physiological homeostasis was identified with physical equilibrium; neurons were treated as logical elements; and thinking was made synonymous with computation.

Wiener played a central role in the attempts to bring together the diverse elements of the cybernetic medley. In particular, he managed to connect Shannon’s concept of entropy as a measure of uncertainty in communication with Schrödinger’s concept of negative entropy as a source of order in living organisms. Shannon’s theory dealt with the precise transmission of messages. In this case, the information transmitted equaled the entropy of the information source: the greater the initial uncertainty (entropy), the greater the amount of information received in the end. For Shannon, greater entropy meant more information; for Schrödinger, however, greater entropy meant less order. Wiener reconciled these approaches by what Shannon called “a mathematical pun”: Wiener considered the case of imprecise transmission, and he ingeniously redefined the notion of entropy: it now referred not to the initial uncertainty (as in Shannon’s definition) but to the degree of uncertainty remaining after the message was received. Greater entropy (noise) now meant less information. Hence, while Shannon’s information equaled regular entropy, Wiener’s information equaled negative entropy.¹¹⁵ Thanks to this simple change of sign, information (order) now opposed entropy both in communication engineering and in biophysics. Thus Wiener was able to generalize the notion of information (negative entropy) as a universal measure of organization, certainty, and order in any “system,” whether technical or living: “Just as the amount of information in a system is a measure of its degree of organization, so the entropy of a system is a measure of its degree of disorganization; and the one is simply the negative of the other.”¹¹⁶ Introducing his seminal

metaphor, "The organism is a message," Wiener wrote: "Organism is opposed to chaos, to disintegration, to death, as message is to noise."¹¹⁷ Through this metaphor, the study of organisms could be translated into communication engineering, and vice versa.

Wiener further linked information with control, describing the mechanism of negative feedback in informational terms: "The information fed back to the control center tends to oppose the departure of the controlled from the controlling quantity."¹¹⁸ He closely tied the notions of control and communication together and generalized them from engineering to human behavior. Wiener interpreted control as communication with feedback:

When I control the actions of another person, I communicate a message to him, and although this message is in the imperative mood, the technique of communication does not differ from that of a message of fact. Furthermore, if my control is to be effective I must take cognizance of any messages from him which may indicate that the order is understood and has been obeyed.¹¹⁹

With equal fluency, Wiener applied the terms *message*, *memory*, and *conditioned reflex* both to the digital computer and to the nervous system. Impulses sent from one neuron to another are called *messages*: "Each neuron has its message fed into it by other neurons."¹²⁰ Memory, of the machine or of the brain, can "hold a message" and is intended for "storing information."¹²¹ Furthermore, Wiener argued, "in the nervous computing machine it is highly probable that information is stored largely as changes in the permeability of the synapses, and it is perfectly possible to construct artificial machines where information is stored in that way," and therefore "there is nothing in the nature of the computing machine which forbids it to show conditioned reflexes."¹²²

Cyberneticians combined concepts from physiology (*homeostasis* and *reflex*), psychology (*behavior* and *goal*), control engineering (*control* and *feedback*), thermodynamics (*entropy* and *order*), and communication engineering (*information*, *signal*, and *noise*) and generalized each of them to be equally applicable to living organisms, self-regulating machines (such as servomechanisms and computers), and human society. In their view, humans and machines were two kinds of *control systems*, which, *operating* in certain *environment*, pursued their *goals* (hitting a target, increasing *order*, achieving better *organization*, or reaching the state of *equilibrium*) by *communicating* with this environment, that is, sending and receiving *information* about the results of their actions through *feedback*.

Cyberspeak provided a set of concepts, which mediated between the worlds of living organisms and man-made artifacts and thus inextricably tied humans and machines together. Through cyberspeak, concepts could travel freely between mathematics, engineering, and biology. (See figure 2.8.)

Cyberspeak became a common language for neurophysiologists, control engineers, and sociologists. Feedback-controlled servomechanisms served as a universal model for physiological homeostasis, locomotion, purposeful behavior, and social equilibrium. Living organisms were viewed as thermodynamic macrosystems; the struggle between life and death was reinterpreted as a battle between negative entropy (the source of stability, organization, and order) and entropy (the force of chaos and disorganization). In cyberspeak, human communication was regarded as a problem of communication engineering (i.e., how to transmit signals accurately in the presence of noise); the intuitive human notions of certainty and uncertainty were translated into the formulas for information and entropy. Both neural cells in the human brain and vacuum tubes in electronic computers were now seen as logical elements of a cybernetic system; thus a structural anal-

MAN	CYBERNETIC SYSTEM	MACHINE
purposeful behavior	control via feedback	target seeking
human communication	information transfer	signal transmission
human brain	self-regulating device	stored-program computer
thinking	information processing	computation
brain cells	logical elements	vacuum tubes/relays
human body	feedback mechanism	servomechanism
homeostasis	equilibrium	homeostat
certainty	information	signal
uncertainty	entropy	noise
life	order	proper functioning
death	disorder	malfunctioning

Figure 2.8

Adapted from Wiener, *Cybernetics*.

ogy between the brain and the computer emerged. Both human memory and the computer storage for programs and data were construed as means for control and self-regulation and thus formed a functional analogy between the computer and the brain. Speaking of humans as control devices and describing computers in anthropomorphic terms became two sides of one coin, brought into wide circulation by cybernetics.

“The essential unity” of control and communication problems in various fields, which Wiener portrayed as a driving force behind cybernetics, may be viewed as the result, rather than a pre-existing condition, of cybernetic studies. This “unity” was deliberately constructed by the “cybernetics group,” a social network of scientists and engineers aspiring to cross their disciplinary boundaries. To borrow Peter Galison’s term, one might describe cybernetics as a *trading zone*, in which specialists from various fields could exchange ideas, experimental results, theories, and hypotheses.¹²³ Cyberspeak emerged as a common language, a pidgin of sorts, for this trading zone—a means to translate the technical terms of one field into the vocabulary of another.¹²⁴ The language of cybernetics quickly overgrew this purely technical function, however, and acquired the aura of a universal language of men, machines, and society. Cyberspeak laid a discursive basis for cyberneticians’ aspirations to create a new, revolutionary interdisciplinary field.

Cyberspeak Becomes Universal

The mood of the Macy meetings was overwhelmingly enthusiastic. As Yehoshua Bar-Hillel, one of the guest participants, recalled, cybernetics and information theory “created among many of us the feeling that the new synthesis heralded in them was destined to open new vistas on everything human and to help solve many of the disturbing open problems concerning man and humanity.”¹²⁵ The scope of metaphors and analogies offered by cybernetics was breathtaking. At the same meeting, the mathematician von Neumann compared digital computers to the brain composed of the McCulloch-Pitts formal neurons, while the neurophysiologist Lorente de Nó spoke of the “computing machine of the nervous system.” The psychologist Klüver offered an explanation of human perception in terms of feedback; the anthropologist Bateson interpreted a transvestite ceremony as a homeostatic mechanism; the psychologist Bavelas described a game of

cards as information transfer, measurable in bits. By adopting cyberspeak, researchers from various disciplines were able to describe highly specific individual phenomena in common terms understood by all participants. From this similarity of descriptions, cybernetics' enthusiasts concluded that the natural, psychological, and social processes underlying these diverse phenomena also had a common basis. For the "cybernetics group," this new description of the world in the language of cybernetics heralded a new and radically different world view.

Cyberspeak undermined some fundamental dichotomies of Western science and philosophy: mind and body, spirit and matter, the organism and the machine, the animate and the inanimate, the natural and the artificial. Classical philosophical questions were proclaimed obsolete and irrelevant. Since cybernetics replaced the old opposition of causality and teleology with the notion of feedback and explained purposeful behavior by circular causality, "the whole mechanist-vitalist controversy," in Wiener's opinion, "has been relegated to the limbo of badly posed questions."¹²⁶ The question whether machines could be considered alive was, in Wiener's view, a "semantic problem." "We are at liberty," he boasted, "to answer it one way or the other as best suits our convenience."¹²⁷ He further suggested that it was "best to avoid all question-begging epithets such as 'life,' 'soul,' 'vitalism,' and the like" and speak merely of the decrease of entropy in both humans and machines.¹²⁸

The new cybernetic terms—*control, communication, information, feedback*—acquired universal meanings that could not be reduced to any particular concepts from the established scientific disciplines. "Information is information, not matter or energy," proclaimed Wiener.¹²⁹ Cyberspeak redefined all concepts it utilized: mathematical and engineering terms were extended to the human sciences, while anthropomorphic notions were generalized to cover mathematical objects and machines. This expansion of meaning was initially metaphorical, but the metaphors gradually stabilized and began to be taken literally. The neurophysiologist Ralph Gerard was among the first to notice this discursive turn; at the seventh Macy meeting, he warned other participants:

We started our discussion in the "as if" spirit. Everyone was delighted to express any idea that came to his mind, whether it seemed silly or certain or merely a stimulating guess that would affect someone else. We explored possibilities for all sorts of "ifs." Then, rather sharply it seemed to me, we began to talk in an "is" idiom. We were saying much the same things, but now saying them as if they were so.¹³⁰

The gradual transformation of Claude Shannon's mathematical analysis of technical communication into a general theory of human communication was a typical example of the universalization of cybernetics' claims. As early as 1928 Ralph Hartley had complained that "as commonly used, information is a very elastic term."¹³¹ He hardly anticipated the degree of elasticity this term would gain in public discourse in the late 1940s. In 1949 Warren Weaver, the former chief of D-2 fire control section of the NDRC, who at the outset of World War II had let Shannon a contract on the study of fire control, wrote a popular account of Shannon's communication theory for *Scientific American*. The same year, Shannon's 1948 paper and an extended version of Weaver's article were published together in a book titled *The Mathematical Theory of Communication*. Weaver helped refashion Shannon's wartime work as a fundamental "information theory" with potentially unlimited applications.

Weaver proposed to interpret any form of communication as transmission of quantifiable information, including not only cases where "one mechanism . . . affects another mechanism" but also "all of the procedures by which one mind may effect another." This would involve, in Weaver's view, "not only written and oral speech, but also music, pictorial arts, the theatre, the ballet, and in fact all human behavior."¹³² Weaver argued that the technical problem of accurate signal transmission, which Shannon's theory had addressed, overlapped in large part with problems emerging at the more general levels of semantics and effectiveness of communication. With a few "minor additions, and no real revision," Weaver proclaimed, Shannon's theory could be extended into semantics. Any distortion of meaning in communication, Weaver suggested, could be viewed as "semantic noise" to be eliminated by proper "adjustment."¹³³ For Weaver, the notion of entropy, which Shannon had associated with uncertainty, became a measure of choice, randomness, and organization, with all the rich cultural connotations of these concepts, including beauty and melody.¹³⁴ Thus, the model Shannon had suggested for technical communication in secrecy systems was extended to include all forms of interaction: between machines, between people, and between people and machines.

Shannon's mathematical model of a "general communication system" served as the conceptual framework for an original linguistic theory elaborated by the Russian-born Harvard linguist Roman Jakobson. In the active intellectual milieu of Cambridge, Massachusetts, Jakobson quickly

established contacts with both Wiener and Shannon.¹³⁵ In a 1949 letter to Wiener, Jakobson called *Cybernetics* an “epoch-making” book and insisted on “the extreme parallelism between the problems of modern linguistic analysis and the fascinating problems you discuss.”¹³⁶ Soon he began speaking of the “feedback process between speaking and hearing.”¹³⁷

Jakobson saw in the language of cybernetics and information theory a “modern, less ambiguous terminology,”¹³⁸ and by blending cyberspeak with traditional linguistic terminology he created his own distinct vocabulary. Instead of Ferdinand de Saussure’s opposition *langue* and *parole*, Jakobson began using *code* and *message*; he substituted *subcodes* for *styles*, and replaced *contextual variations* with *redundant features*; *encoding* and *decoding* supplanted *production* and *comprehension*; and *speaker* and *listener* became *encoder* and *decoder*.

Jakobson eliminated all traces of human/machine distinction from Shannon’s scheme. He combined the *information source* with the *transmitter* in one notion of *addresser*; similarly, the notion of *addressee* embraced both the *receiver* and the *destination*. Jakobson added a new concept of *contact* to denote a “psychological” dimension of communication; the *message* and the *signal* were not distinguished and were subsumed under the rubric of *message*. Instead of the *coding process*, Jakobson preferred to speak of *code*; this code, however, established the correspondence not between the message and the signal, as in Shannon’s scheme, but between the message and its meaning. And finally, most important, Jakobson replaced the notion of *information* with the referential meaning of a message, or *context*, in his terms. Jakobson’s overall scheme of linguistic communication, modeled after Shannon’s, involved the total of six

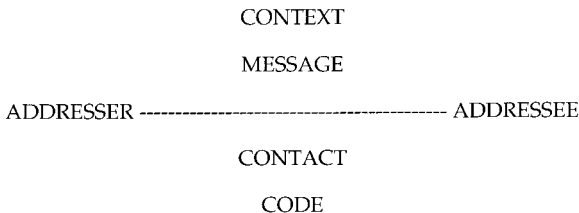


Figure 2.9

Six constituent factors of verbal communication. Adapted from Jakobson, “Linguistics and Poetics.”

“constituent factors”:

- the *addresser* (a speaker and an encoder)
- the *addressee* (a listener and a decoder)
- the *context* (a “referent” in “another, somewhat ambiguous, nomenclature”)
- the *message* sent by the addresser to the addressee
- the *code* that is “fully, or at least partially, common” to them
- the *contact*, a “physical channel and psychological connection” between them.¹³⁹

On the basis of this scheme, Jakobson proposed a new vision of human language. He distinguished six functions of language, each related to the corresponding factor in the scheme above:

- The emotive (expressive) function aims at a “direct expression of the speaker’s attitude toward what he is speaking about.”
- The conative (imperative) function brings focus on the addressee and finds its “purest grammatical expression in the vocative and imperative.”
- The referential (cognitive, denotative) function has an “orientation toward the context.”
- The poetic (aesthetic) function is the dominant function of verbal art, for it sets “focus on the message for its own sake.”
- The metalingual (metalinguistic) function is used “whenever the addresser or the addressee needs to check up whether they use the same code”—for example, when the meaning of words is discussed.
- The phatic function serves “to establish, to prolong, or to discontinue communication, to check whether the channel works,” and so on.¹⁴⁰

Jakobson argued that this new conceptual scheme explicated some aspects of language that were not recognized by traditional linguistics. In his view,

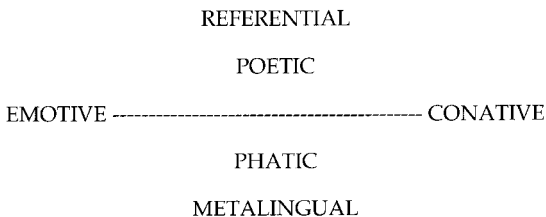


Figure 2.10

Six basic functions of verbal communication. Adapted from Jakobson, “Linguistics and Poetics.”

unlike “the less ‘biased’ communication engineers,” many linguists overlooked the phenomenon of “code variability,” which was pertinent to the use of metaphors: “Metaphoric creations are not deviations but regular processes of certain stylistic varieties, which are subcodes of an overall code.”¹⁴¹ Jakobson did not simply translate an existing linguistic theory into cyberspeak; he used cybernetic concepts to build a new theoretical framework for linguistics.

Wiener, in turn, referred to Jakobson’s work to argue that cybernetics fitted well with modern linguistics. He cited Jakobson and the mathematician Benoit Mandelbrot in describing human communication as a “game played in partnership by the speaker and the listener against the forces of confusion.”¹⁴² In other words, while Jakobson borrowed cybernetic concepts to build a linguistic theory, cyberneticians were “discovering” cybernetic phenomena in his linguistics. Making a formidable example of a closed loop, cybernetic concepts traveled in a circle, extending from cybernetics into linguistics and then coming back as if they were original linguistic notions.

The historian Geoffrey Bowker has described this circular process as a chief feature of the language of cybernetics. It served an important social function by supporting “legitimacy exchange” among scientists: “An isolated scientific worker making an outlandish claim could gain rhetorical legitimacy by pointing to support from another field—which in turn referenced the first worker’s field to support its claims. The language of cybernetics provided a site where this exchange could occur.” In Bowker’s words, the author of the “conditional probability machine,” A. M. Uttley, “used mathematics to support his physiology and physiology to support his mathematics, using cybernetic terminology to spiral between the formal properties of classification machines and the nature of the brain.”¹⁴³

Cybernetics did not merely describe computers metaphorically as brains; the brain itself was conceptualized in logical and engineering terms, and these concepts then returned to computing, serving as a basis for the impressive “discoveries” of man-machine analogies. On the first pages of his *Cybernetics*, Wiener suggested the computer as a model for the nervous system: “It became clear to us [Wiener and Pitts] that the ultra-rapid computing machine, depending as it does on consecutive switching devices, must represent almost an ideal model of the problems arising in the nervous system.”¹⁴⁴ A few pages down, he turned this analogy around and

described the computer itself in neurophysiological terms: "The modern ultra-rapid computing machine was in principle an ideal central nervous system to an apparatus for automatic control."¹⁴⁵ In another example, physiological homeostasis was conceptualized as a feedback-controlled servomechanism, while servomechanisms themselves were described in anthropomorphic terms. The historian Lily Kay argued that "signifying homeostasis as negative feedback and then resignifying such servomechanisms as organismic homeostasis amounted to a circularity."¹⁴⁶ The dialogue between molecular biologists and control engineers in the language of cybernetics, in the words of the historian Evelyn Fox Keller, resulted in a "bootstrap process of modeling organisms and machines, each upon the other."¹⁴⁷ If we allow ourselves to resort to a cybernetic metaphor, we might describe this process as a feedback loop of man-machine metaphors: the more anthropomorphic the machines looked, the more machine-like the human beings appeared.

Whether individual cyberneticians intended this or not, the logic of cybernetic discourse propelled the universalization of their claims. McCulloch and Pitts argued only that even simple formal neural networks could fulfill complex logical functions. In von Neumann's interpretation, however, McCulloch and Pitts had in effect found a means to imitate "any conceivable form of behavior," since such networks could realize any logical function:

McCulloch and Pitts's important result is that any functioning in this sense which can be defined at all logically, strictly, and unambiguously in a finite number of words can also be realized by such a formal neural network. . . . Since the converse statement is obvious, we can therefore say that there is no difference between the possibility of describing a real or imagined mode of behavior completely and unambiguously in words, and the possibility of realizing it by a finite formal network. . . . There is no doubt that any special phase of any conceivable form of behavior can be described "completely and unambiguously" in words. This description may be lengthy, but it is always possible.¹⁴⁸

While McCulloch and Pitts drew from their analysis a pessimistic conclusion about the limitations of our knowledge of the world, von Neumann interpreted the same article as a source of optimistic predictions of the universal capabilities of formal neuron networks. As the neuroscientist Günther Palm has observed, McCulloch and Pitts were so "breathhtakingly arrogant in their use of mathematical formalism" in their 1943 paper that this made "the mathematical portions of the paper, i.e. almost all of it,

quite unreadable even for mathematicians.”¹⁴⁹ As McCulloch himself admitted, this paper might have remained completely unknown to biologists had it not been for von Neumann’s efforts¹⁵⁰; as a result, it was von Neumann’s broad reading of this paper that became its standard interpretation. Similarly, while Turing initially called a particular logical machine “universal” because it could imitate the operations of any other abstract computing machine, this notion of universality was extended to any type of thinking and behavior. In von Neumann’s interpretation, Turing, McCulloch, and Pitts had demonstrated that “what their automata can do can be described in logical terms and, conversely, that anything which can be described rigorously in logical terms can also be done by automata.”¹⁵¹ Later von Neumann tacitly replaced the narrowly defined “logical terms” with a very broadly conceived notion of “a finite number of words,” which could apply to any fragment of human speech. A modest statement of the possibility to implement logical functions in a logical machine thus magically turned into the sweeping claim of the universal capabilities of automata.

First the cyberneticians asked grandiose questions: What is life? How do we know the world? What governs human behavior? Next they translated these questions into cyberspeak, then substituted for them much narrower versions that could be answered within a particular specialized field of study: mathematics, logic, control theory, or communication engineering. Then they said that these grandiose questions had now been “precisely defined.” After obtaining the answer to a “precisely defined” question, they claimed that it could be applied universally, far beyond the original specialized field. Thus cyberspeak became a universal language for answering grandiose questions.

The Cybernetics Bandwagon

Upon its publication in 1948, Wiener’s *Cybernetics* gained enormous popularity, which its author hardly expected. The *Saturday Review of Literature* noted that it appeared “impossible for anyone seriously interested in our civilization to ignore this book.” “It is,” the magazine commented, “a ‘must’ book for those in every branch of science.” A large portion of the book was occupied by complex mathematical chapters, which a broad audience could not possibly understand. These chapters,

although “largely irrelevant,”¹⁵² fulfilled an important rhetorical function: they greatly impressed lay readers, thus conferring legitimacy on the bold claims made in a plain language in the rest of the book. Cybernetics promised solutions to a wide range of social, biological, and technological problems through computer modeling, information processing, and feedback control. Complex social and biological phenomena looked simpler and more manageable when described in cybernetic terms. Masking the differences in the nature and the scale of those phenomena, the common language of cybernetics allowed one to use the same mathematical techniques across a wide range of disciplines. When translated into cyberspeak, biological, technological, and social problems all seemed to have similar—cybernetic—solutions. Taking cybernetic claims seriously, many biologists and social scientists pushed the boundaries of cybernetics even farther than Wiener and his colleagues originally envisioned.

The popular press hailed digital computers as “electronic brains.”¹⁵³ *Scientific American* published an accessible account of cybernetics under the provocative title “Man Viewed as a Machine.”¹⁵⁴ The computer specialist Frank H. George threw a challenge to the readers of the English journal *Philosophy*: “You can’t tell me anything that your wife can do that a machine can’t (in principle).”¹⁵⁵ Political scientists spoke of the “nerves of government.”¹⁵⁶ Engineers, economists, and journalists described the bright technological future populated with intelligent robots.¹⁵⁷ Business consultants began to sell “management cybernetics.”¹⁵⁸ Molecular biologists conceptualized the gene as “the smallest message unit” that could “make a yes-no decision” and measured “the information content and error rate of living things.”¹⁵⁹ Biological specificity was “re-represented through the scriptural tropes of information—*message, alphabet, instructions, code, text, reading, program*. The narratives of heredity and life [were] rewritten as programmed communication systems.”¹⁶⁰

While borrowing cybernetic concepts, biologists and social scientists tended to leave out the complex mathematical techniques of control engineering and communication theory. The author of a 1961 overview observed that biologists “have been aware of information theory and have made qualitative use of some of its concepts” but concluded that “no explicit, and especially no quantitative use of information theory has, however, been made in practice.”¹⁶¹ As Lily Kay put it, information theory functioned in biology “as a discursive rather than mathematical tool.”¹⁶² In

other words, the popularity of cybernetics consisted largely in the spread of cyberspeak. Geoffrey Bowker has argued that cyberspeak turned into “a universal language for a new age” largely because jumping on the cybernetics bandwagon made it easier to attract funding for research:

The advantage for scientists from whatever discipline of being given a remit by cybernetics to use interesting and glamorous words was clear. Anyone tapping into the network of words used by cybernetics would be tapping into the network of problems that cyberneticians were aiming to solve. These, by definition (since cybernetics was the science of the current conjuncture) were at the cutting edge of military and industrial research. Grant applications could follow.¹⁶³

Some pioneers of cybernetics felt uneasy about such unrestrained uses of cyberspeak and about their unruly cultural ramifications. Shannon initially hesitated to apply the thermodynamic term *entropy* to communication engineering and emphasized that his notion of information had nothing to do with the “semantic aspects of communication.”¹⁶⁴ As Shannon’s mathematical theory of communication (packaged as “information theory”) began to spread widely into biology, psychology, and economics, the cultural connotations of his technical terms quickly span out of control. In a 1956 essay titled “The Bandwagon,” Shannon sounded a cautionary note:

[Information theory] has perhaps been ballooned to an importance beyond its actual accomplishments. Our fellow scientists in many different fields, attracted by the fanfare and by the new avenues opened to scientific analysis, are using these ideas in their own problems. . . . It will be all too easy for our somewhat artificial prosperity to collapse overnight when it is realized that the use of a few exciting words like *information*, *entropy*, *redundancy*, do not solve all our problems.¹⁶⁵

Eventually, Shannon withdrew from the public eye and refused to speak about his “information theory.”¹⁶⁶

Wiener also objected to what he saw as a misuse of the words *automation* and *cybernetics* by “a group of eager beavers.” “I cannot protest against the free use by any man of a word [*cybernetics*] that I intended as a common noun,” he wrote in 1954, “but I do protest against the appropriation of words covering a certain philosophy of engineering by many engineers with only a fragmentary idea of what these words mean.”¹⁶⁷ On the one hand, Wiener actively promoted cyberspeak as a universal language; on the other, he tried—apparently in vain—to limit its uses. The tendency toward universality in the cybernetic discourse quickly overpowered his attempts to control the meaning of *cybernetics*.

Though he was hailed as a “prophet” of the new age of automatic machinery, Wiener held ambivalent views about the social implications of cybernetics. He regarded automatic machines as both “threat and promise.”¹⁶⁸ He proclaimed the advent of a “second industrial revolution” that would bring about “the automatic factory and the assembly line without human agents.”¹⁶⁹ This revolution, in his view, carried “great possibilities for good and for evil.”¹⁷⁰ Cybernetic techniques and technologies, he argued, “open to us vistas of a period of greater plenty than the human race has ever known, although they create at the same time the possibility of a more devastating level of social ruin and perversion than any we have yet known.”¹⁷¹ Wiener warned that automation was “bound to devalue the human brain.”¹⁷² “The skilled scientist and the skilled administrator may survive,” he wrote, “[but] the average human being of mediocre attainments or less has nothing to sell that it is worth anyone’s money to buy.”¹⁷³ Wiener was deeply critical of capitalist America. He did not believe in the ability of the “invisible hand” of the free market to establish economic and social equilibrium (*homeostasis*, in cybernetic terms). His social outlook was overtly pessimistic: “There is no homeostasis whatever. We are involved in the business cycles of boom and failure, in the successions of dictatorship and revolution, in the wars which everyone loses.”¹⁷⁴

Cybernetics, in Wiener’s view, provided hope for social change. Two years after *Cybernetics* he published *The Human Use of Human Beings: Cybernetics and Society*, in which he developed a cybernetic critique of the pervasive controls over social communication under McCarthyism in the United States and under Stalinism in Russia. He believed that describing society in cybernetic terms as a self-regulating device would make people realize the danger of “the control of the means of communication” as “the most effective and most important” anti-homeostatic factor, which could drive society out of equilibrium.¹⁷⁵ While Wiener used cyberspeak to emphasize the need for unencumbered communication, his friends among social scientists often interpreted cybernetics as an efficient tool for rational social control. Wiener did not share their “excessive optimism” and argued that “in the social sciences we have to deal with short statistical runs, [and we cannot] be sure that a considerable part of what we observe is not an artifact of our own creation.”¹⁷⁶ Again, his attempts to control the uses of cyberspeak proved futile.

While Wiener hoped to make cybernetics a scientific basis for liberal politics, the language of cybernetics became a convenient rhetorical tool for very diverse political discourses. In December of 1948 *Le Monde* published a review of *Cybernetics* by the Dominican friar Père Dubarle, who contemplated the possibility of an all-powerful “*machine à gouverner*” that would make the state “the only supreme co-coordinator of all partial decisions.” “In comparison with this,” wrote Dubarle, “Hobbes’s *Leviathan* was nothing but a pleasant joke. We are running the risk nowadays of a great World State, where deliberate and conscious primitive injustice may be the only possible condition for the statistical happiness of the masses: a world worse than hell for every clear mind.”¹⁷⁷ While Dubarle viewed cybernetics as the embodiment of a totalitarian spirit, Wiener attempted to make the cybernetic *machine à gouverner* an element of the cybernetic critique of totalitarianism. He employed cyberspeak to criticize both capitalism and communism: “A sort of *machine à gouverner* is thus now essentially in operation on both sides of the world conflict, although it does not consist in either case of a single machine which makes policy, but rather of a mechanistic technique which is adapted to the exigencies of a machine-like group of men devoted to the formulation of policy. Père Dubarle has called the attention of the scientist to the growing military and political mechanization of the world as a great superhuman apparatus working on cybernetic principles.”¹⁷⁸ As it turned out, both Wiener’s desired social equilibrium and the despised “mechanization of the world” were based on the same cybernetic principles. Cybernetics thus could not be tied to a single political agenda; instead, different parties were able to use cyberspeak to express widely divergent social views.

The fate of Wiener’s cybernetics was deeply ironic. Wiener developed a cybernetic critique of the manipulation of social communications for political purposes; however, the same language of cybernetics was employed to develop pragmatic sociological theories within a conservative political framework.¹⁷⁹ After Hiroshima, he became an outspoken critic of the military-industrial complex; yet cybernetic techniques and technologies were widely used by the military.¹⁸⁰ He insisted that information could not be a commodity; nevertheless, the cybernetic techniques of measuring and processing information facilitated the marketing of this product.¹⁸¹ While he spoke against military secrecy, information theory greatly improved cryptography. In Wiener’s view, cybernetic knowledge would liberate rather

than further enslave the individual. Cybernetic analysis of control mechanisms, however, helped create new controls. Wiener hoped that his cybernetic analysis would expose the flaws of both capitalism and communism. Yet capitalism fully embraced cybernetics and quickly appropriated it to its needs.

In the next chapter, I will examine the reaction to cybernetic ideas on the communist side of the Iron Curtain.

Cybernetics in the Service of Communism

[The proposed unified automated control system for the national economy] would make it possible to fully implement the main economic advantages of communism—centralized control and the planned economy. This would ensure full harmony and a match between the political and economic structures of the communist state and the technical means for controlling the national economy.

—Aleksii Liapunov and Anatolii Kitov, “Cybernetics in Technology and in the Economy” (1961)¹

Soviet Cybernetics as a “Science of Government”

Soviet cybernetics began gradually to fashion itself as a universal methodological guide for science, technology, economics, and even politics. This evolution of cybernetic discourse was reflected in the notable change in the translation of Wiener’s favorite term, *control*. In the early 1950s, Soviet critics of cybernetics translated the word *control* in the title of Wiener’s magnum opus, *Cybernetics, or Control and Communication in the Animal and the Machine*, literally as *kontrol’*.² In contrast, Soviet advocates of cybernetics chose a different Russian word—*upravlenie*—to render the term *control* in the 1958 Russian translation of Wiener’s book.³ Both *kontrol’* and *upravlenie* can be translated as *control*, but they differ in their nuances. The word *kontrol’* refers to the processes of checking and examination, while *upravlenie* has such meanings as management, administration, direction, and government.⁴ Russians speak of *kontrol’* when the controller is bound by a preset goal and only monitors the controlled process but does not direct it (e.g., quality control). *Upravlenie*, on the other hand, applies to administrative decision making and often involves setting new policies (e.g., managing a factory or governing a state).⁵ Speaking at Moscow University in November of 1961, Engineer Admiral Aksel’ Berg, the chairman

of the Academy Council on Cybernetics, frankly admitted: “Many people don’t seem to like the word *cybernetics*. I don’t like it either, but we haven’t yet come up with a better one. It would be better to use a Russian word. That’s why we often speak of a new *science of government* [*upravlenie*].”⁶

From its very inception, the Academy Council on Cybernetics interpreted the term *cybernetics* very broadly. Berg argued that cybernetic methods of control were applicable to all “dynamic systems,” including technological processes, living organisms, and human collectives. He believed that even the most complex technological and social systems were amenable to cybernetic control:

There are no unknowable phenomena, only unknown ones; likewise, there are no uncontrollable processes, only those in which the complexity of the task is not yet matched by the methods and means for its solution. Cybernetics broadens the range of controllable processes; this is its essence and its major merit.⁷

The main task of Soviet cybernetics, Berg announced, was “the development of methods and the application of tools for controlling the entire national economy, individual technological processes, and various forms of economic activity to ensure the optimal regime of government [*upravlenie*].”⁸

To underscore this broad interpretation of cybernetic control, the editor of the Russian edition of *Cybernetics*, the mathematician Geliï Povarov, added an extensive footnote in which he traced the genealogy of cybernetics past Wiener to the French physicist André-Marie Ampère.⁹ Ampère had introduced the word *cybernétique* in the 1830s to denote the science of government in his classification of sciences, and listed *cybernétique* among the political sciences.¹⁰ In his initial report on the prospects of cybernetic research to the presidium of the Soviet Academy of Sciences in April of 1959, Berg referred to Ampère in the very first sentence. Berg mentioned Wiener only later in his talk, emphasizing that the latter “reused” the term *cybernetics*. Berg also often linked main ideas of cybernetics to the ideological legacy of Lenin, who in the 1920s had called on Bolsheviks to learn “the fundamentals of the science of government [*upravlenie*].”¹¹ Soviet cybernetics proved to be a very political science indeed.

Berg’s personality, which paradoxically combined the gallantry of the Russian nobility, the iron fist of the Soviet military, the sensibility of a former prisoner of the Gulag, and a naive faith in the bright Communist future, made a decisive impact on the activities of the Cybernetics Council, which he chaired. Berg acted as a great spokesman for progress and packed his



Figure 6.1

Aksel' Berg lecturing. Courtesy of Russian Academy of Sciences Archive, Moscow.

publications with such expressions as “new scientific and technological opportunities,” “new science,” “progressive scientists,” “the reevaluation of values,” “the rejection of obsolete doctrines,” and “the path of progress.” Cybernetics, a new, “young” science, embodied for him this progressive spirit.

Berg pushed cybernetics forward with the thrust of an icebreaker. He had no pity for his worst enemies—“backward people,” “skeptics, dogmatists, and conservatives.”¹² During one of his frequent TV appearances, Berg passionately exclaimed that the conservatives who interfere with scientific progress must be “publicly executed by a firing squad in Red Square.”¹³ In his informal remarks at a 1967 session of his Council, Berg expressed his agenda in unequivocal military terms:

When the computer enters our home . . . there will be no need to call a doctor; the machine will tell you what to do. Students will not have to go some place and listen to hideous lectures of old pensioners, who know nothing; programs will be optimized and you will have connection with a machine, which will come to your home, as water and light did. . . . If someone does not believe it, let him commit suicide. This is the future, and we will fight for it, and we will weed out anybody who would interfere.¹⁴

Propagandizing in favor of cybernetic ideas became of the main activities of Berg's Council. In 1961 the council sponsored "Cybernetics in Our Lives" (a series of twelve 20-minute programs on Moscow radio), "Faster than Thought" (a series of broadcasts on the Moscow TV channel), and more than 200 lectures before various "collectives of workers, clerks, students, engineers, scientists, managers, and the personnel of the Central Committee of the Communist Party."¹⁵

Berg mobilized his council to publish a volume, appropriately titled *Kibernetiku—na sluzhbu kommunizmu* [*Cybernetics—in the Service of Communism*],¹⁶ in time for the opening of the Twenty-Second Congress of the Communist Party in 1961.¹⁷ His efforts paid off when a new Party Program adopted at this Congress mentioned cybernetics among the sciences called upon to play a crucial role in the creation of the material and technical basis of communism. The published draft of the new program stated that it was "necessary to organize the wide use of cybernetics"; the final version adopted at the congress more vigorously asserted that cybernetics and electronic computers and control systems would "be widely applied" in production, research, planning, and management.¹⁸ The popular press began to call computers "machines of communism."¹⁹

The word *cybernetics* began to acquire favorable ideological connotations. The Cybernetics Council adopted *Cybernetics—in the Service of Communism* as the title of its regular proceedings. Berg used any occasion, including a session of the Biological Sciences Division of the Academy, to emphasize publicly the great political significance of cybernetics:

However unusual this may sound to some conservatives who do not wish to comprehend elementary truths, we will be building communism on the basis of the most broad use of electronic machines, capable of processing enormous amounts of technological, economic, and biological information in the shortest time. These machines, aptly called "cybernetic machines," . . . will solve the problem of *continuous optimal planning and control*.²⁰

In the 1960s, "optimal planning and control" became a motto of the cybernetic movement. Soviet cyberneticians assumed that the main problem of the Soviet economy lay in the inefficient mechanisms of data collection, information processing, and control, and offered a solution based on mathematical modeling and computer-aided decision making. They believed that computers produced a politically neutral, "optimal" solution, and hoped that cybernetics would create an opportunity to reform Soviet planning and

management practices without cardinal political change. One memoirist aptly called such cyberneticians *considents*, meaning half-consenting dissidents.²¹ Soviet cyberneticians were looking for a technological solution to an inherently political problem; by its own nature, however, their project was doomed to play a political role.

The “Dialectical Materialization” of Cybernetics

Cyberneticians initially chose cyberspeak as a value-free, “objective” language, hoping to limit their discussions with the authorities to technical terms. Instead, cyberspeak itself became politicized. In the post-Stalin period, Soviet philosophers lost their position as ideological watchdogs and judges of scientific controversies, and cyberneticians played a prominent role in the efforts to discredit dogmatic philosophy. The old order was turned upside down: philosophers could no longer reprimand scientists; quite the opposite, cybernetics now claimed to be a guide to solving philosophical problems. As the historian Alexander Vucinich has put it, “the new orientation demanded a shift in emphasis from the interpretation of modern science in the light of dialectical materialism to the interpretation of dialectical materialism in the light of modern science.”²² Philosophers quickly reorganized: they did not challenge the validity of cybernetic research, but instead adapted philosophical discourse to cybernetic innovations. Well trained in newspeak techniques, some philosophers now adopted cyberspeak as a new ideological language.

While leading Soviet cyberneticians viewed official philosophy as an enemy, Aksel’ Berg, more skilled in the Soviet bureaucratic etiquette, preferred to work out a compromise. When he said that scientists did not have to give a precise definition of cybernetics because it was “the concern of philosophers,” he actually meant it. In January of 1962, at a session of the presidium of the Council on Cybernetics, Berg proposed to add a philosophical section to the council. “Philosophy permeates everything,” he said, “and a philosophical section must be considered most important.” Andrei Markov Jr. immediately remarked that such a section would be superfluous: “If philosophy permeates everything, then everyone must be a philosopher. If we create a separate section, philosophical issues would be isolated from the rest.” The mathematician Anatolii Dorodnitsyn, director of the Computation Center of the Academy of Sciences, realized the political

advantages of having a team of loyal philosophers on staff, and disagreed with Markov: "Mathematics also permeates everything, but the mathematical section exists. Therefore, a philosophical section must exist too." The linguist Sebast'ian Shaumian suggested an elegant linguistic solution: to call the section "methodological" rather than "philosophical." The presidium eventually agreed to organize a "commission," rather than a section, on "methodological foundations of cybernetics," under the condition that non-philosophers would also be included. Berg assured the critics that "philosophy, like other sciences, must obey the order and not be anarchistic."²³ Soon the commission quietly transformed itself into the Philosophical Section.²⁴

The Philosophical Section had a clear mission: to reconcile cybernetics with dialectical materialism by adapting dialectical materialism to cybernetics. Philosophers loyal to cybernetics duly accomplished this task. First, they managed to incorporate the concept of information into the canonical list of categories of dialectical materialism. Wiener's formula, "information is information, not matter or energy,"²⁵ which had provoked so much philosophical criticism during the anti-cybernetics campaign, was reinterpreted in a dialectical light. "Information is not identical with matter," one philosopher wrote, "although it would be wrong to consider information a non-material entity."²⁶ Another author argued that, while negative entropy expressed the orderliness of matter, information referred to the orderliness of one of the attributes of matter: reflection. The inherent connection between negative entropy and information, therefore, confirmed the unity of matter, a basic postulate of dialectical materialism.²⁷ This author identified noise with "disordered reflection" and emphasized the "dialectical rotation of information and noise."²⁸

Philosophers loyal to cybernetics not only reconciled cybernetics with dialectical materialism but also effectively worked out a strategic alliance between the two. Cybernetics no longer posed a challenge to dialectical materialism; it no longer served as a stick with which Liapunov and Sobolev tried to chase philosophers out of the domain of science. Cybernetics was tamed and domesticated; a former rebel turned into a respectable discipline fully compatible with the principles of dialectical materialism.

To propagandize the new alliance between cybernetics and dialectical materialism, the Philosophical Section organized a conference on philosophical problems of cybernetics in June of 1962 in Moscow, attended by more than 1,000 philosophers, psychologists, mathematicians, physicists,

biologists, engineers, computer specialists, and educators from 30 cities. Philosophers employed by the Council on Cybernetics told the participants that cybernetics was “the most important element of the contemporary natural scientific foundation of dialectical materialism.”²⁹ An application of the same cybernetic models to qualitatively different phenomena was fully justified by the principle of the material unity of the world, they argued. By attributing an “objective character” to the notion of purpose, “cybernetics delivered a sharp blow to idealistic teleology.”³⁰ Cybernetics enriched the treasury of dialectical principles of Soviet philosophy by “the principle of unity of the discrete and continuous approaches” and “the principle of unity of the deterministic and probabilistic approaches.”³¹ Not a word of philosophical criticism of cybernetics was raised at the conference.

Soviet authors quickly turned cybernetics—formerly labeled a “handmaid of religion”—into a handmaid of atheism. One journalist argued that “if man can design machines which in a simplified form simulate the workings of the brain, then mental activity must be a natural, material process, and therefore no soul, no supernatural element allegedly put into man by god can really exist.”³² Another author defined God as “information isolated from any signals and existing by itself”; since information could not be carried without signals, he reasoned, the existence of God was also impossible.³³ Books with titles like *Cybernetics is Anti-Religion* and *Information Theory and Religion* appeared.³⁴

Cybernetics occupied a prominent place in the fundamental five-volume *Philosophical Encyclopedia*, published in 1960–1970. The philosopher Aleksandr Spirkin, head of the Philosophical Section, served as Deputy Editor-in-Chief of the encyclopedia, and he secured the publication of an 11-page article on cybernetics. (The article on mathematics was only 6 pages long.) The encyclopedia also included as separate entries such terms as *control systems*, *information theory*, *coding*, *algorithm*, *isomorphism*, and *homomorphism*, thus turning them into philosophical categories.

The encyclopedia article on cybernetics fully reflected the new domination of cybernetic discourse over the old philosophical clichés. The first draft, written by Ernest Kolman, was mildly critical of cybernetic claims, but after a discussion at the Philosophical Section of the Council on Cybernetics it was forcefully rejected. Kolman emphasized the “qualitative differences” between humans and machines, and argued that cybernetic devices did not have consciousness and therefore could not think.

Cybernetics supporters brushed such formulations aside as too limiting. "Why such a sharp boundary between man and machine?" asked one critic. Another remarked: "It is said [in the draft] that cybernetic devices have no consciousness. But so far we have not really figured out what is consciousness." Discussants argued that only cybernetics, not philosophy, could answer the crucial question about the difference between man and machine:

The existence of an essential boundary between man and machine so far has not been proved; this would be a major discovery, not something self-evident. . . . What is stated now as the quantitative differences between machines (of today) and the human brain may be overruled by the development of science.³⁵

Cyberneticians now did not have to submit their articles to philosophers for approval; quite the opposite, it was cyberneticians who corrected philosophers' writings. The Philosophical Section threw Kolman's manuscript out and appointed a group of loyal pro-cybernetics philosophers employed by the council to write a new article for the encyclopedia. The new version, which was eventually published, placed no philosophical limits on cybernetics; it stated that "the question of the possibility of creating automata of a new type that . . . would have the abilities of reduplication, adaptation to the environment, self-improvement, and creativity . . . is currently open to debate."³⁶

Soviet cybernetics gradually assumed the same guiding methodological role that had previously been claimed by official philosophy. In this role, cybernetics employed the same discursive techniques that made Soviet ideological discourse so flexible, adaptable, and virtually universal. Like the language of dialectical materialism, the language of Soviet cybernetics possessed an ambiguous terminology and some flexible rules of reasoning, which offered great freedom to the speaker. Both newspeak and cyberspeak imposed limitations on how to speak but left much room for what could be said, as long as it was said in the right way. Like official philosophy before it, cybernetics now boasted its complete objectivity, political significance, and universal applicability.

Cybernetics in Fashion

Soviet cybernetics became fully legitimized, officially recognized, and almost canonized, but this proved to be a Pyrrhic victory. Cybernetics acquired its place under the sun at the expense of losing its rebellious spirit

and turning into a fashionable trend. As cybernetic concepts acquired the degree of generality characteristic of ultra-flexible categories of dialectical materialism, cyberspeak began closely to resemble newspeak. Cybernetics laboratories and institutes mushroomed throughout the Soviet Union, and many career-minded scientists began using *cybernetics* as a buzzword. It was now trendy to call oneself a cybernetician, and suddenly the cybernetics movement became very crowded.

When former opponents of cybernetics realized that they could not defeat it, they decided to join it. Orthodox Pavlovians, for example, had initially claimed that cybernetics contradicted Pavlov's teaching and was therefore a pseudo-science, but now they completely embraced cybernetics and argued, with equal zeal, that reflex theory fully agreed with cybernetics, and that Pavlov himself was all but the founding father of cybernetics. To support this claim, Iurii Frolov, one of Pavlov's orthodox disciples, unearthed a forgotten 1936 article by the mathematician N. A. Romanov, who had proposed a probabilistic model of Pavlovian reflexes.³⁷ "Of course, this does not mean that cybernetics was 'discovered' in Pavlov's laboratory," admitted Frolov, "but this testifies to the fact that the mathematician Romanov . . . was the first to point out the closest connection between the teaching of Pavlov and the theory of probabilities as the core of modern cybernetics."³⁸

The excessively broad interpretation of cybernetics was facilitated by Aksel' Berg's efforts to expand the authority of his Cybernetics Council indefinitely. The results of the council's 1966 nationwide overview of cybernetic research are reproduced here in table 6.1.³⁹

As the scope of the Cybernetics Council widened to incorporate an ever-growing number of institutions and research fields, the council had to reduce its role from the substantive coordination of research to the simple collection and dissemination of information about ongoing projects. This superficial activity produced little effect. In December of 1967, at a celebratory

Table 6.1

	1962	1963	1964-65	1966	1967-1970
Projects	170	231	374	428	500
Institutions	29	61	96	133	150
Agencies	14	19	22	27	50

session of the Cybernetics Council dedicated to the fiftieth anniversary of the Bolshevik Revolution, Berg made a frank admission later struck from the record: "In practical terms, we are lagging behind, and nothing useful is being done, but we do our work and will keep doing it. I hope we will succeed!"⁴⁰

The expansion of cybernetics came at the expense of meaningful synthesis. Berg was willing to bring under the cybernetics umbrella any study that involved computing or used the terms *control*, *information*, or *communication*, and he stretched the concept of cybernetics almost without limit. In July of 1962 Berg proposed to reorganize the council to cover practically all of Soviet science. He illustrated his idea with a formidable table: eight rows represented basic fields (philosophy, psychology, mathematics, information theory, computing, reliability theory, physics, and chemistry), and twelve columns covered cybernetic applications (in biology, medicine, neuroscience, physiology, the management of the national economy, economics, linguistics, law, the energy sector, manufacturing, chemical industry, and transportation).⁴¹ The total number of cells in the table amounted to 96, and the council, according to Berg's plan, was to coordinate research nationwide in all 96 areas.

Cybernetics specialists began to realize that Berg's enthusiasm was going too far. Anatolii Dorodnitsyn stated bluntly that it was "impossible to implement everything proposed." The mathematician Boris Gnedenko, Deputy Chairman of the council, agreed: "We cannot do everything; we do not have sufficient forces. It is necessary to delineate the most important part, that is, the one that actually relates to cybernetics, and to appoint a commission of 10–15 members to hear all serious proposals."⁴² A few days after this meeting, Sergei Iablonskii wrote to Liapunov:

On the Council. [The situation] has become very difficult. There are almost no results from the Council. Berg only demands paperwork and strives for the expansion of the Council. . . . Berg demands plans, personal assignment of tasks, and so on, while we insist on formulating key problems and focusing all efforts on their solution. Such issues, in our opinion, must be as follows: the organization of the Institute of Mathematical Problems of Cybernetics, the strengthening and expansion of *Problems of Cybernetics*, and the support of certain research groups. Positions on these issues sharply differ. Berg told us to draft plans and all sorts of explanations, and then he would see. In the meantime, all available vacancies [at the Council] already disappeared (probably, wasted on some nonsense). We did not receive a single position for research!⁴³

Without Berg's support, Liapunov, editor of the series *Problems of Cybernetics*, could not obtain the permission to transform this publication

into a regular journal. Liapunov began to distance himself from the fussy activity of the council. Like Karl Marx, who had been so disgusted with the vulgarization of his teaching that he refused to call himself a Marxist, Liapunov, the accepted “father of Soviet cybernetics,” declined to write for the series *Cybernetics—in the Service of Communism*.

The Soviet cybernetics movement spread well beyond Liapunov’s original design and took on a life of its own. After his move from Moscow to Akademgorodok in Siberia in 1961, Liapunov lost his influence over the activities of the Cybernetics Council and numerous research groups that rallied under the banner of cybernetics. As one memoirist put it, after Liapunov’s departure “the center that had unified cybernetics disappeared, and cybernetics naturally split into numerous branches.”⁴⁴ Liapunov, he continued, “had been the living embodiment of cybernetics as a unified movement, and nobody else could play this role.”⁴⁴ Various groups began to pull Soviet cybernetics in various directions in accordance with their particular research agendas and political goals. The more people jumped on



Figure 6.2
Mikhail Lavrent’ev (left) and Aleksei Liapunov with a stack of *Problems of Cybernetics* volumes. Courtesy of Natal’ia Liapunova.

the cybernetics bandwagon, the less certain it was in which direction this wagon was going.

From “Military Cybernetics” to “Economic Cybernetics”

Numerous contributors to the 1961 collection *Cybernetics—in the Service of Communism* enthusiastically called for the introduction of computers and cybernetic methods into industry, biology, neurophysiology, medicine, linguistics, chemistry, power systems, transportation control, and the judicial process. Most important, they claimed that cybernetics could help automate and optimize the management of the national economy. Anatolii Kitov, in particular, wrote that “the national economy as a whole may be regarded as a complex cybernetic system, which incorporates an enormous number of various interconnected control loops with various levels of subordination.” He argued that only the application of cybernetic methods—mathematical modeling and computer simulation—could place economics on a solid scientific foundation: “Computer modeling makes it possible to forecast economic processes and to conduct mathematical experiments in economics. Thereby economics turns into an exact experimental science.”⁴⁵ In January of 1959 Kitov sent the Party Central Committee a copy of his 1958 book *Electronic Computing Machines* along with a proposal to create an automated control system for the national economy that would be based on a unified network of state computation centers. The authorities appointed a commission, headed by Berg, and soon the Central Committee and the Council of Ministers issued a joint decree on the improvement of production and introduction of computers into the economy.⁴⁶

Berg, Kitov, and Liapunov summarized the essence of cyberneticians’ aspirations to reform the Soviet economy in a 1960 joint article in the Party journal *Kommunist*. To increase the productivity and quality of management, they argued, managerial tasks must be automated. Citing positive Western experience, they suggested using computer technology and cybernetic methods of information processing on a scale much greater than in the West:

The dominant trend in the area of management is the transition toward complex automation of all types of information processing with the help of electronic computer technology, aimed at creating “a fully automated managerial office.” . . . In contrast to capitalist countries, where different companies create individual auto-

mated control systems for themselves, in the conditions of socialism it is quite possible to organize a unified complex automated system to control the national economy. It is obvious that the effect from such automation would be much greater than from the automation of control of individual enterprises.⁴⁷

The authors argued that the automation of management would greatly improve the efficiency and productivity of Soviet industry. They wrote, for example, that the use of computers would help reduce procurement planning time from 3–4 months to 3 days, cut the management apparatus by half, and decrease the cost of procurement procedures by a factor of 5. To support the outlined automated control system for the national economy, the authors proposed to build a unified territorial network of information computation centers with centralized control. These centers were to be linked to individual institutions and industrial enterprises, whose computational needs they would serve. The entire network was to provide means for automatic collection of economic data, planning, distribution of resources, banking, and control of transportation.

The initiative in applying computers to the automation of economic management came not from chief Soviet economists and planners, who remained largely conservative, but from cyberneticians working for the military. As a result, the initial visions of automated economic management were heavily influenced by the concept of automated military control and command. Liapunov explicitly compared economic decision making to military command:

In one important type of control algorithms, the purpose of an algorithm is the struggle against another control system or another algorithm. . . . [Such problems] emerge before the reflexologist, who plans experiments, which may be regarded as a struggle against the nervous system of the experimental animal. The same issues arise before the economist, who plans the work of an enterprise or a branch of industry, and calculates the most rational way of overcoming external problems. Finally, the commander on a battlefield faces similar problems.⁴⁸

Soviet specialists in “military cybernetics” viewed weapons systems, including both machines and the people who operated them, as cybernetic systems:

Any combat item (a tank, an aircraft, or a ship) can be seen as a typical cybernetic system with all its characteristic elements. The controlling organs or devices in such systems are the commanders and their control points, while the controlled objects are the weapons and the technical elements of the combat item. . . . With respect to troop control, the controlling organ is the commander and his staff, while the controlled objects are subordinate military units or combat items.⁴⁹

Military cyberneticians proposed that computers and cybernetic algorithms of optimal control be applied to control not only automated weapons but also military units. Computation Center 1 of the Ministry of Defense began working on the principles of automated troop control.⁵⁰ Cyberneticians argued that a “decision reached with the help of an electronic computer is more objective, for it is based not on a subjective opinion of a single person (the commander) and his intuition, but on collective experience, on those operational-tactical views that are generally adopted in a given military area.”⁵¹

Military cyberneticians were eager to apply the same concept of cybernetic control both to military command and to economic management. In 1959–1961 a group of military experts, which included Anatolii Kitov (then Deputy Director of Computation Center 1), Nikolai Buslenko, Lazar’ Liusternik, and Igor’ Poletaev, advocated the creation of a dual-use nationwide network of computation centers for both military and civilian applications. These centers were to be placed underground in secret locations and protected against a direct bomb hit. Then the centers were to be linked by hidden lines of communication to civilian stations in big cities, so that their powerful computers could process economic data in the time left after making all calculations for the military. The Ministry of Defense, however, rejected their proposal outright. The military leaders decided that combining military and civilian functions was “inexpedient.” Kitov later wrote that, in his view, the real reason behind the negative reaction to their proposal was that “people in power were concerned that, as a result of the introduction of computer technology, many of them could prove redundant.”⁵²

Soviet cyberneticians’ proposals for a cybernetic reform of managerial practices posed a direct challenge to the existing power hierarchies. Cybernetic analysis of information flows and managerial procedures exposed serious flaws in Soviet economic management, and cyberneticians believed that the introduction of cybernetic methods would help reform the obsolete managerial practices. For example, citing studies of data-collection procedures in Soviet industry, Liapunov argued that the contemporary managerial practices were clearly inadequate:

Certain data required for expedient control are, in fact, neglected, while much information, which is collected with great effort, has no function to fulfill, for this information does not affect decision making in production management. . . . As a result, many agencies and information channels duplicate one another and make

no real impact on production. A detailed mathematical modeling of production control would help find out which links in a control system are not necessary, and perhaps would help arrive at a more rational system of control in general.⁵³

Cyberneticians explicitly contrasted the promised efficiency of cybernetic control with the ineptitude of the cumbersome Soviet bureaucracy. In December of 1957 the leaders of the Soviet Academy of Sciences wrote to the presidium of the Party Central Committee:

The use of computers for statistics and planning must have an absolutely exceptional significance in terms of its efficiency. In most cases, such a use would make it possible to increase the speed of decision making by hundreds of times and to avoid errors that are currently produced by the unwieldy bureaucratic apparatus involved in these activities.⁵⁴

Cyberneticians' proposals challenged not only the existing administrative hierarchies in the government bureaucracy, but also infringed on the authority of the Communist Party. The officially recognized ideological supervisory function of the Party could hardly be formalized in cybernetic control algorithms. Critics of "military cybernetics" were quick to point out that computers lacked ideological vision:

Outside cybernetics' field of vision, there appear such cardinal questions about controlling military forces as the relationship between objective and subjective factors, the distribution of functions among control organs, the role of ideological and psychological factors, and so on.⁵⁵

The Chief Political Directorate of the Army reportedly played an active role in blocking the cyberneticians' proposal to create a dual-use nationwide network of computation centers. As Poletaev recalled, one official had asked him with disdain: "Where is the leading role of the Party in your [computing] machine?"⁵⁶

The military leaders quickly suppressed the cybernetic trend in the military ranks. In 1961 Computation Center 1 was transformed into Central Scientific Research Institute 27, and its functions were limited to the introduction of computers to automate individual military activities, instead of building automated troop-control systems.⁵⁷ The young reformers Kitov and Poletaev were forced to leave the army; Kitov was also expelled (temporarily) from the Communist Party. They transferred their cybernetic aspirations to other fields: Poletaev began working on mathematical models in biology, and Kitov was hired by Viktor Glushkov, director of the Institute of Cybernetics in Kiev, to supervise the introduction of automated

management systems based on the “method of network planning and control” (the Soviet version of the Western Program Evaluation and Review Technique, or PERT) in the defense industry. Glushkov’s team later tried the same method in civilian enterprises.⁵⁸

Inspired by “military cybernetics,” early proposals for cybernetic reforms in the economy were filled with military imagery. Glushkov argued that without computers “scientifically based control of the economy [would be] just as impossible as, say, military operations without the use of weapons.”⁵⁹ Viktor Belkin, an economist from the Institute of Electronic Control Machines of the State Planning Committee drew an illuminating comparison between controlling the national economy and launching a missile. Before the advent of computers, he wrote, the calculation of a missile’s trajectory took too much time, and this had precluded the construction of guided missiles. Now computers could calculate missiles’ trajectories faster than the missiles could fly, and this made it possible to guide missiles in real time. Similarly, Belkin argued, the introduction of computers in economics would provide the opportunity to calculate and correct economic plans in real time.⁶⁰

In the 1960s, Soviet military cyberneticians’ aspirations to create a computerized “guidance system” for the national economy led to some far-reaching efforts by mathematical economists and “economic cyberneticians” to develop systems of “optimal planning” for the Soviet economy. As with “military cybernetics,” proposals advanced by Soviet “economic cyberneticians” provoked some serious controversy.

“Optimal Decision Making on a National Scale”: Aspirations and Constraints

Under the umbrella of cybernetics, a new trend in Soviet economic thought—“economic cybernetics”—emerged. The participants of Liapunov’s open seminar on cybernetics at Moscow University often discussed the prospects for applying mathematical methods in economics. In May of 1957, for example, the Leningrad mathematician Leonid Kantorovich, the Soviet pioneer of linear programming, presented a paper on mathematical methods in economic planning. Ten days later, two cyberneticians spoke about a proposed “high-speed computer for economic analysis.”⁶¹ The Academy Council on Cybernetics set up an Economics

Section, regularly published papers on mathematical economics in *Cybernetics—in the Service of Communism*, and sponsored several conferences on that topic. In 1958 only a handful of Soviet economists were interested in mathematical models of planning and management; 3 years later, more than 40 institutions were conducting research on mathematical economics.⁶² In 1967 the Council on Cybernetics coordinated cybernetics research in some 500 institutions, and half of them were engaged in applying cybernetic methods to economics.⁶³ “Economic cyberneticians” advanced their ideas in constant struggle with the conservative elite among Soviet economists.

Soviet economists were among the first in the world to apply mathematical methods to nationwide economic planning. In 1926 the State Planning Committee (Gosplan) calculated a balance sheet of the Soviet economy for 1923–24. The idea of balanced economic development did not fit into Stalin’s plans for the forced collectivization of agriculture. In a 1929 speech he dismissed the Gosplan calculations as “a numbers game,” and many of the “players” soon disappeared in purges.⁶⁴ While writing his pioneering 1939 work on linear programming, *Mathematical Methods of Organizing*



Figure 6.3

Leonid Kantorovich (left) receiving the Nobel Prize in economics from King Carl XVI Gustav of Sweden in 1975. From Pospelov and Fet, eds., *Ocherki istorii informatiki v Rossii*.

and *Planning Production*, Kantorovich used the terms *organizing* and *planning* instead of *economics*, since at that time mathematical economics as it had been developed in the West was labeled “anti-Marxist” by Soviet critics; the purpose of using mathematics in economics, they argued, was to defend capitalism.⁶⁵ The publication of Kantorovich’s second book on mathematical economics, *Economic Calculation of the Best Use of Resources*, written in the early 1940s, had to wait for more than 15 years. By the early 1950s, political economists thoroughly dominated Soviet economic studies; mathematical methods had been pushed to the margins of the discipline.

With the political “thaw” and the growth of economic problems, however, Soviet leaders began contemplating an economic reform and permitted an open discussion of its possible directions. A group of prominent economists and mathematicians led by Kantorovich, Vasilii Nemchinov, and Viktor Novozhilov challenged the authority of political economists and proposed to use mathematical methods and computer technology for efficient planning, price regulation, and processing of economic information. Conceptualizing the Soviet economy in cybernetic terms as a giant control system, they aspired to transform the entire Soviet economic system into an optimally functioning one. This formidable task faced serious technical and political difficulties.⁶⁶

Soviet economic cyberneticians interpreted economic management as a form of cybernetic control [*upravlenie*] and brought Western studies on “management science” under the umbrella of cybernetics. Cybernetics was viewed as a guide to the automation of production control and to the automated management of the national economy. In July of 1962 the Academy of Sciences dispatched a group of experts to the United States and England to learn about the latest methods of industrial automation. Upon their return, members of the delegation wrote a confidential report, in which they argued that cybernetics (“management science”) would provide new methods for industrial automation and economic management:

Management science in the United States incorporates the main ideas of automated control of machinery, the principle of feedback, and the methods of application of this principle. The main purpose of this science is the elaboration of methods of optimal planning and industrial management. The delegation believes that now is the time to utilize in a serious way the achievements of management science under the conditions of planned economy in our country in order to place solid scientific principles in the foundation of economic management and production control with the help of operations research methods.⁶⁷

Proposals for cybernetic (“rational,” “objective,” “scientific”) control of the economy had the ear of Party authorities. At a Central Committee plenum in November of 1962, Khrushchev called on his Party comrades to adopt Western “rational” managerial techniques. In the conditions of the planned economy, he argued, these techniques would be even easier to implement than under capitalism.⁶⁸ Also in November of 1962, Aleksei Kosygin, then Deputy Chairman of the Soviet Council of Ministers, met with two leading mathematicians—Mstislav Keldysh and Viktor Glushkov—to discuss the prospects for using cybernetic methods to control the economy. Keldysh and Glushkov proposed to optimize economic decision making on a national scale by creating a nationwide automated system, based on a unified state network of computation centers, for economic planning and management. Soon the Council of Ministers appointed Glushkov to head a commission charged with preparing a detailed proposal.⁶⁹

Glushkov originally envisioned a system that would monitor all labor, production, and retailing. He even proposed to eliminate money from the economy, evoking the Marxist utopian vision of a communist society.



Figure 6.4
Viktor Glushkov and Aksel' Berg at the Institute of Cybernetics in Kiev. Courtesy of Russian Academy of Sciences Archive, Moscow.

Perhaps Glushkov hoped that this idea would appeal to Khrushchev, who in 1961 had announced the Party's goal of building communism in the Soviet Union by 1980. Keldysh, who was much more experienced in top-level bureaucratic maneuvers, reportedly advised Glushkov to drop this radical idea, explaining that it might "arouse unnecessary emotions." Glushkov then excluded this section from his main proposal and submitted it to the Party Central Committee separately. If ideology was to play any significant role in the Soviet top-level decision making, this was its best chance. Glushkov's proposal to eliminate money, however, received neither support nor a formal response from the Party authorities.⁷⁰

In a more acceptable form, economic cyberneticians' proposals gradually made their way through the maze of Soviet government agencies. Eventually, in 1963, the Party Central Committee and the Council of Ministers adopted a joint decree titled *On Improving the Supervision of Work on the Introduction of Computer Technology and Automated Management Systems into the National Economy*. Several major government agencies involved in the management of the national economy established specialized institutions to study problems of automated management: the Academy of Sciences set up the Central Economic Mathematical Institute (hereafter CEMI),⁷¹ the State Planning Committee organized the Main Computation Center, and the Central Statistical Administration created the Scientific Research Institute for Design of Computation Centers and Economic Information Systems. To coordinate all work in this area, the government set up the Chief Administration for the Introduction of Computer Technology into the National Economy.

CEMI, organized with the active support of Berg's Council on Cybernetics, became the hotbed of economic cybernetics. The director of CEMI, Academician Nikolai Fedorenko, also became the chairman of the Scientific Council on Optimal Planning and Management of the National Economy. CEMI put forward an ambitious program of optimal economic planning on a national scale, based on the cybernetic methods of controlling large-scale complex systems. In 1964 CEMI's research agenda was formulated as follows:

- (1) Elaboration of a theory of optimal planning and management, and the construction of a general mathematical model of the national economy;
- (2) Development of a unified system of economic information;
- (3) Development of a unified state network of computation centers;

- (4) Development of mathematical methods for the general model;
- (5) Creation of concrete planning and management systems based on mathematical methods and computer technology; and
- (6) Elaboration of standards and algorithms for planning and management.⁷²

CEMI cooperated closely with Glushkov's Institute of Cybernetics. In 1964 Glushkov and Fedorenko published a joint proposal for a unified system of optimal planning and management on the basis of a three-tier unified nationwide network of computation centers. The proposed network included tens of thousands of local computation centers to collect "primary information," 30–50 mid-level computation centers in major cities, and one top-level center controlling the entire network and serving the government. The structure of the computer network was made flexible enough to provide independence from possible reorganizations of planning and management agencies, which were quite frequent in those days. In the existing economic system, the central planning organs collected primary economic information from individual enterprises by means of four relatively independent parallel channels: the planning system, the material-technical supply system, the statistical system, and the financial system. Glushkov and Fedorenko proposed to replace this scheme with one in which all economic data would be collected only once, stored in data centers, and then made available to all the agencies involved. By this restructuring of economic information flows, the authors hoped to reduce the number of reports submitted by an individual enterprise by a factor of 20–30.⁷³ Glushkov and Fedorenko promised that the proposed unified system of optimal planning and management would provide "optimal decision making on a national scale" by processing "the entire body of primary economic information as a whole."⁷⁴

Economic cyberneticians quickly realized that it was impossible to centralize all economic decision making in Moscow: the mathematical optimization of a large-scale system was simply not feasible. CEMI researchers estimated that complete optimization of the Soviet economy required solving a gargantuan system of equations with 50 million variables and 5 million constraints. They admitted that even a computer performing 1 million operations per second, which was much faster than any available Soviet computers, would require one month to solve a system a billionth as large.⁷⁵ Besides, economic cyberneticians realized that there were some serious conceptual difficulties: linear programming was suited

for the problem of resource distribution, but it did not work well for prospective planning, there were different views on what constituted an economic optimum, and it was difficult to agree on a single criterion for optimization.⁷⁶ In 1967 Fedorenko unequivocally stated that “the full formalization of the functioning of an economic system and the creation of a fully automated centralized system of planning and management of the economy is unwarranted.”⁷⁷

Economic cyberneticians envisioned a hybrid planning system that would provide for some decentralization of decision making while preserving the backbone of the Soviet economic system: the national plan. While some Western observers described their projects in very simplistic terms as attempts at “perfect computation,” Soviet mathematical economists developed a truly sophisticated concept of optimal planning based on the use of the market mechanism through “indirect centralization.”

“Optimal Planning”: A Vehicle of Economic Reform or an Obstacle to It?

The idea of indirect centralization, introduced by Viktor Novozhilov, was based on a mathematical theorem stating that the equilibrium point in a many-person non-coalition game would be an optimum.⁷⁸ Applying the results of game theory to the Soviet economy, economic cyberneticians argued that the central government did not need to impose specific output quotas on individual enterprises; instead, it could set “optimal” prices and investment efficiency norms, then allow individual enterprises to make their own decisions. If the criteria of economic performance were properly formulated, the independent activity of individual enterprises should lead to the fulfillment of the national plan. In contrast to the accepted view, economic cyberneticians argued that the ideal of “optimal planning” could be achieved by a radical decentralization of economic decision making and a regulated use of the market mechanism:

The finding of an optimum may take place in a decentralized way, i.e. the equilibrium point, or optimum, can be found as a result of an exchange of information between economic organs, each of which independently solves the problem of optimization guided by its own individual (local) criterion of optimality. . . . In this way, it is possible to use the market mechanism for organizing the process of the decentralized working out of the optimal plan.⁷⁹

Economic cyberneticians viewed optimal planning as a way to introduce an economic incentive system and to let the economy self-regulate within

some general constraints imposed by the national plan. They sought a rational, “scientific” justification for the ideologically dubious notion of *khozraschet* (a profit-based system of economic incentives for individual enterprises). Describing the Soviet economy in quintessential cybernetic terms, Novozhilov argued that the market mechanism was equivalent to the feedback principle:

By now it is already widely known that cybernetics justifies *khozraschet* as the compensator of randomness in a planned economy. A socialist economy is a very complicated system subject to the activity of a multiplicity of random factors and not lending itself to description in full detail. The control of such systems is possible only on the condition that there exists a self-regulator with feedback, which can speedily compensate for the action of random factors and bring the system back to the target state or the target path of development. A self-regulator with feedback keeps an eye on the values of various variables (e.g. the profitability of production) and acts on the system in such a way as to prevent an excessive deviation of these variables from their normative values. In a socialist economy the market mechanism [tovarno-denezhnye otnosheniia] is such a regulatory mechanism. . . . The basic proportions of the development and the chief controlling normatives must be established by the national economic plan. The detailing, correction and fulfillment of the plan must be regulated by *khozraschet*.⁸⁰

In the absence of real market mechanisms in the socialist economy, computer modeling of these mechanisms was to provide quasi-market stimuli for individual enterprises. In the early 1960s, the Institute of Electronic Control Machines in Moscow, led by the computer designer Isaak Bruk, built the specialized M-5 computer for economic applications and began calculating “optimal” prices for a comprehensive price reform.⁸¹ Kantorovich also proposed a mathematical model for calculating shadow prices on the basis of “objectively determined valuations,” which reflected the relative scarcity of products, thus taking into account the relationship between supply and demand.

Economic cyberneticians strongly emphasized their reliance on “objective” computation and “objective” valuations. Contrasting their approach with the traditional discourse of Soviet political economy, which was loaded with ideological formulas borrowed from the Marxist theory of value, they strongly asserted the discursive autonomy of economic cybernetics from political economy: “[The Marxist concept of] value and objective valuations are two completely different and incommensurable things. Value is a category of political economy and objective valuations are an algorithmic formula for the calculation of equilibrium prices in an optimal plan.”⁸²

Sharply criticizing orthodox economists at a 1959 session of the Academy of Sciences, Kantorovich argued that the impossibility to translate their theories into cyberspeak made the shallowness of these theories self-evident:

The computer cannot digest some of our economists' scholarly products. Their long talks and articles on economic topics, which people listened to and read and thought they understood, proved impossible to use. Any attempt to give them a logical-mathematical, algorithmic form in order to enter them into a computer failed. It turned out that, after removing everything that was said "in general" (sometimes there was substance, but it related to sociology or politics, not to economics per se), and after pouring out all the "water," there was either nothing left, or just one big question mark, the formulation of an unsolved problem.⁸³

The idea of optimal planning not only challenged some dogmas of political economy but also threatened to upset the existing hierarchy of power in the economic sphere. An introduction of optimal planning would have undermined the monopoly of the State Planning Committee on top-level economic decisions; the government bureaucracy would have had to yield some of its power to mathematical economists. Central economic agencies therefore attempted to limit cybernetic experiments to low-level management: they suggested that economic cyberneticians start with the optimization of local planning, and then gradually move up to a nationwide system. Cyberneticians objected, arguing that it was impossible to achieve local optimization without reforming economic mechanisms on a national scale.⁸⁴

Despite the Soviet leadership's general encouragement of their approach, economic cyberneticians faced a stern opposition from the same government agencies that were supposed to help in the implementation of optimal planning projects. Instead of directly opposing the cyberneticians' reforms, these agencies began to scale down and delay these projects. As it made its way through various government committees and commissions, Glushkov's original proposal for a nationwide automated control system for the economy was completely emasculated. The economic reform part disappeared; only a nationwide network of computation centers was preserved, and its functions were reduced significantly. In November of 1964 the Soviet government transferred the responsibility for "finalizing" Glushkov's proposal to the Central Statistical Administration. That agency quickly replaced Glushkov's concept of creating a unified network of computation centers with the idea of installing some computers at the existing statistical data-collecting stations. The State Planning Committee opposed

this idea, fearing that the narrow specialization of computation centers would give an advantage to a rival agency, and insisted on rewriting the proposal to include some planning functions. As the two powerful government agencies struggled, trying to adapt Glushkov's project to their own ends, the prospects for building a unified system of optimal planning and management on the basis of a nationwide network of computation centers quietly withered away.⁸⁵ In 1968 Kitov wrote in a personal letter to Liapunov:

The top leadership realizes the importance of [the introduction of computers into the national economy] but takes no effective measures in support of such work, while responsible officials from the ministries and other government agencies . . . display no interest in the automation of management or the optimization of planning. The problem is apparently rooted not in their personalities, but in their positions [in the bureaucratic hierarchy] and in the overall traditions, which change very slowly.⁸⁶

Kantorovich's attempts to optimize the production of pipes by rearranging the existing network of producers and suppliers ran into a stone wall. One sympathetic ministry official told him: "The Ministry of Metallurgy decides what to produce, and the Ministry of Supplies decides how to distribute it. Neither will yield its powers to anyone. To implement your plan, one must change the entire management system." One factory manager offered a similar explanation: "I cannot reallocate portions of the salary fund; it comes with the state order. This fund is greater for narrow pipes. If you rearrange the orders, this will upset the stability of this fund. To accept your plan, the entire system of management has to be reformed."⁸⁷

While industrial managers and government bureaucrats resisted the computerization of economic planning and management because it could upset the existing power structures, liberal economic reformers opposed Glushkov's proposal because they feared it would further centralize the control of the economy and suppress local initiative. Liberal economists insisted on the radical decentralization of economic planning and management and on the introduction of actual market mechanisms in the Soviet economy. If central planning organs and individual enterprises could arbitrarily manipulate various economic criteria, they argued, computers could only produce "distorted results, though with great speed."⁸⁸

Liberal economists warned that Glushkov's project would divert resources urgently needed for economic reform and merely conserve the obsolete forms of centralized economic management. The economist

Gavriil Popov, who would later play a prominent political role in Mikhail Gorbachev's *perestroika*, severely criticized Glushkov's proposal in a 1970 book:

[The construction of] the pyramids of Egypt was one of the reasons why that fertile ancient country turned into a desert. If one vigorously implements a meaningless economic decision, it ruins the economy. According to the blueprint of a unified state network of computation centers, these centers would spread over this country like those pyramids, designed by talented mathematicians and able engineers with the participation of unqualified economists.⁸⁹

Glushkov indeed admitted that his project for a nationwide network of computation centers would cost more than the space program and the atomic project put together.⁹⁰

Several pilot projects aimed at the development of small-scale computerized systems for production control and information management at individual factories had little success. "Optimal" control yielded poor results when the technology of production was old and obsolete, as was often the case at Soviet factories. At a metallurgical plant in Dneprodzerzhinsk, the use of computers to control a technological process saved minutes, while hours were wasted because of inefficient technology, faulty sensors, and lack of coordination among the stages of production.⁹¹ Glushkov admitted that any potential profit from management-information systems was also lost because of constant interruptions in supply and the inefficient organization of the industry as a whole.⁹² "Optimal planning and control" turned into a pure mathematical abstraction. Elena Markova, a specialist in "chemical cybernetics," has recalled that cyberneticians' suggestion to computerize production control at the Moscow chemical plant was met by plant managers with deep skepticism: "Look, this pipe is leaking, and ammonia leaks out by buckets. What is the point of talking about the optimization of ammonia production? If our machinists were not drunk, the efficiency would have skyrocketed without any computers."⁹³

Having limited political options, Soviet cyberneticians tried to work out a mathematical solution to the political problem of reforming the Soviet economy. Many of the leading Soviet cyberneticians involved in economic projects—Berg, Bruk, Glushkov, Kantorovich, Kitov, Liapunov—were not economists but rather engineers or mathematicians. Their vision of optimal economic planning was based on mathematical optimization, but it worked only on paper. While hoping to use mathematical methods and

computer modeling to advance their reform program, they did not realize that cybernetics could be employed—perhaps even more effectively—to conserve the existing power structures and administrative hierarchies.

Cybernetics in the Service of the Establishment

After October of 1964, when Khrushchev was ousted and Brezhnev came to power, the drive for reform in the Soviet economy quickly withered away. Soviet cybernetics underwent a parallel change: it transformed from a vehicle of reform into a pillar of the status quo. The Soviet political, economic, and academic establishment successfully appropriated cyberspeak and computer technology to serve its own goals.⁹⁴

Under Brezhnev, Khrushchev's decentralized system of regional economic management was quickly dismantled, and the centralized managerial structure of ministries and state committees was restored. Officials at various industrial branch ministries quickly realized that there were many ways to skin the cybernetic cat without necessarily losing their grip on power. Instead of building a nationwide automated management system that would connect all branches of industry within each region, they proposed to develop information-management systems along industrial branch lines. In March of 1966 the Party Central Committee and the Soviet Council of Ministers adopted a joint decree that made this approach an official policy.⁹⁵ It was decided that each ministry would set up a separate automated management system to serve its internal needs. Now the ministries did not have to share information/power with any rival agency; on the contrary, each ministry could use computer technology to strengthen its control over sensitive information.

If computerized management systems were compatible, they could serve to unite different enterprises; if they were incompatible, they would divide enterprises just as effectively. Built without much coordination, branch-based information-management systems proved incompatible with one another and with that of the State Planning Committee, and thus their integration was not possible. By constructing isolated information-management systems, Soviet industrial branch ministries laid a technical foundation for strengthening the centralized control over their subordinate enterprises. "Having different ministries is like having different governments," one contemporary observed.⁹⁶

Instead of facilitating the mechanism of “indirect centralization” envisioned by economic cyberneticians, computer technology was enrolled to strengthen hierarchical structures of centralized control. In 1970 the director of Gosplan’s computation center criticized the supporters of the “theory of optimal planning” for their “nihilistic approach” to centralized planning.⁹⁷ Gosplan officials effectively used computers to shore up their habitual planning practices rather than to reform them.

With the decline of the idea of a nationwide automated management system, CEMI’s mission was significantly reduced. By the end of the 1960s, the nationwide economic management system was not even mentioned in CEMI’s plans. CEMI’s areas of research were defined as follows:

- (1) Elaboration of systems theory for optimal national economic planning;
- (2) Development of automated systems for planning and management; and
- (3) Analysis of problems of national economic development for the period, 1971–75, and forecasting future economic growth.⁹⁸

Mathematics and computing was now “on tap, not on top” of economic decision making. CEMI had to fit its automated systems into the existing managerial structures, instead of developing a new approach to industrial management. In CEMI’s plans, the elaboration of cybernetic algorithms for economic planning and management was replaced with a reference to the then-fashionable “systems theory,” which placed emphasis on qualitative rather than quantitative analysis of control mechanisms.

A new impetus for the idea of a nationwide computer network came from the West when the Soviets learned about the development of the ARPANET, the predecessor to the Internet. Glushkov immediately drew the attention of the Soviet leaders to the similarity between this American project and his own 1964 proposal for a unified state network of computation centers. In his memoirs, Glushkov recalled:

In the late 1960s, the Central Committee and the Council of Ministers received information that Americans had drafted the design of an information network (several information networks, to be precise) as early as 1966, that is, 2 years after us. Unlike us, however, they did not argue but got to work, and in 1969 they already planned to launch the ARPANET network, . . . which linked computers installed in various American cities. Then some concern began to show in our quarters. I came to see [the secretary of the Central Committee] A. P. Kirilenko and handed him a memo that proposed to return to the ideas of my original project. “Write down in detail what has to be done, and we will create a commission,” he said. Then I wrote something like this: “The only thing I ask is not to create a commission. Commissions operate on the principle of subtraction of brains, not summation, and they can wreck any project.” They created a commission all the same.⁹⁹

Glushkov argued that, unless the processing of economic information was automated, by the mid 1980s nearly the entire adult population of the Soviet Union would be engaged in planning, accounting, and management.¹⁰⁰ To solve this problem, he proposed to build a Statewide Automated Management System (*Obshchegosudarstvennaia avtomatizirovannaia sistema upravleniia*, hereafter OGAS)—a network of regional and branch-based management information systems that would include all levels of control, from the top government level to production-control systems at individual factories. Glushkov believed that the larger was the object controlled by an automated management system, the greater was that system's economic effect.¹⁰¹ To coordinate the entire project, he proposed to establish a State Committee on the Improvement of Management, a powerful interbranch agency staffed with scientists who would oversee the construction and operation of a nationwide automated management system. Glushkov lobbied hard, trying to win over the support of the Soviet leaders. In his memoirs, comparing his situation with the position of the leaders of the Soviet space program and the atomic project, Glushkov wrote:

Both Korolev and Kurchatov had a curator among the Politburo members. They could come to him and resolve any issue. Our problem was that we did not have such a person. . . . It was particularly important to have a direct contact with one of the Politburo members, since our task was not only scientific and technical, but also political.¹⁰²

The Soviet leaders also realized that the OGAS project, unlike the space program and the atomic project, had direct political implications, which threatened the established hierarchy of power. The published draft resolution of the Twenty-Fourth Party Congress included the full-scale OGAS project, but shortly before the congress the Politburo decided to scale it down drastically. Instead of creating a powerful State Committee on the Improvement of Management, the Politburo resolved to set up a much more modest agency, the Chief Administration for Computer Technology. The elaboration of mathematical models for the national economy and the implementation of automated management systems were put off, and only the least ambitious, largely technical part of the plan—the construction of a state network of computation centers—survived.¹⁰³

While losing its top-level reformist component, the revised OGAS project acquired a completely different political dimension: it now served to conserve the existing power structures and administrative hierarchies, rather than reform them. Resorting to cyberspeak, one Western observer

has ascribed the failure of a nationwide automated management project to “bureaucratic entropy”:

The massive computerization campaign aimed at alleviating bureaucratic entropy was itself mired in bureaucratic entropy. . . . With each enterprise fending for itself, a confusing variety of incompatible programs, each written in a different language but performing similar and related tasks, appeared. Thus, instead of aiding the integration of administration, computerization actually reinforced existing organizational boundaries.¹⁰⁴

Behind this “entropy,” however, one could find quite deliberate efforts by the Soviet bureaucracy to preserve local control of information.

Unlike central economic organs, Soviet military and intelligence agencies quickly realized the opportunities for large-scale information collection and processing that computer technology had opened up. In September of 1968 the Party Central Committee and the Soviet Council of Ministers issued a joint decree, titled *On the Improvement of Scientific and Technical Information in the Defense Industry and the Exchange of Information Among Different Branches of the Economy*, that instructed the State Committee on Science and Technology to set up a Special Scientific Information Center, equipped with new computers imported from France, to supply scientific and technical information to the Chief Intelligence Administration of the Soviet Armed Forces General Staff, the State Security Committee (the KGB), and other government agencies. The center was created on the basis of the Department of Scientific Institutions of the All-Union Institute of Scientific and Technical Information. In 1963–1968 this department compiled dossiers on 70,000 foreign scientists, engineers, and executives and on 5,000 companies from Western countries. The department produced 284 analytical reports on the leading military industrial and scientific research centers in the United States, England, France, Italy, West Germany, and Japan, and it provided detailed responses to more than 2,000 specific queries from the Chief Intelligence Administration and the KGB. The department distributed its classified publications to 420 addressees, including 35 copies for the Central Committee. With the introduction of computer technology and the organization of the Special Scientific Information Center, it was planned to expand the existing database to keep dossiers on some 500,000 leading foreign scientists, weapons researchers, and company executives, including information on their places of work, their professional activities, their political views, and their visits to the Soviet Union.¹⁰⁵

While downscaling Glushkov's plans for a nationwide automated management system, Party leaders displayed interest in introducing computer technology to rationalize the functioning of the Party apparatus. One Party ideologue argued that the proposed nationwide network of computation centers for economic management "can be used—and it should be used—for gathering, processing and analyzing information on socio-political and ideological processes as well, for the purpose of optimal management [of society]."¹⁰⁶ In September of 1969 CEMI submitted to the Central Committee a proposal to create an Information Computation Center for the Central Committee apparatus. CEMI proposed to design an "information system" that would combine several functions:

- 1) communication system: fast and reliable collection, distribution, and transmission of information;
- 2) computation system: various computation tasks, including the calculation and comparative assessment of different alternatives;
- 3) integrated data-processing system: complex automated processing, preparation, and distribution of documents;



Figure 6.5

Party leaders Leonid Brezhnev and Mikhail Suslov viewing the latest model of the PROMIN' computer at an exhibition in Moscow. Courtesy of Virtual European Museum of Computer Science and Technology.

- 4) information-management system: accumulation and organization of data on various topics, processing of information requests, and preparation of briefs;
- 5) information-gathering system: collection of new information in monitored fields; processing and supply of this information;
- 6) directive-organization system: systematic organization of directives and instructions for various agencies and officials for specific situations, and transmission of these directives at a set time;
- 7) optimization system: evaluation of various alternatives in accordance with set directives, and selection of the optimal variant through mathematical analysis and programming;
- 8) experience-accumulation system: systematic organization of decisions and methods of decision-making, and analysis of the best decision-making methods in specific situations; and
- 9) controlling system: monitoring of all decisions and persons responsible for their implementation, and informing the superiors of the fulfillment (or non-fulfillment) of their decisions at all levels.¹⁰⁷

According to the proposal, the Central Committee Information Computation Center was to keep dossiers on Party, government, and industrial cadres; to process and analyze petitions from ordinary citizens; to process information on population dynamics and on moral, cultural, and political values; to accumulate information on foreign social movements and political organizations; to provide computer simulations of various international crisis situations and select optimal solutions; to analyze the effectiveness of various propaganda and counter-propaganda methods in the media; and to recommend how better to organize ideological work.

Soviet bureaucrats, in a way, learned the lessons of cybernetics better than did some overenthusiastic cybernetic reformers. Instead of creating a nationwide network, Soviet computerization efforts resulted in a patchwork of incompatible information-management and production-control systems. Instead of facilitating the decentralization of power through computer simulation of market mechanisms, computer technology now served to strengthen centralized control within each ministry. The growing power of ministries quickly reduced the autonomy of individual enterprises to a minimum, and economic reforms were effectively buried. The idea to reform the government with the help of a nationwide automated management system was abandoned. Information gathered at secret information-processing facilities, usually designated as “special computation centers,” remained accessible only to the top layer of the Party apparatus and the government bureaucracy. Instead of upsetting the existing power structures, cybernetics was enrolled to reinforce them.

CyberNewspeak: “The Scientific Management of Society”

The Soviet establishment not only put computer technology to its service; it also successfully adapted cybernetic theory to its needs. In the late 1960s, cybernetic ideas were incorporated into the writings of a leading Party theoretician, the philosopher Viktor Afanas'ev, Deputy Editor of *Pravda* since 1968, who later rose to become Editor-in-Chief of the journal *Kommunist* and eventually Editor-in-Chief of *Pravda*. Adopting terms from cyberspeak, Afanas'ev began talking of “social information” and “the scientific management [*upravlenie*] of society.”¹⁰⁸ Instead of exposing the limits on the free circulation of information in the Soviet Union, as Wiener had once hoped, the cybernetic analysis of social processes was now employed to rationalize the management of Soviet society and to ensure its stability.

During the early anti-cybernetics campaign, Soviet critics had attacked cybernetics for being a “technocratic theory.”¹⁰⁹ Now the ideological attitude toward technocratic aspirations of cyberneticians was completely reversed. In 1967 the authors of the fifth volume of *Cybernetics—in the Service of Communism* wrote with pride that “the view of society as a complex cybernetic system with a multi-dimensional network of direct and feedback links and a mechanism of optimization, functioning towards a set goal, is increasingly gaining prestige as the main theoretical idea of the ‘technology’ of managing society.”¹¹⁰ Cyberneticians emphasized Lenin's interest in Taylorism and his attempts to rationalize the organization and functioning of the Soviet apparatus, and portrayed Soviet cybernetics as the heir to this Leninist tradition. Berg's Council on Cybernetics played a crucial role in the ideological rehabilitation of the legacy of Aleksei Gastev and other Soviet pioneers of the “scientific management” movement of the 1920s.¹¹¹

Afanas'ev quickly translated the basic principles of operation of the Soviet government into cyberspeak. He portrayed socialism as “a scientifically controlled society.” The government, the Communist Party, and other political and public organizations constituted the controlling subsystem, while the economy, science, and other social activities made up the controlled subsystem. The Party, “the most important element of the scientific control of socialist society,” played, of course, the role of the chief controller, whose function was to bring “men's subjective activities into line with the requirements of objective laws.”¹¹² Afanas'ev argued that all managerial activities, from running a factory to directing the building of

communism in the Soviet Union, followed the same cybernetic “control cycle”:

The decision has been made, the controlled and controlling systems set in order and regulated so as to maintain and perfect the order, the processes have been checked and corrected, the results summed up by the system of accounting and inspection. On the basis of information about the results another decision is taken and the cycle is resumed.¹¹³

The flexibility of both newspeak and cyberspeak made this translation incredibly easy: the Party principle of “democratic centralism,” for example, could easily be interpreted as control by means of feedback. Afanas’ev formulated the cybernetic approach to the management of society in such a way that it displayed, as the historian David Holloway has put it, “a fundamental congruence” with the Soviet ideological discourse on government:

In both cases, there is an emphasis on control and management: both are purposive and goal-oriented; both claim to be, in some sense, scientific; both underline the need for a systemic or holistic approach; the one seeks policies, the other optimal solutions; the one stresses democratic centralism, the other the hierarchical nature of control systems.¹¹⁴

Afnas’ev’s concept of the “scientific management of society” fitted well with the pragmatic agenda of Brezhnev’s regime. As Holloway has argued, this concept was based on the rationality “appropriate to a society which is regarded as being in a state of equilibrium, and hence differs from the rationality of Stalinist rule, with its goal of transforming Soviet society.”¹¹⁵ Indeed, in Afanas’ev’s interpretation, cybernetics was no longer a vehicle of cardinal reform; it was now a system of analytical and technical tools for maintaining the stability of the Soviet economy and society. Quite understandably, Soviet ideologues embraced the former “reactionary pseudo-science”—not because they had been particularly impressed by cyberneticians’ mathematical models, but because they had transformed cybernetics into a convenient ideological tool that served a useful political purpose. In Afanas’ev’s version, the cybernetic model of society would help make the existing social system function more efficiently without changing its fundamental features. This model promised to find technical solutions for the Soviet Union’s social and economic problems without endangering the hierarchies of political power. As Holloway has argued, the cybernetic model emphasized technocratic rationalism but neglected the question of democratic control over the state apparatus:

Through its scientific claims, its instrumental approach and its congruence with the Soviet concept of government, the cybernetic model legitimates the Soviet system as a whole, in the very process of carrying through partial reforms, precisely by pointing to the possibilities that the system offers for scientific and rational government. Therefore, the cybernetic model promises to rationalize the existing system of power relationships, not to destroy it.¹¹⁶

The appropriation of cyberspeak by Soviet officials became the subject of thorough ridicule in *Ziiaushchiie vysoty* [*The Yawning Heights*], a venomous satire of Soviet society written by Aleksandr Zinov'ev, a prominent Soviet logician and an independent-minded thinker, in the mid 1970s. Afanas'ev, with whom Zinov'ev had served on the editorial board of the journal *Problems of Philosophy*, was portrayed in this book as "a very stupid philosopher."¹¹⁷ Framed as a "scientific" treatise on "socio-mechanics," the first part of Zinov'ev's book parodied Soviet intellectuals' attempts to explain rationally the workings of a socialist regime. If one tried to apply logical deduction to the understanding of society, wrote Zinov'ev, one would quickly make the following discovery:

Deduction is impossible owing to the excess of data, the multiplicity of initial concepts and assumptions, the paucity of deducible consequences, and the practical uselessness of what is deduced. All this has a dispiriting effect on the modern scientist, whose mind has been stuffed full of mathematicization, formalization, model-making, and so on.¹¹⁸

Poking fun at Soviet cyberneticians' hopes to improve the government by introducing efficient procedures for collecting and processing information, Zinov'ev observed with sarcasm that higher authorities did not need information at all:

Information, by definition, could never be truthful and complete. No information at all was needed for the normal functioning of society, and the leadership's instinct was correct: to inflate worthless trivia, hush up important events and do all our rethinking for us.¹¹⁹

When one of the characters mentioned the government's attempts to "perfect society in such a way that it becomes easier to govern scientifically," another immediately replied: "You surely can't be unaware of the attempts that have been made over the last decade to improve and simplify the apparatus of control? How have they ended? Things are now more confused and tangled than they were before."¹²⁰ Zinov'ev explicitly scorned the belief that cybernetics could help bring social harmony by means of some clever control mechanisms. As one of the characters in his book argued, social

stability “is only the resultant of all the various forces at work, and can only be achieved if it is in complete accord with their social nature. It is far from being the realization of some ideal cybernetic system of control.”¹²¹

Zinov’ev dismissed the convergence of cyberspeak and newspeak in the official Soviet discourse as “methodological phraseology.” He scoffed at “generalizing theories” based on such fashionable concepts as “systems, models, structures, functions, and information,” ridiculing their “universal” applicability:

To get some idea of these generalizing theories, the reader might try, for example, to construct a methodological theory of functions which would embody mathematical functions, a functional approach and the function of the trade unions in the textile industry. In the final analysis, the kaleidoscopic use of words like ‘system’, ‘systematic’, ‘information’, ‘structure’, ‘functional’, ‘model’, and so on, in the powerfully running tide of methodological literature, more often than not merely reflects the feel of the age, and not the results of serious research.¹²²

Zinov’ev, the leading Soviet specialist in the philosophical questions of mathematical logic, had once been actively involved with a group of logicians and mathematicians participating in the early cybernetic research. In the late 1950s, he had organized an open seminar on logic at the Academy Institute of Philosophy; several figures prominent in the cybernetics movement had given talks at that seminar, including the philosopher Ernest Kolman, the logician Sof’ia Ianovskaia, the mathematician Aleksandr Esenin-Vol’pin, the linguist Sebasti’an Shaumian, and the mathematician Gelii Povarov.¹²³ Later, however, Zinov’ev distanced himself from the now-fashionable cybernetics. This move was typical of the early generation of cybernetics enthusiasts, who suddenly found their original ideas turned upside down. Instead of equipping academic discourse with mathematical and logical rigor and precision of formulations, as they once had hoped, cybernetics turned into a flexible ideological tool, another newspeak, only now filled with cybernetic terminology.

The End of the Cybernetics Game

By the early 1970s, the cybernetics movement had changed completely in character. It no longer challenged the orthodoxy; instead, tactical uses of cyberspeak overshadowed the original reformist goals that aspired the first Soviet cyberneticians. Igor’ Poletaev (a close associate of Liapunov and the author of the first Soviet book on cybernetics), who had once fought to

legitimize cybernetic research, bitterly told his friends in the 1970s: "Now it is I who will say that cybernetics is a pseudo-science."¹²⁴ Cybernetics became part of the dominant discourse that early cyberneticians despised, and they no longer wanted to call themselves cyberneticians.

Former leaders of the cybernetics movement grew increasingly disillusioned with the claim of universality of cybernetic methods. "Enough talk about the unity of all control systems and the omnipotence of cybernetics," Poletaev told his colleagues. "It's time to start working, building concrete models, and studying concrete problems. Enough philosophy; let's get to work."¹²⁵ The indiscriminate use of cybernetic models without any respect to their validity in specific situations alarmed those mathematicians who recognized the limits of cyberspeak. Citing the infamous project to divert Siberian rivers, Izrail' Gel'fand, a leading specialist in mathematical biology, has argued: "I am convinced that in our computer age only mathematicians can prevent the abuse of 'mathematical methods in . . . ' (biology, medicine, ecology), when only one small, primitive element is taken out of context, and on its basis people try to solve problems which in their essence are not mathematical."¹²⁶ In a 1970 samizdat article, Aleksandr Esenin-Vol'pin, a leading specialist in mathematical logic, called "the rapture over mathematical applications" one of the major obstacles in the way of rigorous scientific thought:

[Natural scientists] realize that their disciplines are insufficiently rigorous, which undermines the reliability of their conclusions, and they are trying, wherever possible, to connect their work with mathematical methods. . . . This is, however, a form of submissiveness. [One must] learn to look at the use of mathematical apparatus in various fields from a critical perspective. . . . Natural scientists' reliance on mathematical rigor has an obvious deficiency: what they constantly use is in fact *applied* mathematics, that is, precisely the area of study where mathematicians say goodbye to rigor, while remaining loyal to mathematical methods in all other respects. Traditional mathematics has achieved its rigor by adopting idealizations, which do not correspond to anything in reality.¹²⁷

After the end of Khrushchev's "thaw," the Soviet scientific community began to polarize. A group of cyberneticians at Moscow University, led by Sergei Iablonskii and Oleg Lupanov, took an openly anti-Semitic stance and almost entirely barred Jews from studying in the Faculty of Mathematics and Mechanics or publishing in *Problems of Cybernetics*, edited by Iablonskii after Liapunov's death in 1973.¹²⁸ In March of 1968, at the other pole, 96 Moscow mathematicians, including such leading cyberneticians as

Gel'fand, Liusternik, and Markov, signed an open letter protesting the forced confinement of Esenin-Vol'pin in a mental institution for his dissident activity. Several of the signers lost their jobs as a result of their defense of Esenin-Vol'pin.¹²⁹

After the 1968 Soviet invasion of Czechoslovakia, Soviet authorities launched a crackdown on independent-minded intellectuals and began “purging” Soviet science of dissidents. Several politically active linguists, including Iurii Apresian, Alexander Zholkovsky, and Lidiia Iordanskaia (Igor' Mel'čuk's wife), lost their jobs at linguistics institutions and found a refuge at Informelektro, the information center for the Moscow Institute of Electrical Engineering, where they began working on a machine translation project.¹³⁰ The consequences were particularly dramatic for Mel'čuk, who was never inclined to play “according to the rules of the game accepted by the greater part of the linguistic community.”¹³¹ Mel'čuk ignored the official hierarchy and deliberately dropped out of the rat race for titles and degrees. His independent style eventually led to a conflict with the political authorities. The publication of his book on the “Meaning—Text” linguistic model, completed in 1968, was delayed 6 years until the book “passed through the censors by accident.”¹³²

The cybernetics movement split into two opposite branches: while cyberspeak was being appropriated by the academic establishment, some independent-minded cyberneticians came into political conflict with the authorities and lost any opportunity to do cybernetic research in the Soviet Union. Many dissident cyberneticians decided to emigrate. The physicist Valentin Turchin, author of the cybernetic treatise *The Phenomenon of Science* and an associate of Andrei Sakharov, moved to the United States.¹³³ The engineer Alexander Lerner, author of *Fundamentals of Cybernetics*, struggled with the authorities for 10 years seeking permission to emigrate to Israel. Unable to continue research after losing his job at the Institute of Automation and Remote Control, Lerner organized a series of weekly home seminars for scientist refuseniks (individuals who had been denied exit permits) on the problems of control and mathematical models in biology and medicine—in other words, on cybernetics.¹³⁴ Mel'čuk, who was similarly denied research opportunities, wrote to Roman Jakobson in July 1974 that his life in the Soviet Union had become “totally unbearable” and asked Jakobson for help in finding a position at an American university.¹³⁵ On 26 January 1976 the *New York Times* published Mel'čuk's open letter protest-

ing against the Soviet persecution of prominent dissident scientists, Andrei Sakharov and Sergei Kovalev. For the authorities, this was the last straw. In March of 1976, despite Mel'čuk's high reputation as a scholar and as the author of numerous first-rate publications, his superiors at the Institute of Linguistics gave him a negative evaluation; shortly afterward, they dismissed him from his job. In a new open letter to his Western colleagues, he wrote:

For several years I have had practically no possibility to publish my papers in leading Soviet linguistic journals. . . . The second volume of my monograph *Toward a Theory of Linguistic Models of the Meaning—Text Type* (the product of many years of painstaking work) waited for publication more than 8 years and failed to be, after all, published. I was and am forbidden to teach, to take part in many scientific conventions, to go abroad for contracts with Western colleagues. Immediately after the appearance of my letter in the *New York Times* Soviet linguistic periodicals and publishing houses began suppressing references to my works, acknowledgements by other authors mentioning my name . . . and even my name as the editor or translator. . . . Under such conditions and having no permanent job I am left without the least possibility for normal continuation of my linguistic research.¹³⁶

Nobody wanted to hire Mel'čuk (for obvious reasons), and soon the police informed him that, as a "parasite" and a "sponger," he was to be deported from Moscow and perhaps imprisoned.¹³⁷ References to his works disappeared from print; his ideas were thus effectively banned. According to one memoirist, Mel'čuk's expulsion from Soviet linguistics marked the end of the "Silver Age" of Soviet structural linguistics.¹³⁸ In May of 1977 Mel'čuk emigrated from the Soviet Union and accepted a position at the University of Montreal. He no longer wanted to play the cybernetics game. He even called one of his own articles on the connection between cybernetics and linguistics "showy and shallow."¹³⁹

In *The Yawning Heights*, Zinov'ev drew a vivid picture of the gradual emasculation of cybernetic discourse and the decline of the cybernetics movement as cybernetics was enrolled to glorify "the ism" (socialism, communism, Marxism-Leninism, etc.):

The Newspaper published an editorial directive *Cybernetics in the Service of the Ism*. . . . [The editorial] made the direct and straightforward statement that the true scientific comprehension of cybernetics was first arrived at by the classic authors of the Ism, who, even though they never heard of cybernetics, had been able to leave some appropriate quotations for posterity. . . . Soon on every street there blossomed Institutes and Laboratories of Cybernetics and the other new sciences which were now of inestimable value in the development of the Ism. . . . The reactionary forces knew from experience that new ideas only begin to bring new

support to the Ism once they are hopelessly out of date and have begun to grow boring. . . . Troglodyte was appointed chairman of cybernetics committee. Under his experienced leadership the progressive forces immediately proved that the new ideas upheld the truth of the Ism on its new stage of development, and were beginning to overtake the West on the cybernetic front. . . . “We are all cyberneticists now,” [said Teacher]. . . . “It’s only recently that we had no more than five or six cyberneticists, and they were all under police surveillance. They’d nearly all done time. Last week we had a symposium and more than a thousand specialists turned up. . . . We are moving into a boom. Things’ll be blown up beyond all measure. All manner of rabble will gather round trying to get in on the act. People will write theses, collect titles, decorations, prizes. . . . And then the boom will begin to blow over. In the meantime any scientists worthy of the name will have been eliminated and crushed. Then there’ll be a period of total disillusionment. Every idea of any unifying ideological significance will have been exhausted. All that will remain will be the usual official mass phenomenon.”¹⁴⁰

“The opposition must stop decking itself out in alien rags and tatters of science, art and economics,” Zinov’ev concluded, summing up the disillusionment of the former reformers. “It must speak out in its own name without resorting to camouflage.”¹⁴¹

In a way, Soviet cybernetics became a victim of its own success. First, Soviet reformers inflated the meaning of cybernetics, trying to push forward a wide range of innovations under the banner of the cybernetization of Soviet science. Later, the establishment inflated this meaning even further, attempting to secure informational and technological means to support their grip on power. Reformist ideas do not remain such forever: after becoming fashionable, they may turn into another dogma and a basis for new conservatism. When cyberspeak first challenged Stalinist discourse, it was a language of opposition and resistance; later, when cyberspeak replaced newspeak in its dominant role, it had nothing new to say.